

When Caring Backfires: Experimental Evidence from Family Caregiving in India *

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Abstract

Does concern for family caregivers discourage elders from obtaining healthcare? We develop a theoretical framework in which elders may withhold information about healthcare opportunities when they care about caregivers and believe their caregivers would be pressured to help. This inefficiently reduces take-up when caregivers would help willingly. We test three model predictions using two field experiments with elder-caregiver pairs in rural Tamil Nadu, India (N=1,175). First, directly informing caregivers about a check-up opportunity for the elder increases attendance by 30% (5.3 percentage points), relative to informing only the elder, suggesting elders withhold information relevant for take-up. The effect is strongest for elders who report being worried about burden at baseline, whose attendance more than doubles. Second, worried elders are less willing to share information about the check-up than not-worried elders and more sensitive to randomized reductions in caregiver costs, suggesting worried elders believe sharing information imposes costly pressure on caregivers. Third, caregivers' decisions to help are not sensitive to randomized observability treatments that vary pressure, suggesting caregivers willingly help out of care for the elder. Our results imply that elders' caring preferences can interact with misperceptions of caregiving burden to result in inefficiently low levels of health investment for older adults.

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1 Introduction

Households are collections of people whose actions and outcomes are highly interdependent – when do they work together and why? There are at least two views of why families help each other that sit in tension with one another. On the one hand, family members genuinely care about one another and value each other’s well-being (Becker, 1974). For example, parental altruism may motivate investments made in children’s education. On the other hand, family also comes with strong social norms and resulting obligations that pressure people to help, such as when workers have some of their financial returns captured by relatives (Jakiela and Ozier, 2015). Both of these are types of social preferences that usually encourage more helping within the family, but they imply very different things about what the help costs the person giving it. A challenge then arises when another family member in need of help cannot tell which motive underlies potential help that they expect. If they are very sensitive to the impact on the help-giver, then this can prevent them from asking for help. The combination of uncertainty about the help-giver’s preferences and caring for the help-giver can prevent them from communicating their request. This can decrease the amount of help given. Further, if the help would have been willingly provided, this would then be inefficient.

We study the tension between caring and pressure motives for cooperation in the context of healthcare decisions of elderly women and their family caregivers in rural India. This is a decision problem of tremendous scale and consequence, given the unprecedented growth of aging populations in many low- and middle-income countries (Banerjee et al., 2023). Take-up of preventative healthcare is low among older adults in India, who exhibit large gaps in awareness and treatment of their chronic conditions.¹ Family caregivers are integral for helping facilitate valuable care for the elder by providing transport, paying for visits, and helping make decisions for the elder (World Health Organization, 2023). At the same time, there is also a backdrop of strong social norms around familial caregiving: 87% of adults in India agree that it is the duty of adult children to care for their aging parents (Haerpfer et al., 2022). We posit that these strong, difficult-to-learn-about social preferences create scope for communication frictions to affect take-up of healthcare. If elders have caring preferences towards their caregivers and over-sensitive to burden they impose, this may lead them to withhold health information and reduce healthcare that otherwise would have been happily facilitated by the caregiver. This dependence is top of mind for elders. When we spoke with elders about their health and key barriers to health take-up, worry about burden was something that often came up. For example, nearly 40% of older women in Tamil Nadu report being

¹In India, an estimated 45% of individuals with hypertension and 40% of individuals with diabetes are unaware of their condition; and even among those diagnosed, less than half have achieved glycemic or blood pressure control (Mohanty et al., 2021; Sekher et al., 2025).

worried about being a burden to family members. Among them, 63% report hiding health problems and 78% report avoiding the hospital out of not wanting to burden others.

We test whether worry about burden leads to elders withholding information about health needs from caregivers, and whether this reduces take-up of health investments for elders. We propose a model that allows us to capture the elder’s strategic decision to share or withhold information. As a result, health take-up is affected by what elders and caregivers know, as well as how much they care about one another. These are dynamics that standard models of group decision-making typically do not let us capture. We test the predictions of the model using two field experiments in rural Tamil Nadu, India with pairs of elders and caregivers.

We first develop a game-theoretic model in which an elder depends on her caregiver to obtain healthcare, but she worries that informing the caregiver may cause them to feel pressured to help. Our model yields predictions about three separate behaviors that vary with the strength of elder and caregiver caring preferences: take-up of the check-up, information sharing by the elder, and willingness to help by the caregiver. In the *household experiment* (N=731), we test whether communication frictions decrease health take-up for the elder, and we test whether communication patterns differ by elder worry about burden. In the *caregiver experiment* (N=444), we explore whether any frictions in communication and take-up found in the household experiment are inefficient by testing whether the caregiver’s decision to help comes at potential costs of inducing pressure-based help or crowd-out of future help.

Model. Our game theoretic model of joint decision-making between elders and caregivers aims to capture the counter-intuitive idea that caring preferences may reduce coordination within the household, in the presence of asymmetric information about preferences. The elder first gets to decide whether to inform their caregiver about a health opportunity. The caregiver has two potential motivations for facilitating a check-up for the elder: genuine care for the elder or external pressure to fulfill the request.² The elder does not know the caregiver’s motivations for helping, so they may be concerned that pressure causes a not-caring caregiver to agree to help even if it makes the caregiver worse off. “Worried elders” are those who both care about caregivers (i.e. they internalize the caregiver’s well-being) and believe there is pressure associated with the information sharing, so they may be dissuaded from asking by these potential costs. The core feature of our model is that the elder’s misperceptions about caregiver motives can lead to inefficiently low communication, which scales with the degree of mutual caring between elder and caregiver.

²We model pressure that can come from several external sources, including social pressure from either the elder or other individuals such as family members or neighbors, as well as psychological pressure that can come from the caregiver’s concern for self-image or guilt.

Our model yields three sets of predictions. First, take-up of the check-up varies with the recipient of the check-up information only when there is information withholding from the elder, and the magnitude of the effect is informative about how binding the information constraint is. Second, elder worry affects willingness to share information in both levels and elasticities: worried elders are (i) less likely to inform when they believe that caregivers are subject to pressure, and (ii) more sensitive to variations in caregiver costs. Third, caring preferences is a key motivation for caregiver helping if the caregiver’s helping behavior is *not* responsive to changes in (i) observability and (i) degree of prior help provided. We then test these predictions with two field experiments.

Household experiment. In the household experiment (N=731 pairs of elderly women and their caregivers), we offer cardiac health checkups for the elderly at a nearby private hospital. The check-up covers heart screening tests that are desirable to elders, not broadly available in the existing public health scheme, and have high market-value, at around \$150-250 USD, adjusted for PPP (Subramaniam et al., 2025). The household experiment tests the first two sets of model predictions about check-up take-up and the elder’s communication decision. Our main intervention randomizes who in the household is informed of the check-up, and we test whether this affects take-up of the check-up. In the *No Inform* group, only the elder is informed about the check-up. In the *Inform* treatment group, both the elder and the caregiver are directly informed about the check-up, closing any communication gaps between elders and caregivers. The intervention is designed to isolate the effect of giving the caregiver information about the check-up. For example, all elders are given official, physical vouchers with invitation details to rule out competing explanations of elders forgetting to inform caregivers or not having credibility when describing the check-up. Further, to test how the elder’s information sharing behavior responds to changes in caregiver costs, we also cross-randomize the free provision of *Transport* to and from the check-up, which reduces the caregiver’s cost of involvement.

Our first result shows that elders withhold information about the check-up from their caregivers that decreases take-up. When only the elder knows about the check-up, take-up is low. Only 26% of households sign up for the check-up by calling a hotline to reserve their spot at the check-up. Even fewer attend the check-up, with only 18% of invited elders actually attending. However, directly informing the caregiver increases take-up significantly: sign-up rates increase by 7.7 percentage points ($p = 0.02$) and attendance by 5.3 percentage points ($p = 0.07$). Viewed through the lens of our model, the positive treatment effect reveals that the elder’s information sharing constraint is indeed “binding” for some households. In other words, there exist households where both the elder and caregiver would be willing to go, but the elder does not initially inform their caregivers about the check-up.

Further, the intervention has the strongest impact for the subset of the sample who identify as *worried* elders. Empirically, we define worried elders as those who report being so worried about being a burden that they have forgone healthcare treatment (26% of elders in our sample).³ Among worried elders, informing the caregiving has a strikingly large impact: attendance increases by 16.9 percentage points, compared to a baseline attendance rate of 16.2%, more than doubling take-up ($p = 0.01$). Meanwhile, the effect for less worried elders is small and statistically insignificant (1.3pp, $p = 0.7$). The difference between the treatment effects is statistically significant ($p = 0.03$). Our results are not explained by differences in demographics or other observables between more- versus less-worried elders.⁴ Taken together, these results suggest that elders' decision to share information with the caregiver has direct consequences for take-up, especially for elders who are most sensitive to the costs imposed on caregivers. We thus directly measure the elder's information-sharing decision, and we characterize how it relates to elder worry.

Our second result is that elders' willingness to share information about the check-up differs by worry, in both levels and elasticities. We directly elicit the elders' willingness to share information about the health check-up among elders in the *No Inform* group. After the invitation is delivered, elders are asked if they are willing to have information about the check-up be shared with their caregiver. We test two model predictions about the elder's willingness to inform. Worried elders' decisions to share information is more elastic to changes in caregiver cost. Consistent with our model, we find that offering transportation support increases willingness to inform by 15 percentage points for worried elders ($p = 0.07$). In contrast, willingness to inform of less worried elders are unchanged by the availability of free transport (-1.1 percentage points, $p = 0.81$). The difference in responsiveness is statistically significant ($p = 0.07$). We also find that worried elders exhibit lower levels of information sharing. Less worried elders are willing to inform the caregiver about the check-up 85% of the time. More worried elders are significantly less likely to be willing to inform caregivers about the check-up, by 14.2 percentage points ($p = 0.04$). Our model allows us to use this comparative static as a diagnostic that reveals elders' beliefs about caregiver motivations for helping: when elders believe that helping is net welfare-decreasing for the caregiver – i.e. that help comes from pressure rather than true caring preferences – communication declines as elder worry increases. Together, our two results on elder information sharing shed light on how caring preferences shapes communication in our context: worried elders are most sensitive to caregiver utility, and they believe caregivers are made worse off from helping. As a result, worried elders are more often withholding information out of worry about burdening others than their less-worried counterparts.

³We are able to rule out several other interpretations of worry about burden, such as sensitivity to rejection, (self-regarding) costs of asking, concerns of pride or ego, uncertainty about whether the caregiver will agree to the request, and lower objective health in the Discussion section.

⁴When we control for imbalanced observables and their interaction with an indicator for the *Inform* treatment status, the heterogeneity effects by worry about burden remain large and statistically significant.

Caregiver experiment. Our caregiver experiment allows us to test model predictions about the caregiver’s motivations for providing help to the elder. To do this, we measure the caregiver’s willingness to help their elder for N=444 caregivers, recruited in pairs following the same method as the household experiment. We cross-randomize two interventions that vary the decision-making environment for caregivers. The first intervention randomizes how observable the caregiver’s decision is, to disentangle the roles of caring preferences and external pressure. The second intervention randomizes whether the caregiver is asked to sign up for a check-up for the elder or not, to test whether the caregiver’s decision to help in one domain leads to crowd-out of help in other domains.

Our third result is that caregiver involvement is not driven by pressure and does not crowd out future help to the elder, suggesting that the additional help from inform caregivers does not come at cost to the elder. We test whether the caregiver’s involvement is driven by external pressure by randomizing each caregiver into one of three arms that vary the observability of their decision to sign up for the check-up for the elder. In the *Public* arm, the caregiver makes their decision verbally, in front of the enumerator, and they are told that their decision may be shared with the elder – meaning the caregiver’s decision is driven by a combination of caring preferences, social pressure (from the enumerator and elder), and psychological pressure (from self-image or guilt concerns). In the *Private* arm, the caregiver makes their decision privately on the enumerator’s tablet, removing the influence of social pressure. In the *Private + Delay* arm, the caregiver can decide privately or the caregiver can forgo the choice by delaying it until later, further removing psychological pressure from even the caregiver’s own self-image or guilt concerns (Dana et al., 2007; Lazear et al., 2012; DellaVigna et al., 2012). To test for pressure-based motivations, we measure whether the caregiver’s willingness to provide help to the elder changes when we experimentally vary the observability of the caregiver’s decision to help the elder. We find no impact of either the *Private* or *Private + Delay* interventions on caregiver willingness to help, relative to the *Public* arm. This suggests that the caregiver involvement observed in our experimental context reflects true caring rather than either social or psychological pressure. Taken together with our result from the household experiment that elders believe caregivers help out of pressure, this evidence suggests that low asking rates from elders in the household experiment are indeed inefficient and reflect a misperception about pressure concerns held by elders. Further, stronger mutual caring then exacerbates these inefficiencies, leading to the largest gaps in take-up precisely when elders and caregivers would benefit the most attending the check-up.

We then test whether dynamic considerations are important for modeling the interaction between elders and caregivers. Specifically, we test whether caregiver involvement for the check-up crowds out future involvement from the caregiver. If caregivers have a binding budget constraint of resources for helping the elder, then caregiver involvement in the health check-up may come at the cost of involvement in other

domains or for future requests. To test for crowd-out, we vary whether the caregiver has already been asked to commit to helping the elder with one task (health check-up) before asking about willingness to commit to helping the elder with other tasks spanning different time frames (willingness to be involved in the case of an urgent health emergency, interest in a health check-up in one year) and non-health domains (signing up for social activities). We find no evidence of crowd-out at any margin: involvement in the check-up does not come at the cost of other support to the elder. As a result, we can conclude that the static model is appropriate and that the increased take-up rates likely do not come at the expense of forgone future care for the elder.

Our work is most closely related to three strands of the literature.

First, we contribute to the literature on information transmission within the household. Recent literature has documented intra-household failures in communication, showing how asymmetric information and hidden actions can lead to inefficiency in household decision-making (Ashraf, 2009; Ashraf et al., 2014; Ashraf et al., 2026; Conlon et al., 2021; Apedo-Amah et al., 2020). While existing work primarily focuses on husbands and wives, our paper is the first to provide experimental evidence on the household decision-making of elders and their caregivers, which are often intergenerational relationships that are less studied (Almas et al., 2023). Because the relationship dynamics differ, the key drivers of communication decisions in our setting are caring preferences and uncertainty about others' preferences, rather than strategic capture of decision-making power.

Second, we contribute to a literature on low take-up of preventative health in low- and middle-income countries (Dupas, 2009; Banerjee et al., 2004; Bai et al., 2021), especially for older adults (Banerjee et al., 2023). Further, several strands of literature have documented that healthcare disparities are a deeply gendered issue in India, with stark gender gaps in healthcare for older women especially (Calvi, 2020; Dupas and Jain, 2024). Our paper provides evidence that lower willingness to communication may be one contributing factor to lower health investments for elderly women, and we show that a simple intervention increasing communication can increase take-up of healthcare, especially for elders who are most hesitant to communicate.

Third, we contribute to the literature on social preferences. Usually, introducing pro-social preferences such as altruism or caring tends to increase the likelihood that decision-makers work together. However, we provide evidence on how mutual other-regarding preferences can exacerbate communication failures between decision-makers in the presence of other frictions, such as asymmetric information about preferences. Fehr and Charness (2025) provide an overview of the literature on social preferences, highlighting the recent class of models of social preferences in which beliefs and signals about intentions matter for

utility, meaning communication and social preferences can play a large role in determining outcomes. On the theory side, our framework of communication is most closely related to Bénabou et al. (2025), who develop a theoretical model to capture why an individual may not communicate their need for help to an altruistic potential helper. In contrast, our model can be viewed as allowing for bi-directional altruism from not only the potential help-sender but also the help-seeker, allowing us to capture a new reason for low communication that is driven by altruism from the help-seeker. Our caregiver experiments also help contribute to a literature on disentangling social pressure and altruism as drivers of pro-social behavior (DellaVigna et al., 2012; Dana et al., 2007; Andreoni and Rao, 2011; Andreoni et al., 2017; Trachtman et al., 2015; Ambler, 2015). We contribute to this literature by testing for the presence of both external social pressure and internal pressure as drivers of pro-social behavior.

2 Theoretical Framework

2.1 Setup

We model the process of an elder e and their caregiver c deciding whether to obtain healthcare for the elder, such as attending a health visit at the clinic. To attend, the elder needs the caregiver to help facilitate the visit (e.g., by providing transportation or paying for the visit). Although we allow elders and caregivers to potentially have caring preferences for one another – i.e. they may internalize each other’s pay-offs, as in Almås et al. (2018) – we model the decision-making process as a non-cooperative game, in which the elder decides whether to inform the caregiver about the health visit.

Our model aims to capture the idea that stronger mutual caring preferences may reduce information sharing within the household when the elder is uncertain about the caregiver’s preferences. An elder who cares about her caregiver may withhold information about her own health needs because she believes that the caregiver would feel obliged to help at a net cost to themselves. In this case, directly informing the caregiver then may increase take-up by removing the elder’s willingness to inform constraint. Whether this increases take-up depends on whether the caregiver would actually help once informed. These types of information frictions within the household are ruled out in standard models of household decision-making, such as the collective or unitary models (Chiappori and Mazzocco, 2017).

2.1.1 Timing

The elder always knows about the health visit. The caregiver knows their own pay-offs. The timing of interactions is as follows:

1. **Nature decides who is informed.** The elder always knows about the health visit opportunity. Nature determines whether the caregiver knows about the visit or not.
2. **(Only if caregiver is informed) Caregiver offers.** If Nature determines that the caregiver knows about the health visit opportunity, the game has an additional stage in which the caregiver decides whether to offer to facilitate the visit.
3. **Elder informs.** If the caregiver has not offered to go (which is always true when Nature does not inform the caregiver), the elder decides whether to inform the caregiver of the health visit.
4. **Caregiver decides whether to attend.** Once informed, the caregiver chooses whether to help facilitate the visit. If she chooses to help facilitate, the elder attends the visit.

We simplify the interaction such that once the caregiver is informed about the visit, the caregiver determines whether the visit occurs or not. We do not include a subsequent stage at which the elder can cancel. This means when the caregiver is directly informed, we assume that the caregiver deciding to help then determines whether the visit occurs.

2.1.2 Pay-offs of health visit

Let Y denote take-up, with $Y = 1$ when the elder attends and $Y = 0$ otherwise.

We let the elder and caregiver have caring preferences towards one another, α^e and α^c , which determine the extent to which they internalize each other's pay-offs (Almås et al., 2018).

Caregiver pay-offs. Facilitating the health visit comes at cost $\kappa > 0$, where κ includes the costs of time, travel, and other caregiving effort. The caregiver values the elder's direct health benefit at rate α^c . Their net pay-off from helping is

$$u^c := \alpha^c B - \kappa. \tag{1}$$

If the caregiver is informed but declines to help, they incur a pressure cost $-\phi \leq 0$. This pressure can reflect internal self-image or guilt concerns, relational pressure from the elder, or social pressure from others (Bénabou and Tirole, 2006; Fehr and Charness, 2025). Suppose that u^c has distribution F among caregivers, and there is a single value ϕ . If the caregiver is not informed, they do not incur any pressure costs of not attending.

Let $K_c = 1$ indicate that the caregiver knows about the visit. Then the caregiver's utility is given

by:

$$U_c = \begin{cases} u^c & \text{if } K_c = 1 \text{ and the caregiver helps,} \\ -\phi & \text{if } K_c = 1 \text{ and the caregiver declines,} \\ 0 & \text{if } K_c = 0. \end{cases} \quad (2)$$

The caregiver helps if their net pay-offs of helping exceed the pressure costs of declining, $u^c \geq -\phi$. Importantly, if the caregiver faces enough pressure, the caregiver may choose to help even when helping yields negative pay-offs, i.e. $u^c < 0$.

Note that in the case where the caregiver directly knows about the visit already, we model the caregiver's decision to offer help as the same as their decision to agree to help when informed by the elder. Put differently, we assume that the pressure term ϕ applies once the caregiver knows about the health visit, regardless of whether it was Nature or the elder who informed them. This assumption reflects the fact that we are modeling the impact of having information about the health visit rather than being persuaded to attend.

Elder pay-offs. The elder receives direct benefit B from attendance and internalizes caregiver pay-offs U_c at rate α^e .

$$U_e = \begin{cases} B + \alpha^e u_c & , \text{ if } u_c > -\phi \\ 0 & , \text{ otherwise} \end{cases}$$

If the elder does not attend the health visit, she gets zero utility.

2.1.3 Information

Asymmetric information. We suppose that when the elder is making the decision to inform, the elder knows neither the caregiver's net pay-offs of attending the visit, u^c , nor the pressure they face ϕ . Thus she decides whether to inform based on her beliefs $\hat{u}^c \sim \hat{F}$ and $\hat{\phi}$.

For a given realization of the caregiver's net pay-offs of attending $u^c = u$, the elder's perceived pay-off from informing is

$$\hat{U}_e(u) = \begin{cases} B + \alpha^e u & \text{if } u \geq -\hat{\phi}, \\ 0 & \text{otherwise.} \end{cases}$$

The elder's decision to inform is determined by the her expectation of this pay-off under belief $u \sim \hat{F}$.

For a given set of costs and benefits κ, B , we can group caregivers into three categories: (1) willing helpers, who agree to help and have $u^c \geq 0$; (2) reluctant helpers, who agree to help and have $-\phi \leq u^c < 0$; and (3) decliners, who do not help and have $u^c < -\phi$.

Let $\hat{\mu}_i = \mathbb{E}_{\hat{F}}[u^c]$ be the elder's belief about the caregiver's expected pay-off of helping facilitate the visit for type i .

Let $\hat{p}_i$ denote the elder's beliefs about the caregiver being in category $i \in \{1, 2, 3\}$.

The elder's expected pay-off from informing is

$$\hat{V}_e = \hat{p}_1(B + \alpha^e \hat{\mu}_1) + \hat{p}_2(B + \alpha^e \hat{\mu}_2) \quad (3)$$

$$= \hat{p}B + \alpha^e \left(\underbrace{\hat{\nu}_1}_{\geq 0} + \underbrace{\hat{\nu}_2}_{\leq 0} \right) \quad (4)$$

where

$$\hat{p} := \hat{p}_1 + \hat{p}_2, \quad \hat{\nu}_i := \hat{p}_i \hat{\mu}_i. \quad (5)$$

The expected pay-off from informing can only be negative when $\alpha^e \hat{\nu}_2$ is sufficiently large and negative. This can happen, for example, when the elder beliefs the caregiver is making a decision under a high degree of pressure $\hat{\phi}$ so is likely to be a reluctant type (i.e. they have negative direct pay-offs of helping but still agree to help), as well as would agree even under reasonably large cost u^c .

The expected pay-off can be negative only if the negative contribution from reluctant help outweighs both the elder's expected attendance benefit and the positive contribution from willing help. Perceived pressure matters because it allows negative-pay-off caregivers to agree to help. Merely believing that pressure is positive need not make informing unattractive.

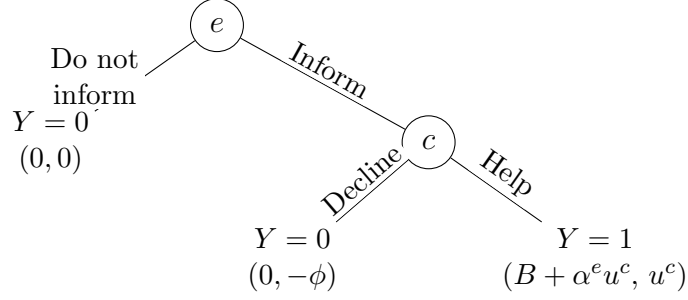
Equilibrium. We solve the model using backwards induction. First we solve the caregiver's choice, then the elder's choice using her subjective beliefs.

2.2 Take-up

2.2.1 Information pooling benchmark

In standard models of the household, it is often assumed that there is frictionless communication among household members, meaning information about the check-up is freely shared between the elder and

Panel A: Only the elder is informed



Panel B: Both elder and caregiver are directly informed

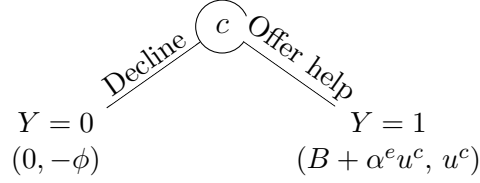


Figure 1: Information sharing and take-up: simplified decision trees.

Notes: Circular decision nodes E and C denote the elder and caregiver. Pay-offs are ordered (U_e, U_c) , and Y is an indicator for whether the elder attends the visit. The caregiver knows (u^c, ϕ) , while the elder uses beliefs $(\hat{F}, \hat{\phi})$ to make decisions. Panel A depicts the scenario in which only the elder is informed. The elder informs when $\hat{V}_e = \hat{p}B + \alpha^e(\hat{\nu}_1 + \hat{\nu}_2) \geq 0$. The caregiver helps when $u^c \geq -\phi$. Panel B depicts the scenario in which both the elder and caregiver are informed. The caregiver helps when $u^c \geq -\phi$.

caregiver (Chiappori and Mazzocco, 2017).⁵ In the information pooling benchmark, information about the visit reaches the caregiver without requiring the elder to initiate by deciding to inform them.

The caregiver helps when

$$u^c = \alpha^c B - \kappa \geq -\phi \quad (C)$$

Let C_h be an indicator for whether this condition holds for household h . Because caregiver help implements the visit, take-up of the visit in the benchmark case is

$$Y_h^{Pooling} = C_h \quad (6)$$

⁵Note that in this section, we use the term “information” in reference to information about the health visit, rather than in reference to information about preferences. At risk of overloading the term, we use the word “information” in both places to match discussions standard in the literature. Throughout the paper, we try to be careful and explicit about what the word refers to, e.g. “asymmetric information about preferences” versus “sharing information about the health visit.”

2.2.2 Case 1: Only the elder is informed

In *Control*, the elder decides whether to inform using her subjective expected pay-off:

$$\widehat{V}_e = \widehat{p}B + \alpha^e(\widehat{\nu}_1 + \widehat{\nu}_2) \geq 0 \quad (\text{I})$$

Let I_h be an indicator for whether condition (I) holds for household h . Take-up of the visit is then:

$$Y_h^{Control} = C_h I_h \quad (7)$$

We say that the elder's willingness to inform constraint *binds* when $I_h = 0$ and $C_h = 1$, i.e. the elder withholds information from a caregiver who would help if informed.

2.2.3 Case 2: Both elder and caregiver are directly informed

In *Inform*, both the caregiver and the elder directly have information about the visit. In this case, the elder does not need to initiate by informing, and instead, the caregiver can offer to facilitate the visit without first receiving a request. Because the source of information does not affect the caregiver's pay-offs, their decision to offer remains the same as their decision to agree once informed by the elder, as in (C).⁶ Thus

$$Y_h^{Inform} = C_h \quad (8)$$

Direct information restores the level of take-up achieved under the information pooling benchmark.

Effect of informing caregiver. Relative to the *Control* case in which only the elder is informed, the *Inform* intervention increases take-up for households where both parties would have been willing to attend, but the elder was not willing to inform given their beliefs about the caregiver's pay-offs. The

⁶Under these assumptions, elders who do not receive an offer will not attend the health visit. Because the caregiver faces the same degree of pressure regardless of whether they are informed by the elder or the research team, this implies that the caregiver offers if and only if they are willing to help facilitate the health visit. As a result, all the caregivers who are willing to go will offer to go, so even if we allow an additional stage of the game in which the elder can still decide to inform, any additional asks that come from elders who did not receive an offer would be rejected by the caregiver.

average treatment effect of the *Inform* intervention can then be written:

$$\begin{aligned}
\text{ATE}_{\text{Inform}} &:= \mathbb{E}[Y_h^{\text{Inform}} - Y_h^{\text{Control}}] \\
&= \mathbb{E}[(1 - I_h)C_h] = \Pr(I_h = 0, C_h = 1) \\
&= \Pr(I_h = 0) \Pr(C_h = 1 \mid I_h = 0).
\end{aligned} \tag{9}$$

We decompose the average treatment effect into two terms. The first term is the share of elders who withhold information. The second is the share of their caregivers who would actually help.

The intervention thus only increases take-up when the elder is withholding *relevant* information about the health visit – the caregiver would have been willing to attend, but the elder was not willing to inform. This gives us our first prediction for take-up of the health visit, which tests for whether the elder’s willingness to inform constraint is binding for any households.

Prediction 1 (Effect of informing caregivers) *Directly informing caregivers weakly increases take-up. The effect is strictly positive if and only if there are elders who withhold information from caregivers who would help if informed.*

Thus the magnitude of the treatment effect depends on the correlation in the elder and caregiver preferences (and beliefs). We summarize this in the corollary below:

Corollary 1 (Heterogeneity in effect of informing caregivers) *For characteristic X that takes value x_h for a subset of households, holding everything else constant, informing the caregiver increases the take-up more in proportion to two quantities:*

1. $\Pr(I_h = 0; X = x_h)$, the share of households where the elder withholds
2. $\Pr(C_h = 1 \mid I_h = 0; X = x_h)$, the share of households in which the caregiver is willing to go if informed, among households where the elder withholds

In other words, we expect there to be higher treatment effects when elders are withholding information from caregivers who are willing to help when informed.

2.3 Elder decision to inform

We now analyze what factors can discourage informing sharing by the elder. Concern for the caregiver can discourage information sharing. We define a *worried elder* as one who both cares about the caregiver

and believes the caregiver faces positive pressure:

$$W_h := \mathbf{1}\{\alpha_h^e > 0 \text{ and } \hat{\phi}_h > 0\}. \quad (10)$$

Positive perceived pressure means that the elder believes it is possible for the caregiver provide help that is net costly to them.

2.3.1 Benchmarks: Elders who are not worried

Elders who are not worried either do not care about the caregiver $\alpha_h^e \leq 0$ or do not believe there is pressure $\hat{\phi}_h$. If either of these holds, the elder will always inform their caregiver.

Benchmark 1: Elder does not care about caregiver pay-offs. If $\alpha^e = 0$, the expected value of informing is

$$\hat{V}_e = \hat{p}B \geq 0 \quad (11)$$

So the elder always informs: she gains whenever the caregiver helps and receives zero otherwise. She will always ask if she thinks there is any chance the caregiver will provide help, i.e. $\hat{p} > 0$, regardless of her beliefs about the caregiver's pay-offs or the pressure.

Benchmark 2: Elder believes there is no pressure. If $\hat{\phi} = 0$, the elder believes the caregiver helps only when they net benefit from doing so $u^c \geq 0$. Then there are no reluctant caregivers who help, so $\hat{p}_2 = \hat{\nu}_2 = 0$ and

$$\hat{V}_e = \hat{p}_1B + \alpha^e\hat{\nu}_1 \geq 0 \quad (12)$$

The elder again always informs. This result holds for any $\hat{F}$: caregivers with negative pay-offs are expected to decline, which is costless to the elder in our model.

2.3.2 When worry prevents informing

Worried elders withhold information precisely when

$$\hat{p}B + \alpha^e(\hat{\nu}_1 + \hat{\nu}_2) < 0, \quad \text{equivalently} \quad -\alpha^e\hat{\nu}_2 > \hat{p}B + \alpha^e\hat{\nu}_1. \quad (13)$$

Thus non-asking requires not only caring and perceived pressure, but a sufficiently negative expected contribution from reluctant help. It requires $\hat{p}_2 > 0$ and $\hat{\nu}_1 + \hat{\nu}_2 < 0$. Neither follows from the definition of worry alone.

A useful necessary condition is $\alpha^e \hat{\phi} > B$. To see this, every perceived helper has $u^c \geq -\hat{\phi}$, so their contribution $B + \alpha^e u^c$ is nonnegative whenever $\alpha^e \hat{\phi} \leq B$. Positive perceived pressure is therefore necessary for non-asking, but pressure must also be sufficiently large relative to the elder's benefit and caring weight.

Prediction 2 (Willingness to inform and worry.) *Worried elders are weakly less likely to inform than not-worried elders, who always inform in the model. The difference is strict if a positive share of worried elders satisfy (13). Holding beliefs fixed, greater elder caring weakly reduces asking when $\hat{\nu}_1 + \hat{\nu}_2 < 0$; if this sum is nonnegative, the elder always informs.*

The comparison with zero perceived pressure does not imply that asking decreases monotonically with every increase in $\hat{\phi}$. Higher perceived pressure expands the set of caregivers expected to help. If $\hat{F}$ has a continuous density $\hat{f}$, then, holding it fixed,

$$\frac{\partial \hat{V}_e}{\partial \hat{\phi}} = (B - \alpha^e \hat{\phi}) \hat{f}(-\hat{\phi}). \quad (14)$$

The new helpers can initially raise the value of informing. At sufficiently high perceived pressure, marginal helpers instead impose losses that outweigh the attendance benefit.

Treatment-effect heterogeneity. For a group g defined before treatment, such as worried elders,

$$\text{ATE}_{\text{Inform}}(g) = \Pr(I_h = 0 \mid g) \Pr(C_h = 1 \mid I_h = 0, g). \quad (15)$$

As above, the joint probability defines the effect if the conditioning event is empty. The effect is zero for model-defined not-worried elders because they already inform. It is positive for worried elders only if some of their withheld requests would elicit help.

We impose no independence between caregiver pay-offs, elder caring, beliefs, and B . Worried elders may have either responsive caregivers or decliners. Across degrees of worry, more non-asking need not imply a larger treatment effect if actual caregiver responsiveness also falls. If all caregivers paired with non-asking elders would decline, direct notification has no effect on take-up.

2.3.3 Reducing caregiver costs

Suppose free transportation reduces caregiver costs by $\tau \geq 0$, so that the net pay-off from helping becomes $u^c + \tau$. The elder understands this reduction. Hold B , actual and perceived pressure, and beliefs over the

original pay-off states fixed. Her informing value becomes

$$\widehat{V}_e(\tau) = \int_{[-\widehat{\phi}-\tau, \infty)} [B + \alpha^e(u + \tau)] d\widehat{F}(u). \quad (16)$$

If the cost reduction leaves the perceived set of helpers unchanged, then

$$\widehat{V}_e(\tau) = \widehat{V}_e(0) + \alpha^e \tau \widehat{p}, \quad (17)$$

where $\widehat{p}$ is the initial perceived probability of help. This expression allows a reluctant helper to become willing: their motivation may change while their participation does not. A previously non-asking elder begins to inform if

$$\tau \geq -\frac{\widehat{V}_e(0)}{\alpha^e \widehat{p}}, \quad \alpha^e > 0, \quad \widehat{p} > 0, \quad (18)$$

provided the unchanged-helper-set condition holds over that reduction.

If the cost reduction also induces perceived decliners to help, there is a composition effect. A marginal entrant has net pay-off $-\widehat{\phi}$ and contributes $B - \alpha^e \widehat{\phi}$ to the elder. This can be negative because the elder previously assigned zero value to refusal. For a continuous density,

$$\frac{d\widehat{V}_e(\tau)}{d\tau} = \alpha^e [1 - \widehat{F}(-\widehat{\phi} - \tau)] + (B - \alpha^e \widehat{\phi}) \widehat{f}(-\widehat{\phi} - \tau). \quad (19)$$

The first term is the gain for existing helpers; the second captures entry. Unlike a model with pressure-free decliners, the present specification does not guarantee that cost relief increases asking at every parameter configuration.

Prediction 3 (Testing for elder caring) *A known reduction in caregiver costs weakly increases asking if it leaves the perceived set of helpers unchanged, or more generally if its welfare gain for existing helpers dominates any adverse entry effect. It strictly increases asking when a positive share of initially non-asking worried elders cross their informing threshold. Not-worried elders already inform and do not increase asking. Under the maintained exclusions, an increase in asking in response to caregiver cost relief requires both $\alpha^e > 0$ and $\widehat{\phi} > 0$. A positive empirical response is consistent with concern for caregiver costs; it is not an unconditional prediction when new reluctant helpers can enter.*

2.4 Caregiver decision to help

2.4.1 Pressure

The caregiver helps if $u^c \geq -\phi$. To distinguish motives, consider interventions that reduce refusal pressure while leaving u^c unchanged. Write $\phi = \phi_{other} + \phi_{self}$. Under public choice, private choice that removes external pressure, and a pressure-free choice environment, respectively, help occurs when

$$u^c \geq \begin{cases} -(\phi_{other} + \phi_{self}) & \text{under } T_{public}, \\ -\phi_{self} & \text{under } T_{private}, \\ 0 & \text{under } T_{out}. \end{cases} \quad (20)$$

These comparisons require the interventions to remove the specified pressure components without changing costs, benefits, or knowledge of the visit. Making a decision private does not by itself establish that self-image concerns have been removed; the interpretation of T_{out} requires that additional exclusion.

Prediction 4 (Testing for pressure-driven help.) *Removing pressure weakly reduces helping. With complete removal, the reduction in the probability of help is*

$$\Pr(C^{public} = 1) - \Pr(C^{out} = 1) = \Pr(-\phi \leq u^c < 0) = p_2, \quad (21)$$

where p_2 is the actual share of reluctant helpers. The public-to-private contrast identifies $\Pr(-\phi \leq u^c < -\phi_{self})$; the private-to-pressure-free contrast identifies $\Pr(-\phi_{self} \leq u^c < 0)$. These are shares whose behavior changes at particular margins, not the level of the pressure parameter.

Misperceptions about burden. If actual pressure is zero, all actual helpers have $u^c \geq 0$ and all other caregivers can decline without guilt. An elder who correctly believes $\hat{\phi} = 0$ therefore always informs, regardless of her beliefs about F . A positive take-up effect in this zero-pressure environment requires elders who care and mistakenly believe pressure is positive, with beliefs sufficiently pessimistic to satisfy (13).

A near-zero behavioral response to removing pressure is a different finding. Under the exclusions above, it implies a small share p_2 of pressure-driven help in the population and decision margin studied; it does not by itself identify $\phi \approx 0$. There may be little probability mass in $[-\phi, 0)$ even with substantial pressure. For comparison, define actual weighted helping pay-offs $\nu_i = p_i \mu_i$ analogously to their perceived counterparts. Elder asking depends on $\hat{\nu}_1 + \hat{\nu}_2$, whereas actual welfare from helping depends on $\nu_1 + \nu_2$.

Since $-\phi \leq u^c < 0$ for reluctant helpers,

$$0 \leq -\nu_2 \leq \phi p_2.$$

A small reluctant-helper share bounds aggregate losses only in conjunction with a bound on pressure or loss severity. Elders can overestimate the incidence of reluctant help, its severity, or both, through $\hat{F}$ and $\hat{\phi}$.

A positive take-up effect among worried elders establishes that some caregivers paired with non-asking elders help once informed. Evidence that those caregivers help willingly is especially informative about whether the withheld requests would have imposed a net burden. Pressure tests on another task or among caregivers who already agreed to help require an additional link to this participation margin; they do not directly establish the preferences of all otherwise-unasked caregivers. Finally, although refusal guilt does not affect elder choices in this specification, it remains part of caregiver welfare when assessing the intervention's overall effects.

2.4.2 Dynamics and crowd-out

Until this point, we model the health visit decision problem as a one-off interaction between the elder and caregiver. However, this does not account for dynamic considerations that would predict low information sharing by the elder. For example, elders may face a budget constraint for goodwill from the caregiver, which might mean that the elder does not inform the caregiver to avoid having caregiver involvement today crowding out involvement with a future task. To analyze this, we extend our model to a two-period interaction in which in each period the elder decides whether to inform the caregiver about a need, and the caregiver decides whether to offer help if informed.

To do this, we introduce a budget constraint faced by the caregiver in providing help, where h_t is an indicator for whether help is provided in period $t \in \{1, 2\}$, q_t is how costly the help is to provide, and M is the overall capacity that the caregiver has for supporting the elder.

$$q_1 h_1 + q_2 h_2 \leq M \tag{22}$$

We let the caregiver's utility be given by U_c , which is the present discounted sum of utilities across both time periods. If the caregiver helps in period t , they get pay-off $\alpha^c B_t - \kappa_t$. If they do not help, they incur pressure cost ϕ_t .

$$U_c = h_1(\alpha^c B_1 - \kappa_1) + \delta h_2(\alpha^c B_2 - \kappa_2) \quad (23)$$

In the first period, the caregiver decides whether to help using the same decision rule as before in the one-period case, helping when the net benefit of helping outweighs the pressure costs:

$$\alpha^c B_1 - \kappa_1 > -\phi_1 \quad (24)$$

In the second period, the caregiver helps if they both find that the task is worth helping with (e.g. $\alpha^c B_2 - \kappa_2$) and they have remaining capacity. If their budget constraint is slack, then help in the first period does not impact the caregiver's decision to help or not in the second period. Only in the case where providing help in both periods would put the caregiver above their capacity M would help in period 1 affect the caregiver's decision in period 2. Crowd-out occurs when:

$$q_2 \leq M < q_1 + q_2 \quad (25)$$

The two-period model thus introduces another reason that elders may not inform caregivers in the first period: if caregiver involvement in the first period crowds out potential involvement in the second period, i.e. providing help in the first period pushes the caregiver over their budget constraint. From the elder's point of view, she only chooses to ask in period 1 when expected benefit today outweighs discounted expected benefit in the second period:⁷

$$B_1 + \alpha^e(\alpha^c B_1 - \kappa_1) \geq \delta \cdot [B_2 + \alpha^e(\alpha^c B_2 - \kappa_2)] \cdot \hat{\mathbf{1}}_{Help} \quad (26)$$

where δ is the discount rate (which can also include some probability that the need in period 2 arises at all) and $\hat{\mathbf{1}}_{Help}$ is an indicator for whether the elder believes that the caregiver would help in the second period.

We summarize the key prediction of the two-period, dynamic model for caregiver's behavior:

Prediction 5 (Crowd-out) *If the caregiver faces a binding budget constraint in providing help to the elder, then providing help in one period will decrease the likelihood of providing help in future periods.*

⁷Note that we do not model misperceptions of the elder regarding crowd-out, based on survey data collected from elders that show that most elders do not report crowd-out as a potential consideration for whether they ask caregivers for help on tasks.

Corollary 2 (Static model) *If there is no evidence of crowd-out, then the caregiver’s decision to provide help in each period is independent from past periods, and a static, one-period model is sufficient to capture decision-making in this context.*

3 Setting

We conduct our study with elderly women and their caregivers in rural Tamil Nadu, India. The study was implemented in partnership with a private charity hospital that serves a predominantly rural area in Ranipet district (Thirumalai Mission Hospital).

The elderly population is growing rapidly in India and globally (United Nations Department of Economic and Social Affairs, Population Division, 2019). Within India, Tamil Nadu is one of the most rapidly aging states, with the population aged 60 and above expected to double between 2011 and 2031 (National Statistical Office, 2021). As in much of India, formal long-term care for the elderly is limited in our setting: even awareness of old age homes is only 28% among rural elders in Tamil Nadu, and consideration of staying in one is even lower (10%) (Burholt et al., 2022; Duflo et al., 2022). Elders are instead generally dependent on family members such as spouses, adult children, and other relatives as the primary providers of support for transport, finances, and health-related decision-making (Duflo et al., 2022).

This dependence on family caregivers is top of mind for elders and has implications for elders’ health-related decisions. In data that we collected for a representative survey of older adults in Tamil Nadu in 2024, we find that 26% of elders report frequently worrying about being a burden on family members, 30% report sometimes not telling family members about pain or sickness, and 29% report having forgone medical care for this reason Table A.1.

We focus on elderly women for several reasons. First, caregivers may play an especially important role for older women. Older women are more dependent on caregivers, since they are more likely than men to have mobility impairments and be financially dependent (Table A.1). Second, these relationship dynamics with family are also a deeply gendered issues. For example, women are much more likely to report being worry about burden: 39% of women reporting having forgone care out of concern of burden compared to 18% of men Table A.1. Third, understanding potential drivers of low take-up is important to addressing gendered disparities in both health outcomes and inputs that have been documented. Research has shown that older women in India have lower rates of health utilization and higher rates of excess mortality compared to their male counterparts (Dupas and Jain, 2024; Calvi, 2020).

3.1 Sample

Elders. Our study sample consists of pairs of elderly women aged 60 to 75 and their caregivers (N=1,175 pairs). Most elders (82%) live in the same household as their caregiver, and 65% live in intergenerational households (i.e., with adult children) Table 1. The mean household size is 3.3 individuals. About half of elders are married (48%) and the remaining are widowed. Although most of the sample has had some kind of check-up in the last year (83%), though most (57%) have never gotten any kind of cardiac check-up, and very few have had a recent cardiac check-up (16% reported having had a heart check-up in the last year). Elders in our sample are quite mobility impaired, with 58% reporting having at least moderate difficulty leaving their home, 77% report difficulty walking a long distance, and 65% report difficulty learning a new task, such as figuring out how to get to a new place. In our sample, 33% report ever worrying about burden, and 26% report ever forgoing healthcare or avoiding the hospital out of this worry about burden.

Caregivers. We define a caregiver as a family member who lives in the same or a neighboring village and provides at least one of the following forms of support to the elder: financial support, transport to medical appointments, transport in emergencies, or someone the elder talks to about her health.⁸ Consistent with caregiving norms in India about the financial provider for elders, we find that caregivers are primarily sons (43%) and husbands (21%) (Mandal and Subaiya, 2023). About a quarter of our sample has daughters (15%) or daughters-in-law (9%) who are caregivers, and the remaining are comprised of grandchildren, siblings, and other family members. Nearly all caregivers are married (87%), and most are working (68%). The average caregiver is 47.8 years old, reflecting the primarily intergenerational relationships between elders and their primary caregivers. Table A.3 presents the full set of demographic, health, and household characteristics for the sample, separately by treatment arm.

3.2 Health check-up

We partner with a private charity hospital in Ranipet, Tamil Nadu to offer free cardiac check-ups to elders in our study. The check-up includes a blood pressure reading, a random blood sugar test, a 12-lead electrocardiogram (ECG), a survey-based cardiovascular risk assessment (QRISK3⁹), and a doctor

⁸When the elder identifies multiple eligible candidates, we select the caregiver by asking, in order: who provides financial support, then transport for check-ups, then transport in emergencies, then who the elder talks to about her health. We take the first person named who lives in the same or neighboring village as the caregiver who we invite to the study.

⁹The QRISK is a risk prediction algorithm for cardiovascular disease, similar to the American Heart Association’s 10-year risk score and is currently used in the UK as recommended by the National Institute for Health and Care Excellence guidelines (Hippisley-Cox et al., 2017; National Institute for Health and Care Excellence, 2023). The QRISK algorithm was recommended by the Chief Medical Officer and ethics review board of our partner hospital.

consultation. Individuals who are determined high risk by the doctor also underwent an echocardiogram or an HbA1c blood sugar test at no additional cost. The full set of services would typically cost approximately 3,000-5,000 INR at a typical private hospital at market rates, equivalent to approximately \$150-250 USD adjusted for PPP.

We chose a cardiac check-up for our empirical setting to help isolate communication and other-regarding preferences as the core mechanisms we are testing in our model. First, the check-up addresses an important source of disease burden with known, effective treatment: cardiovascular disease is the leading cause of mortality among older adults in Tamil Nadu, but treatment for most cardiac conditions diagnosed at this check-up consists of prescription drugs that are available free of charge via the public health system (Indian Council of Medical Research et al., 2017; Kondal et al., 2022). As a result, the primary cost of the check-up is the caregiver’s time and effort in providing transport and accompanying the elder to the hospital – a cost that is clear and commonly known to both parties. This avoids a confound that may be present for other health conditions, where elders and caregivers may hold divergent views of treatment and management costs. Second, elders in our setting rarely undergo cardiac-specific screening, even when they have undergone general check-ups in the past year¹⁰. Together with careful piloting around what services to provide during the check-up, this ensured the check-up was attractive enough to most elders so that there is scope for them to consider whether to communicate about the check-up with caregivers. Third, the check-up is valuable but non-emergency, so the elder’s decision to attend is more likely to be shaped by the kinds of communication and cost frictions our experiment is designed to identify, rather than by acute medical need.

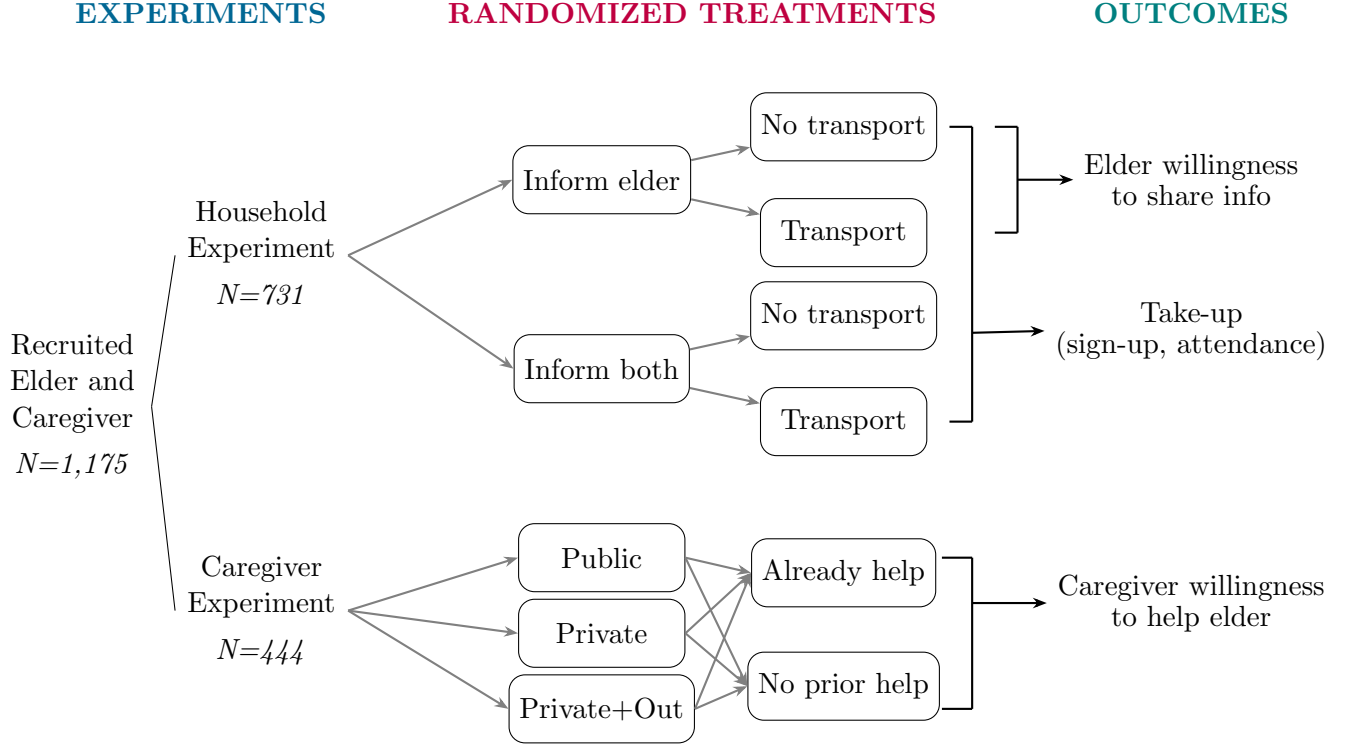
4 Study overview and sampling

Our main empirical results come from two complementary field experiments, with participants drawn from the same sampling frame. First, in the household experiment, we randomize who in the household is directly informed about the check-up and measure how directly involving caregivers affects take-up of a health check-up for elders. We also cross-randomize providing free transportation to understand how elders’ willingness to share information responds to changes in costs to the caregiver of bringing the elder to the check-up. Second, in the caregiver experiment, we randomize the direct observability of caregivers’ investment choices for elders to measure the role of pressure in influencing caregiver decisions to help the elder. In the same experiment, we also cross-randomize how much the caregiver has already been asked

¹⁰Existing community health programs to address non-communicable diseases in Tamil Nadu primarily focus on diabetes and hypertension (Subramaniam et al., 2025)

to help to understand whether elders face a “budget constraint” for favors from the caregiver, in which asking for more commitment now reduces caregiver’s willingness to commit to helping the elder later. A summary of our experiments and design is presented in Figure 2 below.

Figure 2: Overview of Design: Household and Caregiver Experiments



4.1 Sample construction and household listing

We construct our sampling frame from our partner hospital’s administrative records, which contain basic identifying information (name, village, street) for approximately 10,000 elderly women aged 60–75 from the 315 villages in the hospital’s catchment area. We sample primarily from the closest areas of the hospital districts, and we exclude the smallest villages with fewer than 15 eligible households in the administrative data in order to minimize travel costs for enumerators.

The study populations for both experiments are drawn from two separate rounds of household listing exercise with elders: one in November 2025 and one in February–March 2026. During the listing, enumerators from J-PAL South Asia visit households on the administrative list to locate the elder, invite them to participate in a household study about health care, and elicit consent to conduct a brief baseline survey. Enumerators also recruit additional eligible elders not in the administrative list from the same study villages, who comprise 35% of the sample.

We frame the study as a general household survey on elder health and well-being. In the listing survey with elders, we collect baseline information for the elder about basic demographics (age, marital status), household structure (number of children, household members), mobility, and relationship to family (concern of burden). We also collect contact information for one caregiver, who is a family member living in the same or neighboring village who provides support to the elder for health-related needs. See Section 3.1 for the details about the caregiver elicitation protocol.

At the end of the survey, we collect a measure of demand for the health check-up through a series of incentivized survey choices among gift bundles, described in Section B.2. Respondents were informed that one randomly selected choice per household is implemented as a thank-you gift for participating in the survey, which served during recruiting as an incentive to participate in our study. These choices were also used as a measure of valuation for the check-up used in our heterogeneity analysis.

Of the $N=2,351$ elders who participated in the listing survey, $N=1,805$ were eligible, i.e. elder was not physically or cognitively impaired, consented to the study, and had a caregiver living nearby for whom they could share contact information. Of these, we were able to reach $N=1,175$ households where both the elder and caregiver were reached for the study.

5 Household experiment

5.1 Household experiment: Design

In the household experiment, we cross-randomize pairs of elders and caregivers into two interventions that may affect take-up of the check-up¹¹. First, our main experimental treatment varies who is informed of the check-up to understand how information transmission affects take-up of the check-up. In the control group (*No Inform*), only the elder is informed about the check-up. In the treatment group (*Inform*), both the elder and caregiver are directly informed about the check-up. Second, we cross-randomize whether there is free transport provided to and from the check-up to understand whether alleviating caregiver costs affects elder willingness to share information about the check-up with the caregiver. We also measure the impact of transport on take-up rates to benchmark the effect sizes for the *Inform* treatment.

The *Inform* intervention is randomized at the household-level, stratified by panchayat.¹² The *Transport* intervention is cross-randomized at the geographic cluster level, stratified by panchayat.¹³

¹¹We refer to a pair of elder and her caregiver as a household in exposition, though only 82% of pairs are living in the same household in our sample.

¹²A panchayat is a village or cluster of villages at which rural governments are organized.

¹³We randomize transportation at the cluster-level to minimize spillovers of the transportation offer between

5.1.1 Intervention: Who is informed

Our main intervention varies whether the caregiver is directly informed about the free check-up. Elders are always directly informed about the check-up. We randomize pairs of elders and caregivers into one of two arms:

1. *No Inform* (50%). Only the elder is informed about the check-up.
2. *Inform* (50%). Both the elder and the caregiver are directly informed about the check-up.

We compare take-up between these two arms to identify the causal effect of removing the information friction that may exist between elders and caregivers about the check-up. In the full-information benchmark of our model, the two arms should yield identical sign-up and attendance rates. When the elder has perfect information about the caregiver’s pay-offs, an elder who values the check-up sufficiently will pass the information along, and a caregiver who would help if informed will help in both arms. In our model with uncertainty about the caregiver’s preferences, the difference in take-up between *No Inform* and *Inform* therefore identifies the share of households in which the elder’s information-sharing decision is the binding constraint.¹⁴

We designed our experiment to distinguish information sharing from alternative explanations for why take-up would differ between the treatment groups. First, we minimize mechanical reasons why information may not be shared, such as the elder forgetting or not being able to explain the details of the check-up. When delivering the check-up invitation, we give each elder a paper voucher with the camp date and location, as well as the sign-up deadline and phone number (Figure A.1). Second, we minimize additional social or psychological pressure from the surveyors when delivering the invitation to the caregiver, since we are interested in the impact of having information about the check-up, rather than explicitly persuasive content to convince the caregiver to attend. Towards this end, we design the check-up invitation script to be as neutral in tone as possible. The same check-up invitation process is followed for elders and caregivers: the invitee is read a scripted invitation explaining the purpose of the check-up, date and location, the tests included, and the sign-up deadline. The script is accompanied by a 2-minute video describing the check-up and a poster with images (see Section B.5 for details).

neighboring households. Clusters are comprised of on average 2.6 households that are geographically close to one another.

¹⁴Similarly, if there were perfect information pooling about the check-up across household members, and the elder does not make a decision to share information or not (she always shares), then we would expect there to be no difference in take-up between the two arms.

Check-up Invitations. The check-up invitations are delivered privately and separately at the end of each participant’s survey. Neither the caregiver nor the elder is explicitly told that the other has been (or will be) informed.¹⁵ To deliver the invitation, enumerators informed participants that they would receive a free health check-up for cardiac health. They described what services are included in the check-up, as well as location and timing details. To sign up for the check-up, the caregiver of the elder would have to make a call to the research team hotline, before the deadline of two days before the check-up.¹⁶ The experimental study was conducted in weekly cycles of invitations and check-ups. The deadline for sign-ups was 1-4 days after the initial invitation¹⁷, and the check-up occurred 3-6 days after the initial invitation. Details about the experimental timeline are described in Section B.4.

Outcome: Take-up. We collect two measures of take-up:

1. **Sign-up for check-up by caregiver.** We measure whether the caregiver calls the J-PAL hotline by the deadline to confirm a check-up slot.
2. **Elder attendance at check-up.** Among households who sign up by the deadline, we measure whether the elder shows up at the hospital for the check-up.

5.1.2 Intervention: Free transportation

We cross-randomize a second intervention that varies the caregiver’s cost of helping the elder attend the check-up. The *Transport* intervention is randomized at the geographic-cluster-level. We organize the households in the sampling frame into small clusters of households that are geographically close to one another (average size of 2.6 households), and half of the clusters are offered free transport to and from the hospital for the check-up.

¹⁵The team of enumerators was split so that each enumerator was either an elder surveyor or a caregiver surveyor, and for most of the experiment, surveyors independently sought out the elders and the caregivers. The protocol for reaching out to elders and caregivers from the listing exercise was adjusted after the first two weeks of the household experiment to accommodate field logistics. Initially, elders were surveyed only once the caregiver had been reached and agreed to participate in the study. After the first two waves of the study, the protocol was adjusted so that surveyors independently sought out the elders and caregivers. Of the households in our experimental sample (N=731), in 59% of cases the caregiver was surveyed before the elder. This is balanced across treatment arms (Table A.3).

¹⁶Visits to deliver invitations to elder and caregivers happened in the first one to three days of the week, and the sign-up deadline happened on the fifth day, so all households had two to four days to discuss and sign up.

¹⁷We also conducted phone follow-ups with caregivers after the sign-up deadline had passed, before the check-up occurred. During these phone calls, we administered a brief survey that measured whether the elder communicated the information about the check-up to the caregiver, and for households where the caregiver knew about the check-up, we also asked about how the conversation with the elder went, as well as asked if they would still be interested in attending. In these cases, we also allowed these households to still actually sign-up for and attend the check-up.

1. *No Transport* (50%). The household is not offered transport assistance and must arrange their own transport for the elder.
2. *Transport* (50%). The household is offered free, private transportation to and from the check-up. The transport is arranged and paid for by the research team, so that the caregiver does not need to arrange, pay for, or accompany the elder to the check-up.

The free transport intervention is designed to exogenously lower the costs to the caregiver of facilitating the check-up. Although the transport may lower elder costs directly in some cases – e.g. by providing a higher quality form of transport than public transit – for most of the elders in our context, caregivers are ultimately the payers of any costs associated with the elder’s health-related needs. This is especially true in our study sample because (1) many elderly women face mobility and financial constraints that would prevent them from visiting a health facility alone; and (2) we intentionally selected caregivers to be those who bear financial and transportation costs of elders.

The *Transport* intervention serves two purposes in our study. First, it allows us to measure how exogenous variation in caregiver costs affects elder willingness to share information with their caregiver. To do this, we will compare elder willingness to share information across the *No Transport* and *Transport* arms among elders in the *No Inform* group (in which the caregiver is not informed about the check-up directly). We can also estimate this effect separately for worried versus not-worried elders to test whether this responsiveness differs by elder-reported worry about burden. Second, it allows us to measure how the decrease in caregiver costs affects actual take-up rates. This provides a benchmark for the magnitude of the main treatment effects of the *Inform* intervention.

Outcome: Willingness to share information. The primary outcome of interest for the *Transport* analysis is the elder’s willingness to share information with the caregiver. For elders in the *No Inform* arm, our primary measure of willingness to share information about the check-up is measured after delivering the invitation to the elder during the elder visit. The elder is asked whether she would like the research team to call the caregiver on her behalf two days before the check-up to confirm attendance.

If we don’t hear from [Caregiver NAME] by [DEADLINE], would you like us to call or send them a follow-up message about the check-up?

The goal is to measure the elder’s willingness to share information, so we were careful to word the question to be precise and naturalistic towards this end. The question is worded so that it is clear that the elder always has the option to inform the caregiver themselves before the deadline. This is to minimize concerns

that some elders are willing to inform the caregiver directly but unwilling to do so indirectly – for example, to avoid seeming passive aggressive. The question is also worded to minimize any perceptions of pressuring the caregiver into the check-up, so that the act of informing is primarily about sharing information about the check-up with the caregiver, rather than acting as a persuasive nudge.

To address the concern that elders may not share information simply because they are willing and able to attend on their own, we design the sign-up process to ensure that the caregiver was necessary in order for the elder to attend the check-up. We note that this decision did not detract much from the naturalistic setting, since most of the elderly women in our study population already need to rely on their caregivers in order to visit a health facility. The dependence on caregivers is reflected in the high rates of mobility impairment reported in our sample: 78% of elders report having at least moderate difficulty with walking a long distance, 64% with learning a new task such as going to a new place, and 58% with leaving home at all. To further ensure that caregivers must be involved in the context of the study, we tell households upfront that the caregiver must make the check-up appointment by making a phone call to the research team before the sign-up deadline.¹⁸

As a secondary measure of information sharing, we also collect caregiver-reported rates of the elder mentioning the check-up in the caregiver follow-up surveys conducted after the sign-up deadline has passed¹⁹, as described in more detail in Section B.4.

5.2 Household experiment: Results

In this section, we first describe the subset of elders who express being worried about burdening their families. We then test the take-up and elder information sharing predictions from our model by measuring the impact of the *Inform* intervention on take-up of check-up and the impact of the *Transport* intervention on elder willingness to inform.

¹⁸In our analysis, our main results use measures of take-up that include all sign-ups (and subsequent attendances) that happened before the deadline, of which 75% came from caregivers and 25% came from other callers (e.g. a different member of the family, neighbors, or elders themselves). In Section A.12, we show that our results are robust when we restrict our analysis to sign-ups and attendance that follow from only the caregiver calling.

¹⁹We elicit both an unprompted and prompted measure of whether the elder has informed the caregiver. First, we have the enumerators mark whether caregivers themselves bring up the check-up (unprompted). If they do not, we have the enumerators explicitly probe for whether the elder mentioned a check-up (prompted). Finally, we elicit whether the elder explicitly asked or expressed interest in attending the check-up.

5.2.1 Elder worry about burden

Our main heterogeneity analysis are along this dimension of elder worry. Specifically, we refer to those who report ever having forgone medical treatment out of concern of being a burden as “worried” elders²⁰. Among the N=731 households in the household experiment, 26% of respondents are classified as “worried elders.”

We report differences in observables between worried and not-worried elders in Table 1. Worried elders have less immediate support available to them, compared to their not-worried counterparts. Although worried elders have a similar number of children on average (2.53 vs. 2.61, $p = 0.47$), they live in smaller households (average household size of 3.13 vs 3.33, $p = 0.04$) and are less likely to live in the same house as their caregivers (77% vs. 84%, $p = 0.04$). Notably, worried elders are more mobility constrained across several measures of impairment, which may reflect worse physical health such that they are objectively more of a burden for caregivers. They are more likely to report having difficulty leaving their home (70% vs. 54%, $p < 0.01$), walking a long distance (86% vs. 74%, $p < 0.01$), or learning a new task (83% vs. 59%, $p < 0.01$).

Consistent with lower rates of co-residence with their caregiver, caregivers of worried elders are less likely to be sons (37% vs. 45%, $p = 0.08$), who are generally the primary caregiver in our patrilocal context (Mandal and Subaiya, 2023). Instead, worried elders are more likely to report having daughters (19% vs. 14%, $p = 0.07$), sons-in-law (3% vs. 1%, $p = 0.03$), or other individuals²¹ as their main caregiver (9% vs. 4%, $p = 0.01$). Caregivers are otherwise very similar in other demographics, such as age (average 46.7 vs. 48.2 years, $p = 0.25$), marital status (86% vs. 87%, $p = 0.77$), whether they are working (65% vs. 69%, $p = 0.27$), and monthly income (10,950 vs. 11,660 INR per month, $p = 0.52$).

When we compare measures of relationship closeness between households with worried versus not-worried elders, we include controls for the imbalanced demographics (elder mobility impairment, elder household size, co-residence, whether caregiver is daughter). We use a 5-question subset of the Zarit Burden Interview scale to measure caregiver’s feeling of being burdened by their caregiving duties (Zarit et al., 1980). We find that caregivers of worried elders are actually *less* likely to report feeling burdened (67% vs. 52%, $p = 0.02$). Caregivers of worried elders also tend to be more involved with and informed about their

²⁰Elders were asked the following question as part of the listing survey conducted during recruitment: *Now, we will ask how much you think about others in your household. We will read a couple of statements and ask you how often you felt this way in the **last 30 days**. “I sometimes refuse treatment or to go to the hospital to not burden my household or family.”* Responses were elicited on a 5-point scale, where 1=Not at all, 2=Rarely, 3=Sometimes, 4=Often, 5=All the time. Those who responded with a “2” or higher were classified as “worried elders”, and those who responded “1” were classified as “not-worried elders.”

²¹The other categories of caregivers includes neighbors (4.3% vs. 2.4%), siblings (3.7% vs. 1.3%), and other relations (1.6% vs. 0.7%)

elders, as measured by a standardized index capturing agreement in three questions asked to both elders and caregivers about the elder’s health status, health needs, and communication style.

5.2.2 Balance

Elder and caregiver characteristics are generally balanced across treatments, as reported in Table A.3. To test for balance, we compare each of the three separate treatment arms (*Inform*, *Transport*, and *Inform + Transport*) to the *Control* arm (with neither *Inform* nor *Transport* implemented). For the *Inform* and *Transport* arms, we fail to reject the overall F-tests of joint significance ($p = 0.23$ and $p = 0.74$, respectively). For the *Inform + Transport* arm, the p-value of the F-test is marginally significant ($p=0.09$), reflecting some imbalance in the fraction of elders who had a check-up in the last year (81% in *Control* compared to 88% in *Inform + Transport*) and elder interest in the check-up (92% vs. 96%). Our results are robust to controlling for these imbalances, as shown in Section A.13.

5.2.3 Elder willingness to share information

Empirical strategy We analyze willingness to share information among elders in the control group to test the two information sharing predictions from our model. In our framework, worried elders are caring, so they more strongly internalize the caregiver’s welfare. This has two testable implications. First, the *Transport* arm allows us to test Prediction 3, which predicts that worried elders are generally more sensitive to exogenous changes in the caregiver costs of facilitating the check-up.²² Second, we can test ??, which tells us that worried elders should have lower asking rates.

To test these predictions, we run the following regression among households in the *No Inform* arm:

$$z_i = \omega_0 + \omega_1 T_i^{Transport} + \omega_2 Worry_i + \omega_3 T_i^{Transport} \cdot Worry_i + X_{imbal} + X_{imbal} \cdot T_i^j + X_i + \mu_s + \epsilon_{si} \quad (27)$$

where z_i is an indicator for whether the elder is willing to let the research team send information about the check-up to the caregiver, $T_i^{Transport}$ is an indicator for whether the household is randomized into the *Transport* arm offering free transport to the check-up, $Worry_i$ is an indicator for when the elder is

²²The sign of the effect is technically ex-ante ambiguous, as there are two opposing effects of changing caregiver costs: the direct effect (caregiver costs κ are lower, increasing the net pay-offs of facilitating the check-up, which increases an worried elder’s willingness to ask) and the selection effect (shifting the costs affects who the marginal caregiver is – if they have preferences such that they would help primarily out of pressure, leading to negative utility, then this effect on a worried elder’s willingness to ask is negative). If the direct effect of lowering caregiver costs outweighs the selection effect, then we will see a positive effect of providing free transportation on elder willingness to share information. In the case that this is true, then elders who are more worried about burdening their caregivers should be indeed more responsive to the free transport than those who are not worried.

categorized as worried about burden.²³ In our preferred specification, we include a vector of controls X_{imbal} that includes a set of variables that are statistically significantly different in value between the elders who are worried about burden ($Worry_i = 1$) and the elders who are not ($Worry_i = 0$).²⁴ In our preferred specification, we also include as controls the set of LASSO-selected variables X_i and randomization strata fixed effects μ_s . Standard errors are clustered at the geographic cluster-level to reflect the level at which the *Transport* intervention was randomized.

We test Prediction 3 by testing whether $\omega_3 > 0$, which captures the degree to which worried elders are more sensitive to caregiver costs than not worried elders.

We test ?? about how willingness to inform varies with worry about burden by testing whether ω_2 is negative.

Results In Table 4, we report the results on elder willingness to inform their caregiver about the check-up.

First, we find that when transportation is not provided, elders worried about burden are less willing to inform caregiver about the check-up by 14.2 percentage points ($p = 0.04$), compared to a baseline of 85% of their not-worried counterparts. According to ??, this demonstrates that elders believe that caregivers who agree to help in response to being informed would do so primarily out of pressure rather than true caring preferences towards the elder.

Next, we look at how willingness to inform responds to an exogenous shock to the caregiver’s costs (free transportation), separately for not-worried and worried elders. Not-worried elders are not at all responsive to transport ($\omega_1 = -0.011$, $p = 0.81$), but worried elders are, with free transport increasing their willingness to inform by 15 percentage points ($p = 0.04$) more than not-worried elders. We can conclude that among worried elders, the potentially negative selection effects of inducing different types of caregivers to help do not outweigh the positive, direct effects of lowering costs. Importantly, we find evidence consistent with Prediction 3, which predicts that worried elders are more responsive to an exogenous shock to caregiver costs ($\omega_3 = 0.16$, $p = 0.07$).

²³The survey question that corresponds to our measures of worry about burden asks respondents to describe how often they refuse treatment or avoid the hospital out of worry of burdening their family. The possible responses are *Not at all*, *Rarely*, *Sometimes*, *Often*, and *All the time*. This indicator is equal to one when the elder reports ever engaging in this behavior.

²⁴As a robustness check, we also control for the interaction of these imbalanced variables with the treatment indicator of interest $X_{\text{imbal}} \cdot T_i^j$ to isolate the heterogeneity in treatment effect that is attributable to worry about burden, after controlling for other differences in observables between worried and not-worried elders. These regression results are reported in the Appendix in Table A.7.

5.2.4 Take-up of check-up

Empirical strategy

Overall effect. For both the *Inform* and *Transport* interventions, we separately run the following household-level regression to assess the impact of that intervention on take-up of the check-up:

$$y_i = \beta_0^j + \beta_1^j T_i^j + X_i + \mu_s + \epsilon_{si} \quad (28)$$

where y_i is a measure of take-up (either *sign-up*, captured by an indicator for whether the caregiver called to sign the elder up for the appointment or *attendance*, an indicator for whether the elder showed up at the check-up), indicator T_i^j is 1 when household i is assigned to intervention j (either *Inform* or *Transport*), μ_s is a strata (panchayat) fixed effect, X is a vector of controls²⁵, and ϵ_{si} are standard errors clustered at the randomization level²⁶ (Belloni et al., 2013; Abadie et al., 2022). We test Prediction 1 about whether informing the caregiver increases take-up by testing whether $\beta_1^{Inform} > 0$. To benchmark the magnitude of this treatment effect, we compare it to $\beta_1^{Transport}$ in our analysis.

Heterogeneous effects by worry about burden. To investigate whether the effect of the intervention is heterogeneous by elder worry about burden, we run the following regression:

$$y_i = \gamma_0 + \gamma_1 T_i^j + \gamma_2 Worry_i + \gamma_3 T_i^j \cdot Worry_i + X_{imbal} + X_{imbal} \cdot T_i^j + X_i + \mu_s + \epsilon_{si} \quad (29)$$

where indicator $Worry_i$ is 1 when the elder in household i reported ever forgoing treatment out of worry about being burden in the baseline elder survey. For the analysis of heterogeneity by worry about burden, our preferred specification also includes a vector of controls X_{imbal} that includes a set of variables that are statistically significantly different in value between the elders who are worried about burden ($Worry_i = 1$) and the elders who are not ($Worry_i = 0$).

We test Corollary 1 about whether the effect of informing the caregiver differs by elder worry by testing whether γ_3 differs from 0.

²⁵In our preferred regression specification, we include controls selected with post-double selection LASSO as well as an indicator for the other cross-randomized intervention (e.g. an indicator for whether the household was part of a cluster randomized to receive free transportation, when the main treatment indicator is for whether the caregiver was informed of the check-up directly).

²⁶Standard errors are not clustered for analysis of the *Inform* intervention and clustered at the geographic cluster-level for analysis of the *Transport* intervention

Results In Table 2, we report the treatment effects of the *Inform* and *Transport* interventions on take-up for the whole sample. In columns (1) and (2), we report the effects on sign-up rates, which is a measure of whether the caregiver called to make an appointment. In columns (3) and (4), we report the effects on attendance, which is whether the elder ultimately showed up to the check-up at the hospital. Columns (1) and (3) report the coefficients from the specification controlling only for randomization strata. Columns (2) and (4) report the differences, including controls LASSO-selected baseline variables. Overall, sign-up rates are higher than attendance rates, with 25.8% of households in the control group signing up but only 17.5% attending.

In Panel A, we report the effect of the *Inform* intervention, pooling across the transportation arms. We find that the intervention of informing both the caregiver and elder directly increased sign-up rates by 7.7 percentage points ($p = 0.02$) and attendance rates by 5.3 percentage points ($p = 0.07$). Compared to the control group take-up rates, these amount to 30% increases in both sign-up and attendance rates.

According to Prediction 1, the finding of a positive treatment effect implies that the elder’s willingness to inform constraint is binding for some households in the study. Viewed through the lens of the model, this means there exist households where the elder does not inform the caregiver about the check-up, but the elders and caregivers would both agree to attend if they knew each other’s preferences. Further, ?? tells us that this rejects the perfect information benchmark in which the elder knows the caregiver’s pay-offs.

In Panel B, we show that the *Transport* intervention also increases take-up. Providing free, arranged transport support for the check-up increased sign-ups by 11.1 percentage points ($p < 0.01$) and attendance by 8.6 percentage points ($p = 0.01$). We can benchmark the magnitude of this effect against the effect of the *Transport* intervention. Our results show that a light-touch messaging intervention can have meaningful effects on take-up, amounting to around 60 – 70% of the efficacy as an intervention that provide free, private round-trip transportation to the hospital. Back-of-the-envelope cost calculations suggest that the *Inform* intervention is 7.8 – 9.1 times as cost-effective as the *Transport* intervention.²⁷

In Panel C, we report the estimates of each of the separate three treatment groups to test for interactions. The coefficient on the indicator for the *Inform + Transport* arm is small and statistically insignificant

²⁷Our cost calculations calculate that informing the caregiver costs 48 INR per respondent. This includes enumerator wages (one-tenth of a day’s worth of work at 425 INR per day) and the cost of the printed materials (5 INR per voucher). We calculate that providing transportation costs 850 INR per respondent. This includes the auto fare (on average, 425 INR per trip per elder) and enumerator wages to accompany the elder (one-half of a day’s worth of work). Taken together, the *Transport* intervention costs about 13 times as much as the *Inform* intervention per participant. Normalizing for the effect sizes of each treatment, we end up with the final calculation of the *Inform* intervention being 7.8–9.1 times more cost-effective.

(for sign-up rates, we obtain a point-estimate of -0.027 , $p = 0.69$; and for attendance, we obtain a point estimate of 0.016 , $p = 0.78$). We thus conclude that our data suggests that the *Inform* and *Transport* interventions are neither complements (which would yield a significant, positive coefficient on the interaction) nor substitutes (which would yield a significant, negative coefficient on interaction term). This suggests that the channels of providing transportation and informing caregivers of the check-up are separate, operating primarily independently.

Heterogeneity In Table 3 we report heterogeneity analysis of our main treatment effects. In columns (1) and (4), we report raw estimates controlling only for strata fixed effects. In columns (2) and (5), we include LASSO-selected baseline variables as controls. In columns (3) and (6), we report the main results from our preferred specification, which further includes the controls for the imbalanced characteristics across worried versus not-worried households discussed in Section 5.2.1.

In Panels A and B of Table 3, we separately report the effect of the *Inform* intervention for worried and not-worried elders, respectively. We see that the effect of the intervention is strikingly large and statistically significant for worried elders, but small and insignificant for not-worried elders. For example, attendance increases by 16.8 percentage points in response to the *Inform* intervention among worried elders, relative to a baseline attendance rate of 16.2% in the control group – more than doubling attendance. Sign-up rates also increase dramatically, increasing by 20.1 percentage points from 28.8% in the *No Inform* group. In contrast, among not-worried elders, the estimated coefficient for the *Inform* intervention is 0.03 ($p = 0.37$) for sign-up and 0.01 ($p = 0.67$) for attendance.

Panel C reports the formal test of comparing effect sizes of the *Inform* intervention between households with worried versus not worried elders. Consistent with Corollary 1 and our finding that worried elders inform less, we find that the *Inform* intervention is most effective among households where the elder is worried about burden. The difference between the treatment effect for worried versus not-worried elders is large and statistically significant.

We note that the results in Columns (3) and (6) demonstrate that our point estimate for γ_3 remains large and significant, even when adding controls for the elder characteristics that were imbalanced by worry about burden (e.g. for attendance, $\gamma_3 = 0.154$, $p = 0.03$).

Robustness. In the Appendix, we conduct two robustness checks for our heterogeneity analysis, reported in Table A.7. First, we isolate the heterogeneity in treatment effect that is attributable to worry about burden, after controlling for other differences in observables between worried and not-worried elders. To do this, we include controls for not only the imbalanced variables, but also the interaction of

these imbalanced variables with the treatment indicator of interest $X_{\text{imbal}} \cdot T_i^j$. Results are reported in Column (4) of Table A.7. We find that including these controls leaves our results qualitatively unchanged, and does not significantly attenuate our estimate of heterogeneity γ_3 . When adding controls interacted with the treatment indicator, the point estimate for γ_3 goes from 0.154 ($p = 0.03$) to 0.145 ($p = 0.04$) for attendance, and the point estimate for γ_3 virtually unchanged for sign-up rates (from 0.167 to 0.166, with p-values of 0.034 and 0.036, respectively). Second, in Column (5), we include proxies for caregiver care for the elder as controls to account for the correlation between elder and caregiver caring preferences, and we find that our main results are also robustness to include these controls.²⁸ For example, our estimate of γ_3 shifts from 0.154 ($p = 0.03$) to 0.150 ($p = 0.04$) when include controls for caregiver care for the elder.

In the Appendix in Table A.5, we also report whether the *Transport* intervention is differentially effective for worried versus not-worried households. While the point estimates qualitatively suggest that transportation is more effective for households where the elder is worried about burden, the coefficient on the interaction term is generally not statistically significant for either the sign-up or attendance outcomes.

6 Caregiver experiment

6.1 Caregiver experiment: Design

While the *household experiment* examines the elder’s decision to inform and demonstrates how this decision can matter for take-up of health investments for the elder, the *caregiver experiment* sheds light on the caregiver’s motives and preferences. To do this, we conduct an experiment with a sample of N=444 caregivers, recruited in an identical manner to caregivers in the household experiment, from the same study areas.²⁹

The caregiver experiment allows us to test two key model predictions around the caregiver’s motives for providing help. First, we test whether the model should have dynamic considerations by testing whether caregiver decisions to be involved are independent of their involvement in prior periods. In other

²⁸This is important because Corollary 1 technically assumes that all other variables are held fixed in estimating the comparative static with respect to elder worry. At the same time, the magnitude of the treatment effects for the *Inform* intervention depends on not only the elder’s worry (which mediates their decision to inform or not) but also the caregiver’s caring preferences (which mediates their decision to agree or not). Different correlations between elder and caregiver caring preferences could affect our interpretation of the coefficients.

²⁹Elders were recruited to the study via the listing exercise, and we used the same protocol to identify and then follow up with their primary caregivers.

words, we test for crowd-out, which may mean that caregivers may reduce help in other domains as a result of being involved in facilitating the check-up. Second, we test the degree to which the caregiver’s agreement to help facilitate the check-up is borne of pressure, as opposed to true care for the elder. If social or psychological pressure factors are important, then it is likely that caregivers who agree to help with the check-up do so only because they feel pressure to do so. In this case, observed help overstates true willingness to be involved and may actually result in welfare losses overall. We cross-randomize two experimental treatment conditions to test for crowd-out and pressure concerns. Both interventions were randomized at the caregiver-level, stratified by panchayat.

6.1.1 Intervention: Testing for crowd-out

We test whether having already committed to help the elder for one task has an impact on the likelihood of the caregiver agreeing to additional commitments to help. To do this, we randomize the order in which we offer to the caregiver a free health check-up for the elder, relative to other measures of willingness to help, which are our outcomes. The idea is that offering the health check-up before measuring the outcomes exogenously induces some caregivers to already pre-commit help for the elder, which may crowd out other willingness to help. For caregivers interested in the check-up, these caregivers are asked to set aside a portion of their survey payments (20 INR out of a 200 INR survey payment) in order to reserve a spot at the group activity, if available. Among invited caregivers, 68% express interest in the check-up. A random subsample of respondents who sign up for the check-up are then selected to receive the check-up.

1. *Already help.* (50%) Caregivers are first asked whether they would commit to signing up the elder for a check-up. After the invitation is delivered, we measure the three other outcomes of willingness to help.
2. *No prior help.* (50%) Caregivers are asked about willingness to help in the three outcomes of willingness to help. After measuring outcomes, the invitation to the check-up is delivered.

If caregiver help is subject to a fixed favor budget, then we would expect a crowd-out effect, meaning those asked about the check-up first report lower willingness to be help in the other domains. Otherwise, we would expect to see no effect of pre-committing help on subsequent willingness to help.

Outcomes. We have three measures of willingness to help that we study as outcomes, which span different time frames (to vary the gap in timing between commitments) and domains (to vary health versus non-health-related commitments): (i) signing up for a social group activity event for the elder, (ii)

interest in learning more about a general check-up in one year, and (iii) self-reported willingness to help the elder in a medical emergency (measured on a 5-point scale).

6.1.2 Intervention: Testing for pressure

In the same experiment, we test whether caregiver decisions to commit resources to help the elder is driven by pressure. We consider three sources of pressure that may impact caregivers decisions: experimenter demand from the enumerators, social pressure from the elder, and self-image concerns from the caregiver themselves.

Outcomes. Focusing on the sample of caregivers who agree to the check-up, we offer all caregivers the option to select which type of transportation support they would prefer if selected for the check-up. The outcome of interest is whether the caregiver selects an option that is more convenient for themselves but less preferred by the elder (arranged transport) or an option that is the opposite (transport reimbursement). When explaining the transportation support options, the enumerator explicitly emphasizes that prior surveys have indicated that most caregivers prefer the former while elders prefer the latter.

We cross-randomize caregivers offered transport into one of three experimental conditions, which progressively shut down different sources of pressure.

1. *Public.* The caregiver chooses on the spot whether to help, and we inform her that the elder may be told of her choice. Caregivers who choose to help are exposed to pressure from the experimenter, elder, and themselves.
2. *Private.* The caregiver chooses on the spot on a tablet, with the enumerator looking away. We assure the caregiver that her choice will not be shared.
3. *Delay.* The caregiver is offered the *Private* option plus two delay options: choose now privately, delay and call us back, or delay and have us call her back. The caregiver is informed that if they are not reached later (in either call-back scenario), they will be defaulted into the choice that elders prefer (transport reimbursement).

The *Private* treatment is designed shut down social image concerns in the caregiver’s decision, since their choice will not be revealed to the enumerator or the elder (Bursztyn and Jensen, 2017). The *Delay* treatment aims to further shut down self-image concerns that may make the caregiver feel pressured to decide in favor of the elder, following the spirit of the experimental designs in Dana et al. (2007), DellaVigna et al. (2012), and Lazear et al. (2012). By offering the caregiver the option to delay their

choice, we allow them to introduce some plausible deniability into the process of making their choice - not only from others, but also potentially from themselves.

If caregiver willingness to help is driven by any kind of pressure that reduces their ability to decline a request face-to-face, we expect help rates to be highest in the *Public* condition and lowest in the *Delay* condition. If caregivers help because they genuinely want to be involved, help rates should be similar across the three conditions.

6.1.3 Balance.

We are generally balanced on elder and caregiver characteristics, as reported in Table A.4.

6.2 Caregiver experiment: Results

6.2.1 Results: Caregiver crowd-out

Empirical strategy To test whether prior commitment to help crowds out future willingness to help, we run the following regression:

$$y_i = \delta_0 + \delta_1 T_i^{AlreadyHelp} + X_i + \mu_s + \epsilon_i \quad (30)$$

where indicator y_i is 1 when the caregiver chooses an option indicating willingness to help the elder, indicator $T_i^{AlreadyHelp}$ is 1 when the caregiver is randomized into the arm where they have already been invited to the check-up before measuring other willingness to help outcomes, X_i is a vector of LASSO-selected controls, and μ_s are strata fixed effects. Finding an effect of already helping $\delta_1 < 0$ would suggest crowd-out occurs, meaning caregiver help is rationed.

Results In Panel A of Table 5, we show the effect of the caregiver already having committed to help the elder on several other outcomes measuring willingness to help for additional future tasks. In the *No prior help* arm, the caregivers are directly asked about willingness to help on outcome tasks after the initial baseline survey. In the *Already help* arm, the caregiver has first been invited to sign the elder up for the check-up before being asked about the outcome tasks. In theory, the intervention invites caregivers to voluntarily commit to helping the elder rather than directly forcing them to do so, but in practice most caregivers in the *Already help* arm do actually sign up for the check-up (67%).

Across all our outcome tasks, asking the caregiver to already commit to helping the elder does not significantly impact the likelihood of the caregiver being willing to help the elder in the other outcome

tasks. For example, among those not already asked to help, 50% of caregivers agree to sign the elder up for group activities, which they are told involves them potentially providing transport for the elder to and from the activity as well as committing a small portion of their survey payment to sign up. Asking caregivers to help before has a negative but statistically insignificant effect on likelihood of agreeing to help ($\delta_1 = -0.04$, $p = 0.38$), suggesting that agreeing to the check-up does not crowd out assistance in non-health domains. Similarly, the likelihood of expressing interest in being contacted for a general check-up in one year is similar across both groups ($\delta_1 = 0.04$, $p = 0.19$), so long-term health investments are also not affected. Finally, caregiver-reported likelihood of helping the elder if there were an urgent health issue in the near term is similar for both likelihood of reporting “probably or always” able to help ($\delta_1 = 0.04$, $p = 0.31$) and likelihood of reporting “always” able to help ($\delta_1 = -0.04$, $p = 0.41$). Taken together, we find no evidence of large, negative crowd-out effects when involving the caregiver in the elder’s health take-up.

Further, on the elder’s side, we do not find strong evidence that elders are worried about crowd-out as a key reason for not asking for help. Among a subsample of elders responded to an open-ended question about reasons they hesitate to ask for help from their caregivers (N=98), the most frequently mentioned reason for not asking is not wanting to be a burden (69% of responses), followed by not wanting to upset the caregiver (37%) and thinking the caregiver will say no anyway (22%). Only 8% of elders report crowd-out (i.e. potentially “needing help for something urgent later”) as a reason.

6.2.2 Results: Caregiver helping from pressure

Empirical strategy To test whether caregiver involvement is due to pressure, we run the following regression:

$$y_i = \nu_0 + \nu_1 T_i^{Private} + \nu_2 T_i^{Delay} + X_i + \mu_s + \epsilon_i \quad (31)$$

where indicator y_i is 1 when the caregiver chooses arranged transportation (which elders generally prefer) and 0 when the caregiver chooses transport reimbursement (which caregivers generally prefer), indicator $T_i^{Private}$ is 1 when the caregiver is randomized into the arm where they make their choice privately, indicator T_i^{Delay} is 1 when the caregiver is randomized into the arm where they can choose to delay their choice or make their choice privately on-the-spot, X_i is a vector of LASSO-selected controls, and μ_s are strata fixed effects. Finding an effect of privacy $\nu_1 < 0$ would suggest that caregiver decision to help is partly driven by external social pressure from the elder or experimenter, and finding an effect of giving caregivers an out $\nu_2 < \nu_1$ would suggest that caregiver decision to help is partly driven by internal pressure from the caregiver themselves, such as guilt or self-image concerns.

Results In Panel B of Table 5, we show the estimated effects of the choice environment on whether the caregiver choose the transportation option preferred by elders (reimbursed transport). In the baseline *Public* condition, in which the decision is subject to potential influence from the experimenter, the elder, and the caregiver’s own self-image or guilt concerns, caregivers choose the option preferred by the elder 21% of the time. The *Private* condition, in which the decision is made privately on the tablet (and explicitly explained as totally private), leads to no statistically significant change in likelihood of choosing the elder-preferred option ($\nu_1 = 0.03$, $p = 0.62$). This suggests that caregiver decisions to be involved is not driven by elder- and experimenter demand in this context. Finally, in the *Delay* condition, in which the decision is made privately or the caregiver has the option to make a choice later, the likelihood of choosing the elder-preferred option is still not significantly different from the baseline *Public* condition ($\nu_2 = -0.03$, $p = 0.68$). This suggests that caregiver decisions are not driven by any pressure-based motivations from the elder, experimenter, or the caregiver themselves.

In the context of our theoretical framework and decision-making context, we conclude that the true pressure-based distortion to decision-making ϕ is minimal. Viewed through the lens of our model, this suggests that interventions that directly surface the check-up to caregivers – as well as additional informing rates from elders – are generally welfare-improving, as there are no large, negative welfare consequences that would arise if many caregivers were acting out of pressure rather than genuine care for the elder.

7 Discussion

7.1 Alternative explanations

In this section, we discuss potential alternative explanations for our empirical results. First, we note that alternative explanations would have to capture our key empirical findings of stark heterogeneity by elder worry about burden. First, elders worried about burden more strongly take up check-ups in response to an intervention that informs their caregivers about a check-up. Second, elders worried about burden are more sensitive to caregiver costs, responding to higher costs by reducing willingness to communicate about the check-up.

Rejection sensitivity In Appendix Section C.1, we discuss an alternative framework proposed in Bénabou et al. (2025), which captures the idea that elders may not inform caregivers about the check-up because doing so is essentially an uncertain ask that makes them vulnerable to being rejected. A key prediction of this model is that there should be positive demand for technology that allow elders to surface

the ask without fully internalizing the rejection costs, such as the experimenter being the one to deliver the invitation to the check-up, and that this demand should increase for the most rejection-sensitive elders. We test and reject this alternative model using data that shows elders’ willingness to allow the experimenter to inform the caregiver are quite similar to elder’s actual asking rates in the control group (as reported by a sample of caregivers in a follow-up survey). Further, the gap between these measures of willingness to ask is not higher for more worried elders.

Asking costs We find that an intervention informing caregivers of the check-up increases take-up, which could be consistent with a simpler model of communication frictions in which elders face a cost of asking the caregiver for help (that is not other-regarding). However, this alone would not explain our findings of heterogeneous take-up effects and asking rates by worry about burden. In Appendix Section C we discuss a model of communication frictions in which a necessary assumption is that asking frictions scale with elder caring preferences α^e . In the model, asking costs that are “fixed” in that they are incurred regardless of whether the caregiver agrees or not (potentially as a reduced form representation of elder self-image (e.g. pride or ego), awkwardness, or actual communication costs such as traveling to the caregiver or making a phone call), and they scale with elder caring preferences. However, even beyond the fact that our experiment design intentionally minimized logistics-based frictions associated with asking, we also find limited empirical evidence of them – as elders are willing to have the experimenter inform the caregiver on their behalf at similar rates as actually informing themselves, as reported by caregivers.

Misperception of caregiver costs Another reason for low information sharing is that elders may misperceive the actual costs of caregivers facilitating the check-up κ . Because elders internalize caregiver costs in proportion to their care for their caregiver, a model in which elders overestimate caregiver costs can also generate the main results of our paper. We discuss the model with misperceived costs in detail in Appendix Section C.

Objective Burden When we compare elder characteristics by their worry about burden, we do find that worried elders report more functional impairment for tasks such as difficulty learning new tasks, walking far distances, and leaving the home (Table 1). However, our results are not explained by differences in objective burden. First, when we control for these differences in functional impairment (as well as their interactions with the treatment indicator), this does not change our key finding of heterogeneity in treatment effects of the *Inform* intervention. Second, we collected objective measures of cardiac risk using handheld ECG devices for a subsample of elders, and this objective measure is balanced between worried and not-worried elders. Finally, from the caregiver’s point of view, the elders worried

about burden have caregivers who report feeling *less* burdened.

Uncertainty Another potential interpretation of worry about burden would be the elder’s degree of uncertainty about whether the caregiver will agree to the request. In our model, a mean-preserving increase in uncertainty about the caregiver’s pay-offs – i.e. the caregiver’s caring or benefit or cost of attending – would not change the elder’s willingness to inform, because the elder is taking an expectation over linear pay-offs. Future extensions of the model could introduce non-neutral risk preferences, in which case different degrees of uncertainty may have first-order effects on the elder willingness to inform. Empirically, we can also check whether there is heterogeneity in treatment effect of the *Inform* intervention on the attendance of the check-up by whether the caregiver or the elder was invited first to the check-up. In the field, we conducted separate visits via independent teams of enumerators to survey and deliver check-up invitations to elders and caregivers. Cases where the caregiver is visited first may have less uncertainty facing the elder about the caregiver’s willingness to attend the check-up, if it gives the caregiver a chance to initiate the conversation (or the elder a chance to observe the absence of a conversation about the check-up). However, we find no evidence of heterogeneity in treatment effects by who was visited first, suggesting uncertainty may not play a large role in explaining our effects.

8 Conclusion

In this paper, we combine a framework of elders and caregiver communication with mutual caring and uncertainty with two randomized experiments to study how household communication affects take-up of healthcare for elderly women in India. We find that an intervention directly informing caregivers about a check-up increases take-up. The effects are especially strong for households with elders who are caring and worry about being a burden to their family members. These worried elders are also less willing to communicate about the check-up and more sensitive to caregiver costs, suggesting elder care for their caregivers can interact with beliefs about imposed burden to cause low levels of communication. Using the caregiver experiments, we show that caregiver decisions to help are generally not driven by pressure and do not crowd out future support for the elder. As a result, the low take-up rates reflect misperceptions, and increasing involvement from caregivers is likely to be welfare-improving for both elders and caregivers.

Our findings emphasize the challenges of nuanced communication in the real-world, even between two individuals with a longstanding relationship and ongoing, repeated interactions. In our framework, we show that genuine concern for another person can interact with uncertainty about another person’s motiva-

tions for helping to lead to inefficiently low communication. Our empirics suggest that a low-communication equilibrium can persist despite mutual caring between elders and caregivers who share a common goal of keeping the elder healthy. Future work may shed light on whether these dynamics of mutual caring and low communication apply to other types of relationships within and across households and contexts.

Further, our study focuses on a single decision about whether to take up preventative care that is offered for free, which has the potential of being long-term beneficial to both the elder and caregiver, saving them both future costs. We leave it to future work to test for communication breakdowns in other aspects of the caregiving relationship and decisions, which may involve different strategic motivations and trade-offs for household members.

Our results also directly suggest potential solutions in policy for communicating to households about medical results and opportunities. Family caregivers play an important role in the health decisions and outcomes of the elderly, especially for older women who face especially severe healthcare access constraints and low household bargaining power in India (Dupas and Jain, 2024; Calvi, 2020). Clear, direct communication to caregivers advocating health opportunities for women and elders may be beneficial for many reasons (e.g. higher medical literacy, reduced memory concerns) including reducing the load of communicating health needs which weigh disproportionately on older women. At the same time, these types of interventions aimed at improving information of the household as a whole can be cost-effective, with our finding that the simple intervention of informing the caregiver directly can have about two-thirds of the impact of providing private, arranged transportation.

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Tables and Figures

8.1 Tables

Table 1: Characteristics by Worry about Burden

Variable	Overall (1)	Not worried (2)	Worried (3)	Difference (4)	p (5)
<i><u>Elder Characteristics</u></i>					
Age over 65	0.59	0.58	0.62	0.04	0.36
Widowed	0.52	0.51	0.56	0.04	0.3
Elder living alone	0.11	0.10	0.13	0.03	0.22
Household size	3.28	3.33	3.13	-0.20	0.04**
Number of children	2.59	2.61	2.53	-0.08	0.47
Lives with caregiver	0.82	0.84	0.77	-0.07	0.04**
Checkup last 1y	0.83	0.83	0.82	-0.01	0.72
Heart checkup ever	0.43	0.45	0.39	-0.05	0.23
Heart checkup last 1y	0.16	0.16	0.17	0.01	0.71
High cardiac risk (ECG)	0.50	0.48	0.56	0.07	0.22
Has blood pressure	0.61	0.59	0.67	0.09	0.03**
Has diabetes	0.42	0.41	0.48	0.07	0.09*
Has cholesterol	0.15	0.15	0.14	0.00	0.92
Has cardiac	0.15	0.15	0.15	0.00	0.87
Has chronic pain	0.10	0.09	0.11	0.02	0.42
Difficulty leaving home	0.58	0.54	0.70	0.16	0***
Difficulty walking long dist	0.77	0.74	0.86	0.12	0***
Difficulty new task	0.65	0.59	0.83	0.24	0***
Interest in check-up	0.93	0.93	0.91	-0.02	0.44
Chose checkup over cash	0.60	0.62	0.57	-0.05	0.24
<i><u>Caregiver Characteristics</u></i>					
Is Husband	0.21	0.23	0.18	-0.05	0.15
Is Son	0.43	0.45	0.37	-0.07	0.08*
Is Daughter	0.15	0.14	0.19	0.05	0.07*
Is Son-in-law	0.02	0.01	0.03	0.02	0.03**
Is Daughter-in-law	0.09	0.09	0.06	-0.03	0.21
Is Grandchild	0.05	0.04	0.06	0.02	0.19
Is Other	0.06	0.04	0.10	0.05	0.01***
Average age (years)	47.78	48.15	46.71	-1.44	0.25
Married	0.87	0.87	0.86	-0.01	0.77
Household size	4.38	4.38	4.37	0.00	0.99
Working	0.68	0.69	0.65	-0.04	0.27
Monthly Income Per Capita (1000 Rs)	2.99	2.97	3.03	0.06	0.86
Social Desirability Score (Pct)	0.26	0.27	0.26	-0.01	0.49
High NCD knowledge	0.72	0.72	0.72	0.00	0.95
Interest in checkup for eld	0.85	0.84	0.87	0.03	0.48
CG present for EL invite	0.14	0.14	0.13	-0.01	0.65
<i><u>Closeness Measures</u></i>					
Agreement index (SDs)	-0.90	-0.92	-0.83	0.09	0***
CG does not feel burdened	0.56	0.52	0.67	0.15	0.03**
CG speak at least weekly	0.36	0.36	0.34	-0.02	0.65
EL comf asking transport	0.39	0.40	0.36	-0.04	0.33
EL comf asking cash	0.34	0.36	0.29	-0.07	0.1
EL no pressure asking cash	0.04	0.04	0.05	0.01	0.49
N	730	543	187		

Notes: This table compares baseline characteristics of households by whether the elder is worried about being a burden for the N=730 households in the Household Experiment. Elders are classified as “Worried” if they report having ever forgone care or avoided the hospital out of worry of being a burden to family members. One observation has missing value for worry about burden, because the respondent refused to respond to the question. Column (1) reports the full-sample mean of each row variable. Columns (2) and (3) report means for the not-worried and worried groups. Column (4) reports the difference estimated from a regression of the row variable on the worried indicator without any controls, and column (5) reports the p-value on the difference. Elder characteristics come from the elder listing survey, and caregiver characteristics come from the caregiver baseline survey. The exceptions are the self-reported diagnoses measures, which are indicators equal to one if either caregiver or elder (from the elder follow-up survey) reports the elder having the condition. Among closeness measures, caregiver-reported variables are denoted with the prefix “CG” and elder-reported variables are denoted with the prefix “EL”. The agreement index is a standardized index where higher values indicate more agreement between elder and caregiver responses on three questions: multiple-select list of elder health diagnoses,

Table 2: Effect of Interventions on Take-up

	Signed up		Attended	
	No controls (1)	Controls (2)	No controls (3)	Controls (4)
<u>Panel A: Inform both</u>				
Inform	0.069** (0.033) <i>0.035</i>	0.077** (0.033) <i>0.020</i>	0.046 (0.029) <i>0.112</i>	0.053* (0.029) <i>0.069</i>
Control mean	0.258	0.258	0.175	0.175
<u>Panel B: Free transport</u>				
Transport	0.105*** (0.035) <i>0.003</i>	0.111*** (0.036) <i>0.002</i>	0.089*** (0.031) <i>0.005</i>	0.086*** (0.033) <i>0.010</i>
Control mean	0.227	0.227	0.143	0.143
<u>Panel C: Interacted</u>				
Inform	0.073* (0.043) <i>0.093</i>	0.090** (0.045) <i>0.046</i>	0.035 (0.034) <i>0.310</i>	0.044 (0.036) <i>0.216</i>
Transport	0.110** (0.047) <i>0.019</i>	0.124*** (0.048) <i>0.010</i>	0.079* (0.042) <i>0.060</i>	0.077* (0.044) <i>0.080</i>
Inform $\times$ Transport	-0.011 (0.065) <i>0.864</i>	-0.027 (0.067) <i>0.685</i>	0.018 (0.055) <i>0.738</i>	0.016 (0.058) <i>0.779</i>
Control mean	0.198	0.198	0.132	0.132
<i>p: Inform = Transport</i>	<i>0.443</i>	<i>0.478</i>	<i>0.288</i>	<i>0.435</i>
<i>p: Inform+Tran = Inform</i>	<i>0.043</i>	<i>0.053</i>	<i>0.019</i>	<i>0.032</i>
<i>p: Inform+Tran = Transport</i>	<i>0.197</i>	<i>0.185</i>	<i>0.208</i>	<i>0.161</i>
Observations	731	731	731	731
Strata FE	X	X	X	X
Controls (LASSO)		X		X

Notes: This table presents estimated treatment effects of the two interventions (*Inform* and *Transport*) on take-up (sign-up and attendance rates). Panel A reports coefficients from regressing take-up on an indicator for assignment to the *Inform* treatment. Panel B on an indicator for the *Transport* treatment. Panel C reports coefficients from a regression of take-up on both treatment indicators and their interaction. Columns (1) and (2) show the effects on a sign-up indicator, which is equal to 1 when the household called to make an appointment at the check-up. Columns (3) and (4) show the effects on an attendance indicator, which is equal to 1 when the elder shows up for the health check-up after their household called to make an appointment. All regressions include strata fixed effects. Columns (2) and (4) add baseline controls selected via post-double-selection LASSO and, in Panels A and B, an indicator for assignment to the other (cross-randomized) treatment. Standard errors are reported in parentheses, clustered at the transport-randomization cluster level in Panels B and C. P-values are reported in italics below. The bottom rows report p-values from Wald tests based on the interacted specification in Panel C. *** p<0.01, ** p<0.05, * p<0.1.

Table 3: Effect of Information Intervention on Take-up by Worry about Burden

	Signed up			Attended		
	(1)	(2)	(3)	(4)	(5)	(6)
<u>Panel A: Inform effect for Worry</u>						
Inform	0.126* (0.068) <i>0.062</i>	0.202*** (0.068) <i>0.003</i>	0.201*** (0.068) <i>0.003</i>	0.119* (0.061) <i>0.053</i>	0.169*** (0.062) <i>0.007</i>	0.168*** (0.062) <i>0.007</i>
Control mean	0.288	0.288	0.288	0.162	0.162	0.162
<u>Panel B: Inform effect for Not Worry</u>						
Inform	0.045 (0.037) <i>0.228</i>	0.033 (0.038) <i>0.388</i>	0.034 (0.038) <i>0.370</i>	0.018 (0.032) <i>0.577</i>	0.013 (0.033) <i>0.696</i>	0.014 (0.033) <i>0.674</i>
Control mean	0.256	0.256	0.256	0.179	0.179	0.179
<u>Panel C: Inform $\times$ Worry</u>						
Inform	0.045 (0.037) <i>0.228</i>	0.033 (0.038) <i>0.388</i>	0.034 (0.038) <i>0.370</i>	0.018 (0.032) <i>0.577</i>	0.013 (0.033) <i>0.696</i>	0.014 (0.033) <i>0.674</i>
Worry	0.032 (0.053) <i>0.544</i>	0.009 (0.055) <i>0.869</i>	0.006 (0.056) <i>0.914</i>	-0.017 (0.047) <i>0.711</i>	-0.035 (0.048) <i>0.465</i>	-0.042 (0.049) <i>0.397</i>
Inform $\times$ Worry	0.081 (0.078) <i>0.296</i>	0.169** (0.079) <i>0.033</i>	0.167** (0.079) <i>0.034</i>	0.100 (0.070) <i>0.150</i>	0.156** (0.071) <i>0.029</i>	0.154** (0.071) <i>0.030</i>
Observations	730	730	730	730	730	730
Strata FE	X	X	X	X	X	X
Controls (LASSO)		X	X		X	X
Controls (Worry predictors)			X			X

Notes: This table presents estimated treatment effects and treatment effect heterogeneity of the *Inform* intervention on take-up (sign-up and attendance rates). Panel A and B report coefficients from regressing take-up on an indicator for assignment to the *Inform* treatment arm, separately for Worried and Not-worried elders, respectively. Panel C reports coefficients from regressing take-up on an indicator for assignment to the *Inform* treatment arm, an indicator for whether the elder is worried about being a burden, and the interaction between the two. Worry is an indicator for whether the elder reported ever forgoing healthcare or avoiding the hospital out of worry of being a burden to family members. One observation has missing value for worry about burden, because the respondent refused to respond to the question. Columns (1) - (3) show the effects on a sign-up indicator, which is equal to 1 when the household called to make an appointment at the check-up. Columns (4) - (6) show the effects on an attendance indicator, which is equal to 1 when the elder shows up for the health check-up after their household called to make an appointment. All regressions include strata fixed effects. Columns (2) and (5) add baseline controls selected via post-double-selection LASSO. Columns (3) and (6) further add controls for characteristics that are imbalanced by worry about burden (indicators for above-median functional impairment, whether the caregiver is a daughter, large household, and whether elder is co-residing with the caregiver). Standard errors are reported in parentheses. P-values are reported in italics. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4: Effect of Transport Intervention on Elder Willingness to Inform Caregiver By Worry about Burden

	Elder willing to inform		
	(1)	(2)	(3)
<u>Panel A: Transport effect, Not worried</u>			
Transport	0.010 (0.043) <i>0.816</i>	-0.002 (0.048) <i>0.968</i>	-0.011 (0.048) <i>0.814</i>
<u>Panel B: Transport effect, Worried</u>			
Transport	0.122* (0.071) <i>0.086</i>	0.136* (0.071) <i>0.058</i>	0.150** (0.071) <i>0.036</i>
<u>Panel C: Transport $\times$ Worry</u>			
Transport	0.010 (0.043) <i>0.816</i>	-0.002 (0.048) <i>0.968</i>	-0.011 (0.048) <i>0.814</i>
Worry about burden	-0.059 (0.069) <i>0.390</i>	-0.114* (0.069) <i>0.100</i>	-0.142** (0.070) <i>0.042</i>
Transport $\times$ Worry	0.111 (0.084) <i>0.187</i>	0.138 (0.088) <i>0.121</i>	0.161* (0.088) <i>0.069</i>
Control mean (Worried)	0.787	0.787	0.787
Control mean (Not worried)	0.852	0.852	0.852
Observations	364	364	364
Strata FE	X	X	X
LASSO controls		X	X
Worry predictors			X

Notes: This table presents the estimated treatment effect of the *Transport* intervention on the elder's willingness to inform their caregiver about the check-up. Panel A and B report coefficients from regressing willingness to inform on an indicator for assignment to the *Transport* treatment arm, separately for Worried and Not-worried elders, respectively. Panel C reports coefficients from regressing willingness to inform on an indicator for assignment to the *Transport* arm, an indicator for whether the elder is worried about burden, and the interaction of the two. The sample restricted to households in which only the elder was informed about the check-up (N=364). The dependent variable is an indicator for the elder agreeing to let the study team inform the caregiver about the check-up, elicited at the end of the elder survey. Worry about burden is an indicator for the elder reporting an above-median frequency of forgoing healthcare out of worry of being a burden to family members. All columns include strata fixed effects. Column (2) adds baseline controls selected via post-double-selection LASSO. Column (3) further adds controls for characteristics imbalanced by worry about burden (above-median

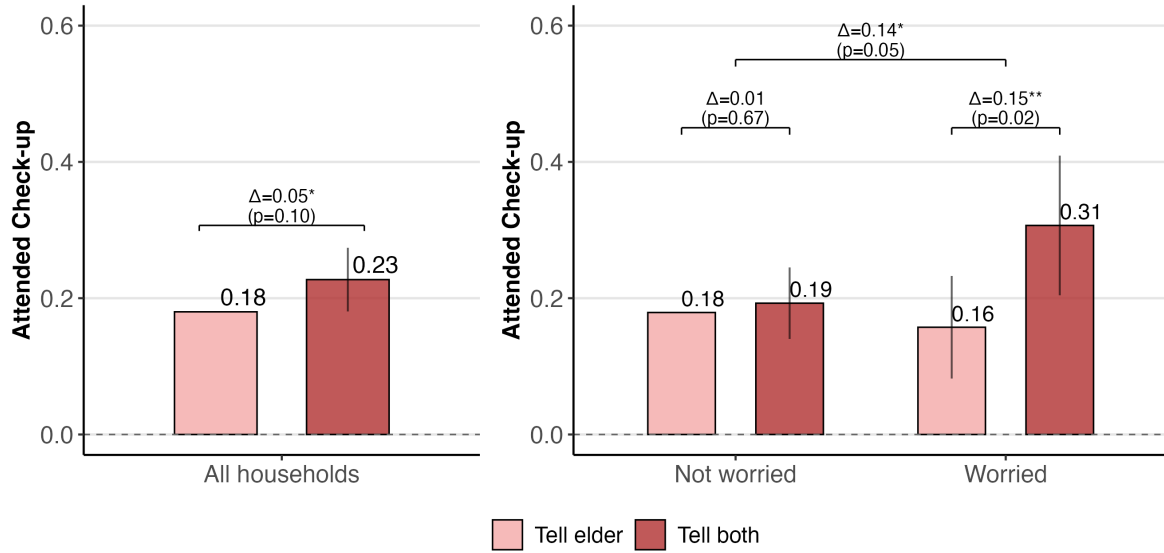
Table 5: Effects of Invitation Timing and Choice Condition on Caregiver Behavior

Panel A: Test for crowd-out						
	Social group sign-up		Interested in checkup 1y		Likely help in emerg	
	(1)	(2)	(3)	(4)	(5)	(6)
Invited to Checkup Prior	-0.042 (0.048) <i>0.380</i>	-0.038 (0.049) <i>0.442</i>	0.039 (0.030) <i>0.186</i>	0.005 (0.030) <i>0.880</i>	0.043 (0.042) <i>0.308</i>	0.062 (0.042) <i>0.142</i>
Control Mean	0.500	0.500	0.871	0.871	0.634	0.634
Observations	444	444	444	444	444	444
Strata FEs	X	X	X	X	X	X
Controls (LASSO)		X		X		X
Panel B: Test for pressure-based help						
	Chose elder-preferred option					
	(7)	(8)				
Private	0.024 (0.060) <i>0.683</i>	0.064 (0.067) <i>0.344</i>				
Delay	-0.030 (0.061) <i>0.623</i>	0.029 (0.071) <i>0.677</i>				
Control Mean	0.211	0.211				
Observations	278	278				
Strata FEs	X	X				
Controls (LASSO)		X				

Notes: This table presents the results of the caregiver experiment. Panel A reports the coefficients from a regression of caregiver willingness to be involved in non-check-up tasks on an indicator for whether they were assigned to the treatment arm in which they were already invited to sign the elder up for a check-up before (*Invited to Checkup Prior*). Panel A uses the full sample of caregivers in the caregiver survey experiment (N=444). *Invited to Checkup Before* is an indicator for the caregiver being randomly assigned to receive the elder check-up invitation before (rather than after) the outcome questions (the *Already help* arm). Dependent variables: an indicator for signing the elder up for the social group activity (Columns 1-2); an indicator for expressing interest in a general check-up for the elder in one year (Columns 3-4); and an indicator for reporting they would probably or always help the elder in a medical emergency, from a five-point scale (Columns 5-6). Control mean is the mean of the dependent variable among caregivers in the *No prior help* arm. Panel B reports the coefficients from a regression of caregiver choosing the elder-preferred transportation support on indicators for which observability treatment arm they were assigned to. Panel B uses the subsample of caregivers who signed up for the check-up earlier in the survey (N=278). *Private* and *Delay* are indicators for the corresponding cross-randomized treatment arms. The omitted treatment group is *Public* arm. The dependent variable is an indicator for whether the caregiver chose the option the elder prefers. All regressions include strata fixed effects. Even columns (2, 4, 6, and 8) include Lasso-selected controls. Standard errors in parentheses, and p-values are reported in italics below. * p<0.1, ** p<0.05, *** p<0.01.

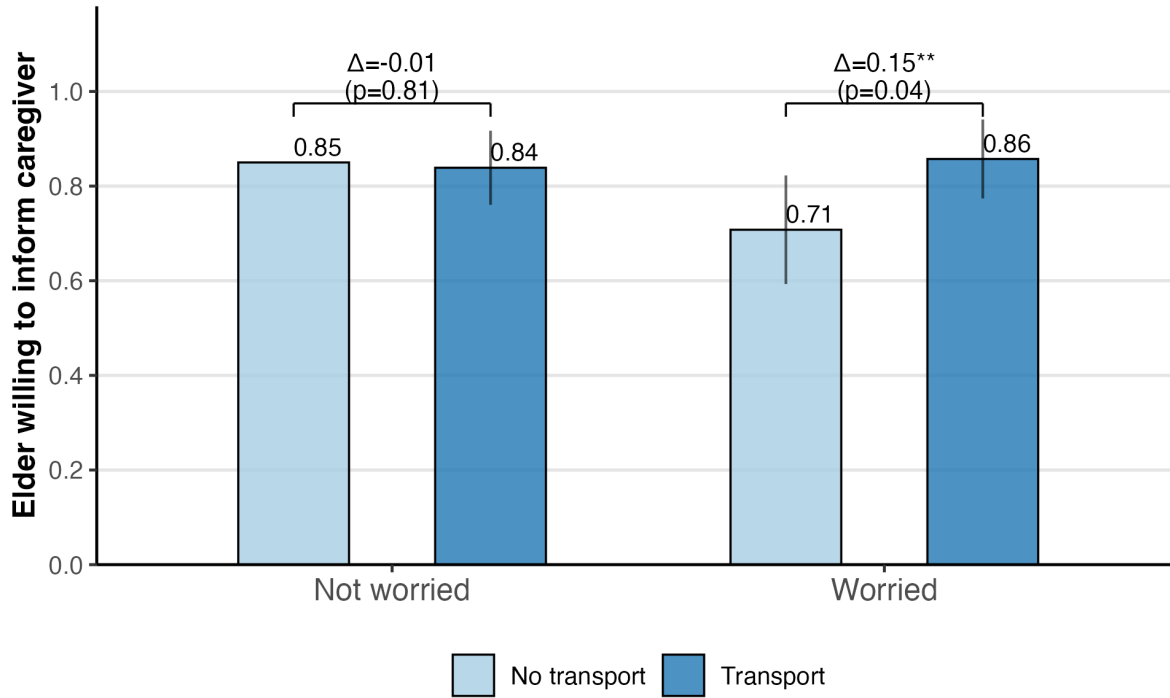
8.2 Figures

Figure 3: Take-up by Information Treatment and Worry about Burden



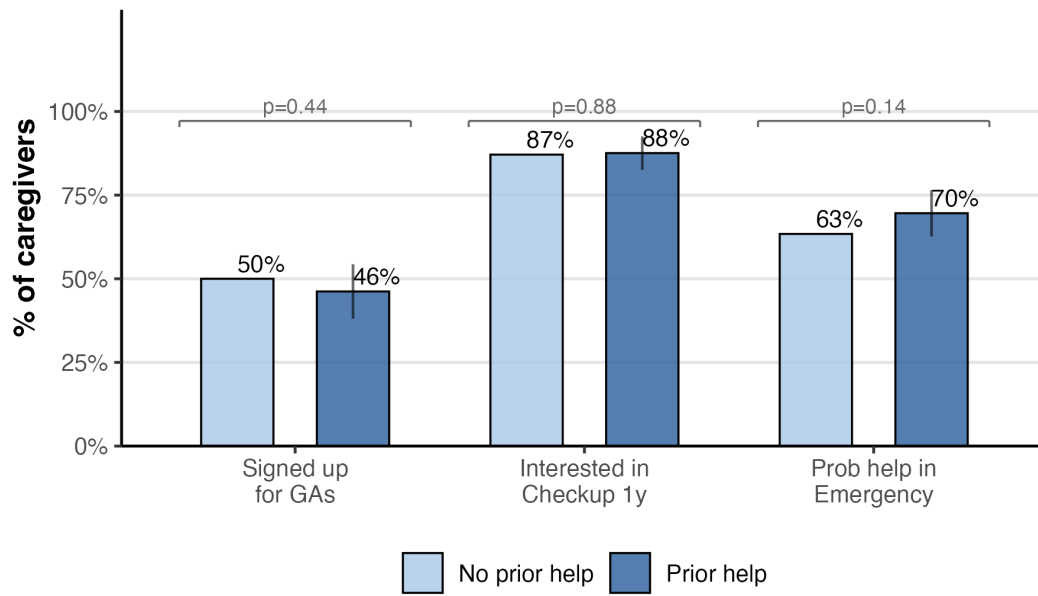
Notes: This graph shows results from Equation 28 and Equation 29, which estimates the effect of the *Inform* intervention that directly informs the caregiver about the check-up on the fraction of households whose elder attended the check-up. The left panel shows the overall effect across all households (N=731). The right panel shows the effect separately for “Worried” and “Not worried” households; a household is classified as “Worried” when the elder reported ever forgoing treatment or avoiding the hospital out of concern of being a burden in the baseline (N=187) and classified as “Not worried” otherwise (N=543). Each bar is sum of the control group mean (*Control*, in which only the elder is informed) and the relevant regression coefficients. The estimates come from the regression that includes controls for randomization strata and LASSO-selected variables. Displayed are 90% confidence intervals.

Figure 4: Elder Willingness to Inform by Transport Treatment and Worry about Burden



Notes: The figure displays the rates of elder willingness to inform their caregiver about checkup eligibility. The rates are predicted means estimated from an OLS regression of an indicator for whether the elder is willing to inform the caregiver of the check-up on transport assignment, an indicator for above-median worry about burdening one's caregiver (based on pre-treatment elder recruiting survey responses), and their interaction. Controls include strata fixed effects, LASSO-selected baseline controls, and controls for variables imbalanced by elder concern of burden. The sample is restricted to elders assigned to the *No Inform* control arm. Bars show predicted means for each cell of the 2×2 worry-by-transport design: the Not Worried / No Transport bar equals the control mean; other bars add the relevant OLS coefficients. Brackets report the effect of free transport separately for not-worried and worried elders, with p-values computed using cluster-robust standard errors clustered at the village level. Error bars show 90% confidence intervals.

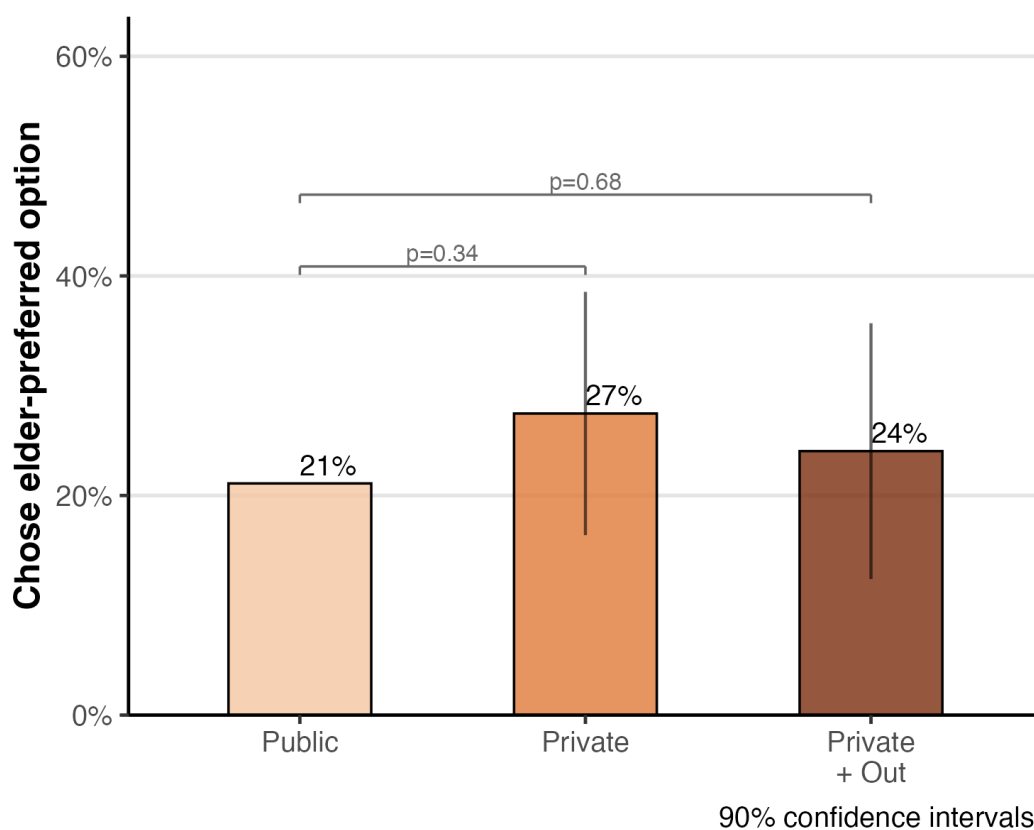
Figure 5: Caregiver Experiment: Test for Crowd-out



Treatment = invited to checkup before GAs. 90% confidence intervals.

Notes: The figure displays OLS estimates of the effect of prior checkup invitation on caregiver outcomes. "No prior help" bars show the mean outcome among caregivers assigned to the GA-first order; "Prior help" bars show the predicted mean for caregivers assigned to the checkup-first order (control mean plus OLS coefficient). Regressions include strata fixed effects, treatment choice fixed effects, and LASSO-selected baseline controls. Brackets report the p-value for the treatment coefficient from the underlying regression. Error bars show 90% confidence intervals based on heteroskedasticity-robust standard errors.

Figure 6: Caregiver Experiment: Test for Pressure-motivated Involvement

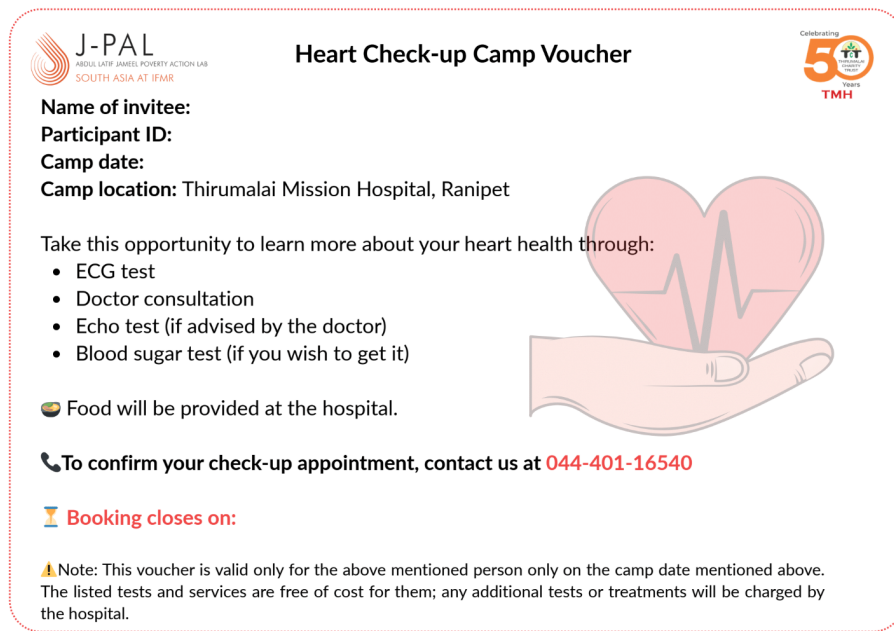


Notes: The figure displays OLS estimates of the effect of transport framing on caregivers' willingness to choose the elder-preferred transport option. "Public" shows the mean outcome among caregivers randomly assigned to the public framing arm (reference group); "Private" and "Private + Out" show predicted means for the private and delay arms respectively (control mean plus OLS coefficient). Regressions include strata fixed effects, treatment order fixed effects, and LASSO-selected baseline controls. Brackets report p-values for pairwise comparisons against the public arm. Error bars show 90% confidence intervals based on heteroskedasticity-robust standard errors.

A Additional Tables and Figures

A.1 Study materials

Figure A.1: Check-up voucher



The image shows a 'Heart Check-up Camp Voucher' with a red dotted border. At the top left is the J-PAL logo (ABDUL LATIF JAMHEEL POVERTY ACTION LAB, SOUTH ASIA AT IFMR). At the top right is a 'Celebrating 50 Years TMH' logo. The title 'Heart Check-up Camp Voucher' is centered at the top. Below the title, there are fields for 'Name of invitee:', 'Participant ID:', 'Camp date:', and 'Camp location: Thirumalai Mission Hospital, Ranipet'. To the right of these fields is an illustration of a hand holding a heart with an ECG line. Below the fields, it says 'Take this opportunity to learn more about your heart health through:' followed by a bulleted list: 'ECG test', 'Doctor consultation', 'Echo test (if advised by the doctor)', and 'Blood sugar test (if you wish to get it)'. Below the list, it says 'Food will be provided at the hospital.' with a fork and knife icon. Then, it says 'To confirm your check-up appointment, contact us at 044-401-16540' with a phone icon. Below that, it says 'Booking closes on:' with a calendar icon. At the bottom, there is a note: 'Note: This voucher is valid only for the above mentioned person only on the camp date mentioned above. The listed tests and services are free of cost for them; any additional tests or treatments will be charged by the hospital.'

J-PAL
ABDUL LATIF JAMHEEL POVERTY ACTION LAB
SOUTH ASIA AT IFMR

Heart Check-up Camp Voucher

Name of invitee:
Participant ID:
Camp date:
Camp location: Thirumalai Mission Hospital, Ranipet

Take this opportunity to learn more about your heart health through:

- ECG test
- Doctor consultation
- Echo test (if advised by the doctor)
- Blood sugar test (if you wish to get it)

🍴 Food will be provided at the hospital.

📞 To confirm your check-up appointment, contact us at **044-401-16540**

📅 **Booking closes on:**

⚠️ **Note:** This voucher is valid only for the above mentioned person only on the camp date mentioned above. The listed tests and services are free of cost for them; any additional tests or treatments will be charged by the hospital.

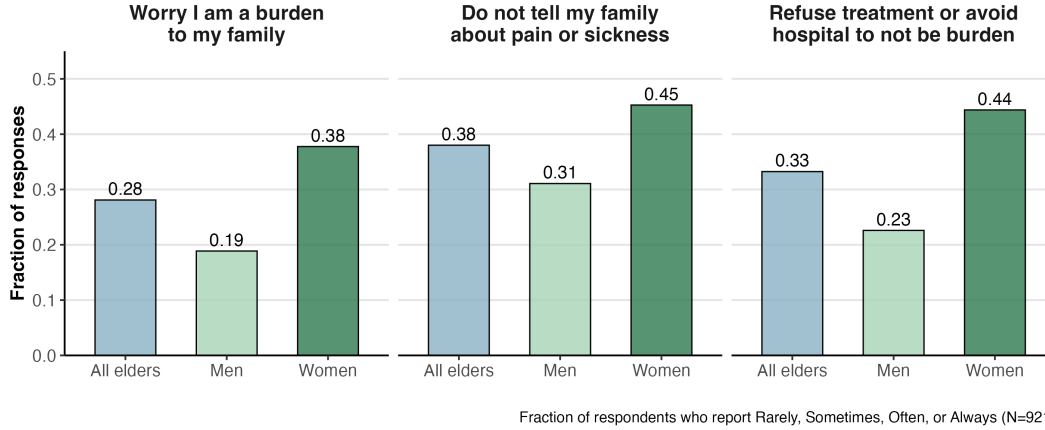
Notes: This image shows the English version of the printed voucher that was given to elders when they were invited to the check-up as part of the household experiment. The vouchers used during the study were translated into Tamil.

A.2 Panel data

Table A.1: Health behaviors, attitudes, and outcomes by gender

	All		Men	Women	Women - Men		
	Mean	SD	Mean	Mean	Diff	P-val	N
Checkup in last year	0.78	0.41	0.78	0.79	0.01	0.58	3761
Monthly med expenses (1000 INR)	3.29	46.10	6.13	1.56	-4.57	0.03**	3373
Has hypertension	0.62	0.49	0.61	0.62	0.02	0.36	3076
Has diabetes	0.20	0.40	0.21	0.20	-0.02	0.21	2894
Has undiagnosed hypertension	0.74	0.44	0.71	0.76	0.05	0.04**	1693
Has undiagnosed diabetes	0.61	0.49	0.57	0.64	0.07	0.11	580
Difficulty walking long distance (1km)	0.65	0.48	0.55	0.70	0.15	0.00***	3543
Difficulty leaving home	0.43	0.49	0.33	0.49	0.16	0.00***	3688
Difficulty getting to places outside home	0.53	0.50	0.40	0.61	0.22	0.00***	3675
Difficulty learning new task	0.46	0.50	0.34	0.53	0.18	0.00***	3651
Food and supplies provided by family	0.46	0.50	0.42	0.48	0.06	0.00***	3761
Med expenses payer: Family	0.51	0.50	0.45	0.55	0.10	0.00***	1528
Difficulty to get tested NCDs	0.17	0.38	0.12	0.22	0.10	0.00***	816
Difficulty to get tested: Need help	0.40	0.49	0.17	0.51	0.34	0.00***	143
Difficulty to get tested: Money	0.64	0.48	0.70	0.61	-0.09	0.30	143
Worry I am a burden to my family	0.26	0.44	0.16	0.37	0.20	0.00***	694
Do not tell family about pain or sickness	0.29	0.46	0.25	0.35	0.10	0.00***	692
Refuse treatment or avoid hospital to not be burden	0.28	0.45	0.18	0.39	0.21	0.00***	692

Figure A.2: Concern of Burden by Gender



Notes: For each of the three behaviors asked about in the survey, respondents were asked how frequently they engaged in the behavior on a five-point scale ($1=Never$, $2=Rarely$, $3=Sometimes$, $4=Often$, or $5=Always$). In the figure, we report the fraction of individuals whose responses were 3 or higher. This data comes from a survey with a representative sample of $N=921$ adults ($N=434$ men and 487 women) aged 55 and older from Tamil Nadu, India, collected in parallel with the third wave of the Tamil Nadu Elder Panel Survey (Duflo et al., 2022)

Table A.2: Open-ended Responses for Reasons to Take Care of Health

Reason	Pct Agree
To live a comfortable / pain-free life	56
To care for / reduce the burden on my family	55
To reduce future healthcare expenses / pain	30
No one to take care of me, so I need to stay healthy	19
I do not worry about my health	6
Do not know / Other	4

Notes: This table lists the responses reported by elders in response to an open-ended question about why they take care of their health (*I would like to know how you approach your health. How important do you feel it is to take care of your health? Why (or why not) do you think it's important to take care of your health? What motivates you to take care of your health?*). Responses were coded as multiple-select question by enumerators. The question was administered as part of a free-response survey module among adults aged 55 and older from Tamil Nadu India, collected in parallel with the third wave of the Tamil Nadu Elder Panel Survey (Duflo et al., 2022). Among the N=921 adults who participated in the broader survey, N=776 respondents agreed to participate in the free-response section, whose responses are reported here.

A.3 Balance

Table A.3: Characteristics by Treatment Assignment in Household Experiment

Variable	Control		Inform		Transport		Inform + Tran		Inform=Tran (9)	Inform=Both (10)	Tran=Both (11)	N (12)
	Mean	SD	Mean+ β	p	Mean+ β	p	Mean+ β	p				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)				
<i>Elder Characteristics</i>												
Age over 65	0.58	(0.50)	0.68	0.03**	0.56	0.75	0.56	0.69	0.02**	0.02**	0.93	731
Widowed	0.49	(0.50)	0.51	0.71	0.57	0.12	0.55	0.22	0.25	0.41	0.72	731
Elder living alone	0.11	(0.31)	0.11	0.94	0.09	0.6	0.12	0.66	0.54	0.71	0.36	731
Household size	3.37	(1.20)	3.25	0.3	3.23	0.23	3.27	0.41	0.84	0.85	0.7	731
Number of children	2.63	(1.22)	2.56	0.57	2.64	0.97	2.55	0.59	0.53	0.94	0.53	731
Lives with caregiver	0.83	(0.38)	0.81	0.57	0.83	0.97	0.83	0.95	0.61	0.55	0.92	731
Checkup last 1y	0.80	(0.40)	0.81	0.81	0.80	0.94	0.88	0.05*	0.89	0.09*	0.08*	659
Heart checkup ever	0.43	(0.50)	0.46	0.51	0.37	0.27	0.44	0.84	0.08*	0.67	0.18	659
Heart checkup last 1y	0.19	(0.40)	0.19	0.86	0.12	0.08*	0.15	0.25	0.12	0.34	0.5	659
High cardiac risk (ECG)	0.56	(0.50)	0.48	0.32	0.50	0.41	0.51	0.43	0.85	0.7	0.87	357
Difficulty leaving home	0.58	(0.49)	0.60	0.67	0.58	0.93	0.55	0.5	0.74	0.28	0.44	730
Difficulty walking long dist	0.78	(0.42)	0.81	0.41	0.79	0.85	0.74	0.3	0.54	0.08*	0.23	731
Difficulty new task	0.64	(0.48)	0.72	0.09*	0.64	0.94	0.61	0.56	0.05*	0.02**	0.58	727
Often worry burden	0.27	(0.44)	0.22	0.33	0.24	0.45	0.27	0.99	0.8	0.31	0.47	730
Often forgo treatment	0.18	(0.39)	0.21	0.54	0.14	0.33	0.17	0.77	0.11	0.34	0.48	730
Uncomf asking for cash	0.67	(0.47)	0.65	0.64	0.68	0.96	0.67	0.99	0.64	0.67	0.96	626
Uncomf asking for transport	0.62	(0.49)	0.60	0.73	0.64	0.6	0.67	0.32	0.45	0.24	0.61	626
Chose checkup over cash	0.65	(0.48)	0.59	0.23	0.59	0.23	0.60	0.3	1	0.85	0.83	659
Interest in check-up	0.91	(0.28)	0.92	0.78	0.94	0.44	0.96	0.09*	0.57	0.12	0.36	729
<i>Caregiver Characteristics</i>												
Is Husband	0.21	(0.41)	0.23	0.81	0.20	0.79	0.22	0.9	0.6	0.91	0.67	731
Is Son	0.45	(0.50)	0.41	0.5	0.44	0.98	0.40	0.42	0.53	0.9	0.43	731
Is Daughter	0.16	(0.37)	0.16	0.92	0.14	0.65	0.14	0.66	0.73	0.74	0.99	731
Is Son-in-law	0.02	(0.15)	0.02	0.89	0.01	0.44	0.01	0.2	0.36	0.15	0.56	731
Is Daughter-in-law	0.09	(0.28)	0.06	0.28	0.10	0.73	0.10	0.58	0.18	0.15	0.83	731
Is Grandchild	0.03	(0.16)	0.06	0.06*	0.06	0.12	0.04	0.48	0.91	0.3	0.35	731
Average age (years)	47.66	(14.08)	48.98	0.42	46.90	0.61	47.70	0.98	0.2	0.46	0.59	731
Married	0.87	(0.34)	0.86	0.76	0.84	0.36	0.89	0.42	0.58	0.29	0.08*	731
Household size	4.27	(1.71)	4.52	0.24	4.25	0.92	4.57	0.15	0.15	0.8	0.08*	731
Monthly Income Per Capita (1000 Rs)	2.92	(2.44)	2.83	0.84	2.77	0.64	3.30	0.48	0.91	0.45	0.26	469
Social Desirability Score (Pct)	0.27	(0.15)	0.26	0.44	0.27	0.93	0.26	0.62	0.35	0.79	0.56	720
Invited CG First	0.64	(0.48)	0.58	0.17	0.57	0.13	0.60	0.36	0.87	0.73	0.55	731
F-stat p.			0.23		0.7		0.09*					
Observations	182		175		183		191					731

Notes: This table shows randomization balance for baseline observable variables by each of the four treatment groups. In each row, the columns (1) and (2) report the mean and standard deviation in the *Control* group (only elder informed, no transport). For each treatment arm (*Inform*, *Transport*, and *Inform + Transport*), the table reports the control mean plus the arm coefficient β from a regression of the row variable on indicators for the three arms with strata fixed effects, along with the p-value on β (in italics). Columns (9)-(11) report p-values from Wald tests of pairwise equality of the three arm coefficients. Standard errors are clustered at the transport-randomization cluster level. The N column reports the number of observations with non-missing values of the row variable. Elder variables come from the elder listing survey, and caregiver variables come from the caregiver baseline survey. The F-stat row reports the p-value from a joint F-test of all baseline characteristics in a regression of the arm indicator on all listed variables (omitting categories that would be collinear). *** p<0.01, ** p<0.05, * p<0.1.

Table A.4: Characteristics by Treatment Assignment in Caregiver Experiment

Variable	Checkup Order				Transport Choice				
	No prior help (1)	Prior help (2)	<i>p</i> (3)	N (4)	Public (5)	Private (6)	Delay (7)	<i>p</i> (8)	N (9)
<u>Elder Characteristics</u>									
Age over 65	0.65	0.63	<i>0.65</i>	444	0.62	0.61	0.67	<i>0.56</i>	278
Widowed	0.52	0.46	<i>0.23</i>	444	0.53	0.43	0.42	<i>0.30</i>	278
Elder living alone	0.17	0.17	<i>0.88</i>	444	0.22	0.15	0.17	<i>0.39</i>	278
Household size	3.12	3.14	<i>0.86</i>	444	2.87	3.11	3.14	<i>0.23</i>	278
Number of children	2.92	2.84	<i>0.57</i>	444	3.06	2.86	2.65	<i>0.23</i>	278
Checkup last 1y	0.85	0.88	<i>0.30</i>	444	0.90	0.88	0.87	<i>0.85</i>	278
Heart checkup ever	0.41	0.39	<i>0.68</i>	444	0.39	0.38	0.40	<i>0.96</i>	278
Heart checkup last 1y	0.18	0.15	<i>0.36</i>	444	0.18	0.17	0.09	<i>0.23</i>	278
Often worry burden	0.32	0.31	<i>0.97</i>	444	0.37	0.38	0.29	<i>0.36</i>	278
Often forgo treatment	0.25	0.22	<i>0.48</i>	444	0.21	0.30	0.27	<i>0.42</i>	278
Chose checkup over cash	0.62	0.66	<i>0.56</i>	198	0.79	0.64	0.68	<i>0.36</i>	115
<u>Caregiver Characteristics</u>									
Is Husband	0.25	0.25	<i>0.98</i>	444	0.26	0.22	0.25	<i>0.84</i>	278
Is Son	0.39	0.36	<i>0.60</i>	444	0.37	0.37	0.29	<i>0.44</i>	278
Is Daughter	0.12	0.13	<i>0.61</i>	444	0.11	0.12	0.17	<i>0.50</i>	278
Is Son-in-law	0.00	0.00	<i>0.33</i>	444	0.00	0.00	0.01	<i>0.44</i>	278
Is Daughter-in-law	0.10	0.07	<i>0.24</i>	444	0.07	0.09	0.11	<i>0.53</i>	278
Is Grandchild	0.06	0.07	<i>0.54</i>	444	0.04	0.07	0.07	<i>0.66</i>	278
Is Other	0.08	0.09	<i>0.80</i>	444	0.13	0.10	0.07	<i>0.36</i>	278
Average age (years)	49.33	48.71	<i>0.69</i>	444	49.92	49.15	49.18	<i>0.93</i>	278
Married	0.81	0.86	<i>0.12</i>	444	0.83	0.81	0.87	<i>0.55</i>	278
Household size	4.15	4.20	<i>0.79</i>	444	3.98	4.00	4.32	<i>0.44</i>	278
Working	0.60	0.54	<i>0.22</i>	444	0.62	0.52	0.56	<i>0.42</i>	278
Monthly Income Per Capita (1000 Rs)	3.16	3.02	<i>0.74</i>	177	2.61	3.15	3.74	<i>0.41</i>	114
Social Desirability Score (Pct)	-0.26	-0.75	<i>0.55</i>	444	-1.00	-0.95	0.25	<i>0.59</i>	278
High NCD knowledge	0.82	0.92	<i>0.00***</i>	444	0.87	0.86	0.87	<i>0.99</i>	278
F-stat p.			<i>0.89</i>					<i>0.71</i>	
Observations	224	220		444	89	94	95		278

Notes: This table shows randomization balance for baseline observable variables across the two cross-randomized experiments. Columns (1)–(4) report balance for the checkup order experiment (*No prior help* vs. *Prior help*). Columns (1) and (2) report the mean of the row variable in each arm, pooling across transport choice assignment. Column (3) reports the p-value on the arm coefficient β from a regression of the row variable on the order indicator with transport choice arm fixed effects (in italics). Column (4) reports the number of observations with non-missing values of the row variable. Columns (5)–(9) report balance for the transport choice experiment (*Public*, *Private*, *Delay*), restricted to caregivers who chose a checkup for their elder. Columns (5)–(7) report the mean of the row variable in each arm, pooling across checkup order assignment. Column (8) reports the p-value from a joint F-test of the two arm coefficients from a regression of the row variable on transport choice indicators with checkup order fixed effects (in italics). Column (9) reports the number of observations with non-missing values of the row variable in this subsample. The F-stat row reports the p-value from a joint F-test of all baseline characteristics in a regression of the arm indicator on all listed variables (omitting categories that would be collinear). When a variable is missing more than 10% of its observations, we impute the average value for the missing observations and include a dummy for whether the variable is missing a value in the joint F-test regression. Elder variables come from the elder listing survey, and caregiver variables come from the caregiver baseline survey. *** p<0.01, ** p<0.05, * p<0.1.

A.4 Effects on Take-up: Tables

Table A.5: Effect of Transport on Take-up by Worry about Burden

	Signed up			Attended		
	(1)	(2)	(3)	(4)	(5)	(6)
<u>Panel A: Transport effect for Worry</u>						
Transport	0.195*** (0.070) <i>0.005</i>	0.210*** (0.070) <i>0.003</i>	0.205* (0.106) <i>0.054</i>	0.163** (0.065) <i>0.013</i>	0.153** (0.065) <i>0.019</i>	0.151 (0.095) <i>0.114</i>
Control mean	0.252	0.252	0.252	0.137	0.137	0.137
<u>Panel B: Transport effect for Not Worry</u>						
Transport	0.081** (0.040) <i>0.045</i>	0.081* (0.042) <i>0.052</i>	0.092 (0.096) <i>0.336</i>	0.067* (0.034) <i>0.053</i>	0.065* (0.037) <i>0.082</i>	0.088 (0.079) <i>0.267</i>
Control mean	0.227	0.227	0.227	0.144	0.144	0.144
<u>Panel C: Transport $\times$ Worry</u>						
Transport	0.081** (0.040) <i>0.044</i>	0.081* (0.042) <i>0.052</i>	0.092 (0.096) <i>0.336</i>	0.067* (0.034) <i>0.052</i>	0.065* (0.037) <i>0.082</i>	0.088 (0.079) <i>0.267</i>
Worry	0.025 (0.054) <i>0.639</i>	0.026 (0.059) <i>0.655</i>	0.029 (0.059) <i>0.617</i>	-0.007 (0.047) <i>0.877</i>	-0.008 (0.050) <i>0.879</i>	0.005 (0.050) <i>0.918</i>
Transport $\times$ Worry	0.114 (0.079) <i>0.147</i>	0.129 (0.080) <i>0.109</i>	0.113 (0.080) <i>0.161</i>	0.095 (0.071) <i>0.182</i>	0.089 (0.072) <i>0.221</i>	0.063 (0.073) <i>0.386</i>
Observations	730	730	730	730	730	730
Strata FE	X	X	X	X	X	X
Controls (LASSO)		X	X		X	X
Controls (Worry predictors)			X			X

Notes: This table presents estimated treatment effects and treatment effect heterogeneity of the transport intervention on take-up (sign-up and attendance rates). Each panel reports coefficients from regressing take-up on an indicator for assignment to the *Transport* arm, an indicator for whether the elder is worried about being a burden, and the interaction between the two. Worry is an indicator for whether the elder reported ever forgoing healthcare or avoiding the hospital out of worry of being a burden to family members. Columns (1) - (3) show the effects on a sign-up indicator, which is equal to 1 when the household called to make an appointment at the check-up. Columns (4) - (6) show the effects on an attendance indicator, which is equal to 1 when the elder shows up for the health check-up after their household called to make an appointment. All regressions include strata fixed effects. Columns (2) and (5) add baseline controls selected via post-double-selection LASSO. Columns (3) and (6) add controls for characteristics that are imbalanced by worry about burden (indicators for above-median functional impairment, whether the caregiver is a daughter, large household, and whether elder is co-residing with the caregiver) and their interactions with the treatment indicator. Standard errors are reported in parentheses, clustered at the transport-randomization cluster level. P-values are reported in italics below. The control mean is the mean of the dependent variable among untreated households with not-worried elders. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.6: Effect of Transport on Take-up by Worry about Burden - Only free transport

	<i>Dependent variable:</i>					
	Sign-up			Attendance		
	(1)	(2)	(3)	(4)	(5)	(6)
Tell both	-0.005 (0.055) p = 0.922	-0.023 (0.055) p = 0.680	-0.155 (0.136) p = 0.257	-0.028 (0.048) p = 0.567	-0.049 (0.050) p = 0.325	-0.064 (0.135) p = 0.639
Worry about burden	0.046 (0.079) p = 0.565	0.004 (0.083) p = 0.959	0.016 (0.083) p = 0.850	-0.037 (0.070) p = 0.597	-0.102 (0.077) p = 0.185	-0.071 (0.080) p = 0.375
Tell both*Worry about burden	0.170 (0.109) p = 0.120	0.285** (0.112) p = 0.012	0.265** (0.111) p = 0.017	0.226** (0.100) p = 0.025	0.338*** (0.106) p = 0.002	0.295*** (0.107) p = 0.007
Control mean	0.312	0.312	0.312	0.225	0.225	0.225
Strata FEs (village)	Y	Y	Y	Y	Y	Y
Controls (LASSO)		Y	Y		Y	Y
Controls (Burden predictors)			Y			Y
Observations	373	373	373	373	373	373
R ²	0.187	0.357	0.378	0.214	0.363	0.394

Note:

*p<0.1; **p<0.05; ***p<0.01

Notes: This table presents estimated treatment effects and treatment effect heterogeneity of the Inform intervention on take-up (sign-up and attendance rates), estimated using only the subsample of households who were offered free transport (N=373). Each panel reports coefficients from regressing take-up on an indicator for assignment to the *Transport* arm, an indicator for whether the elder is worried about being a burden, and the interaction between the two. Worry is an indicator for whether the elder reported ever forgoing healthcare or avoiding the hospital out of worry of being a burden to family members. Columns (1) - (3) show the effects on a sign-up indicator, which is equal to 1 when the household called to make an appointment at the check-up. Columns (4) - (6) show the effects on an attendance indicator, which is equal to 1 when the elder shows up for the health check-up after their household called to make an appointment. All regressions include strata fixed effects. Columns (2) and (5) add baseline controls selected via post-double-selection LASSO. Columns (3) and (6) add controls for characteristics that are imbalanced by worry about burden (indicators for above-median functional impairment, whether the caregiver is a daughter, large household, and whether elder is co-residing with the caregiver) and their interactions with the treatment indicator. Standard errors are reported in parentheses, clustered at the transport-randomization cluster level. P-values are reported in italics below. The control mean is the mean of the dependent variable among untreated households with not-worried elders. *** p<0.01, ** p<0.05, * p<0.1.

Table A.7: Effect of Inform on Attendance - Robustness

	(1)	(2)	(3)	(4)	(5)
<u>Panel A: Sign-up</u>					
Inform	0.045 (0.037) <i>0.228</i>	0.033 (0.038) <i>0.388</i>	0.034 (0.038) <i>0.370</i>	-0.011 (0.095) <i>0.907</i>	-0.091 (0.105) <i>0.387</i>
Worry	0.032 (0.053) <i>0.544</i>	0.009 (0.055) <i>0.869</i>	0.006 (0.056) <i>0.914</i>	0.004 (0.056) <i>0.937</i>	0.012 (0.055) <i>0.832</i>
Inform $\times$ Worry	0.081 (0.078) <i>0.296</i>	0.169** (0.079) <i>0.033</i>	0.167** (0.079) <i>0.034</i>	0.166** (0.079) <i>0.036</i>	0.161** (0.079) <i>0.041</i>
Control mean	0.256	0.256	0.256	0.256	0.256
<u>Panel B: Attendance</u>					
Inform	0.018 (0.032) <i>0.577</i>	0.013 (0.033) <i>0.696</i>	0.014 (0.033) <i>0.674</i>	0.051 (0.080) <i>0.526</i>	-0.044 (0.092) <i>0.636</i>
Worry	-0.017 (0.047) <i>0.711</i>	-0.035 (0.048) <i>0.465</i>	-0.042 (0.049) <i>0.397</i>	-0.037 (0.049) <i>0.453</i>	-0.033 (0.048) <i>0.498</i>
Inform $\times$ Worry	0.100 (0.070) <i>0.150</i>	0.156** (0.071) <i>0.029</i>	0.154** (0.071) <i>0.030</i>	0.145** (0.071) <i>0.042</i>	0.150** (0.071) <i>0.036</i>
Control mean	0.179	0.179	0.179	0.179	0.179
Observations (Sign-up)	730	730	730	730	730
Observations (Attendance)	730	730	730	730	730
Strata FE (village)	X	X	X	X	X
Controls (LASSO)		X	X	X	X
Controls (Worry predictors)			X	X	X
Controls (Worry predictors $\times$ Inform)				X	
Controls (Caregiver altruism)					X

Notes: This table presents estimated treatment effects and treatment effect heterogeneity of the Inform intervention on attendance rates, with different sets of control variables included. All specifications include randomization strata fixed effects and an indicator for whether the *Transport* treatment. Column (1) specification only includes strata controls. Column (2) specification includes Lasso-selected controls. Column (3) specification includes Lasso-selected controls and variables that are imbalanced by elder worry about burden (indicators for whether the elder is above median mobility-impaired, whether the elder has daughter as caregiver, whether elder has above-median household size, whether elder co-resides with caregiver). Column (4) includes Lasso-selected controls, imbalanced variables, and imbalanced variables each interacted with the treatment indicator for *Inform*. Column (5) includes Lasso-selected controls, imbalanced variables, and controls for caregiver altruism (i.e. an indicator for whether the caregiver was asked to respond to the Zarit Burden Index (ZBI) subscale comprised of 5 questions about the degree to which the respondent feels burdened by their caregiving duties; the ZBI score when present; a variable interacting the two; an indicator for whether the caregiver was offered a choice between a check-up for the elder and a cash transfer; an indicator for whether the caregiver chose the check-up over the cash transfer; and a variable interacting the two).

A.5 Effects on take-up: Figures

Figure A.3: Effect of Transportation on Attendance By Worry about Burden

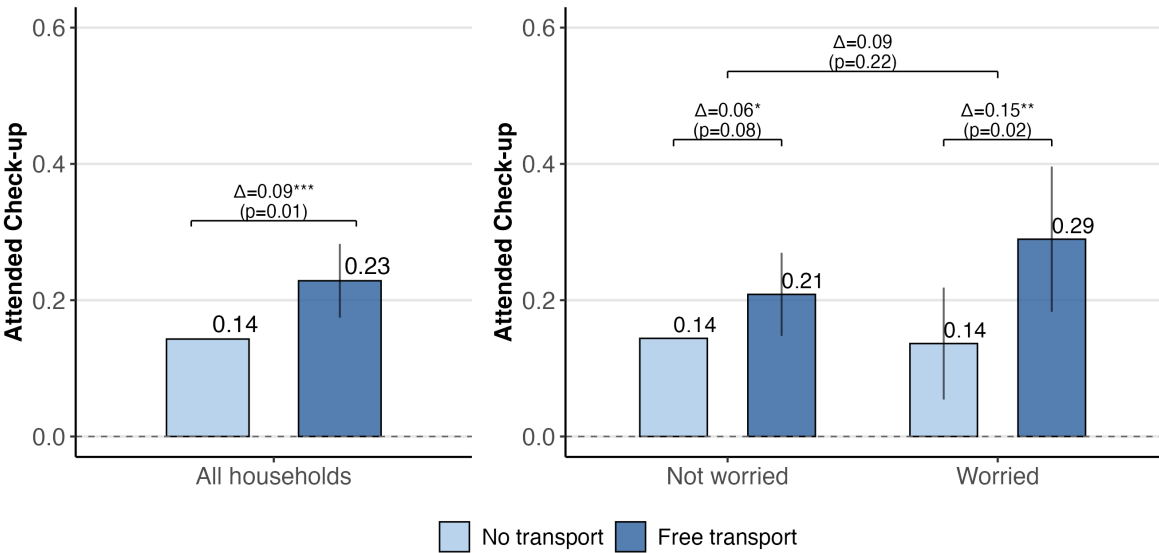
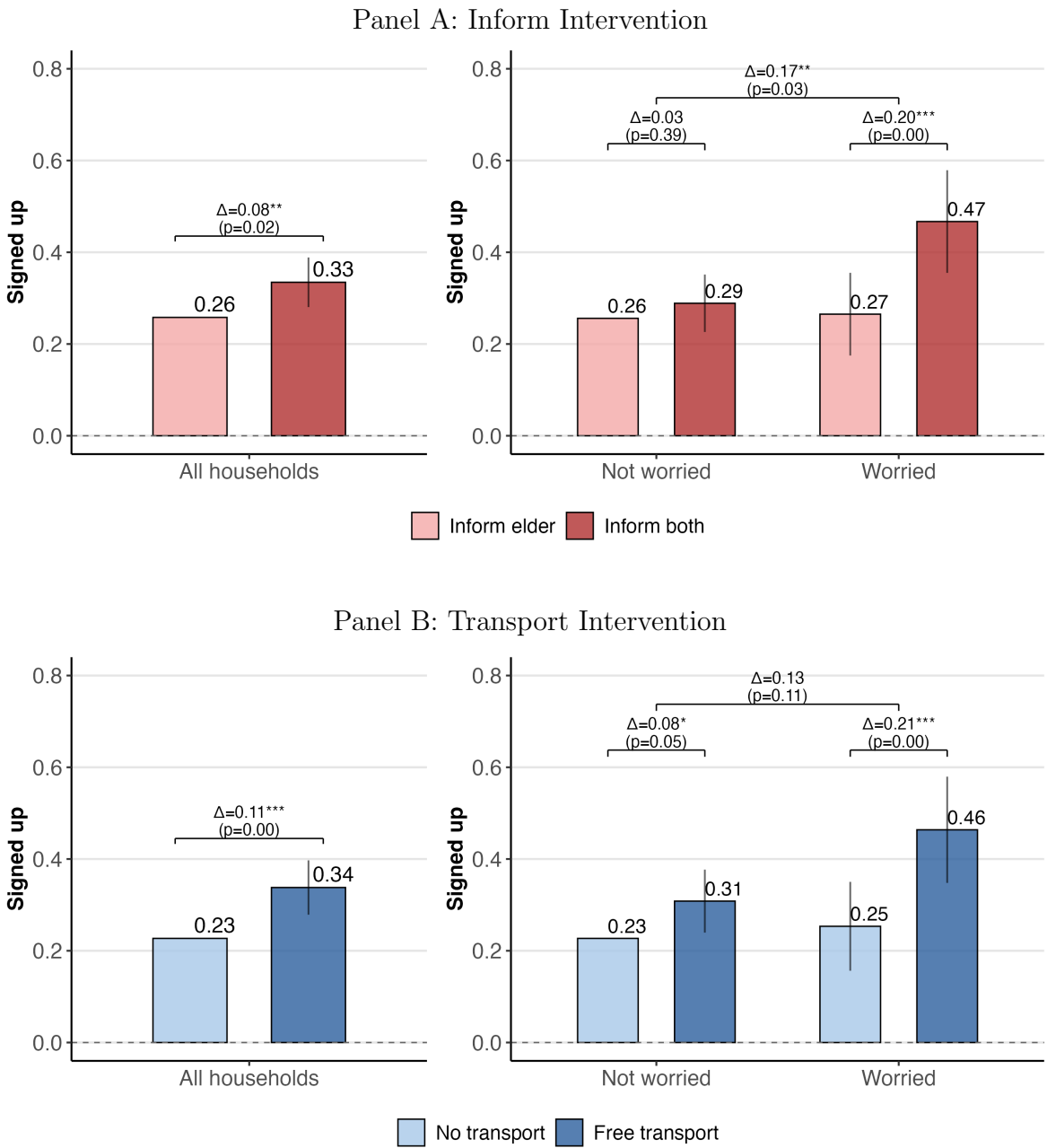


Figure A.4: Effect of Interventions on Sign-up Rates By Worry about Burden



A.6 Elder willingness to inform: Figures

Figure A.5: Elder Willingness to Inform Caregiver by Worry about Burden

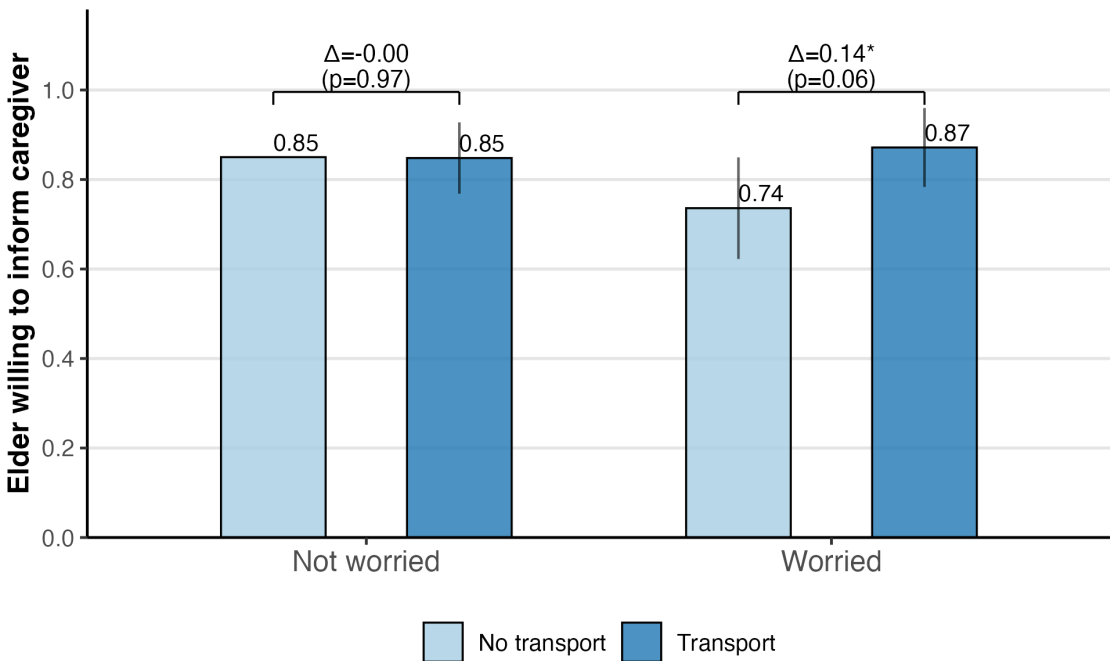
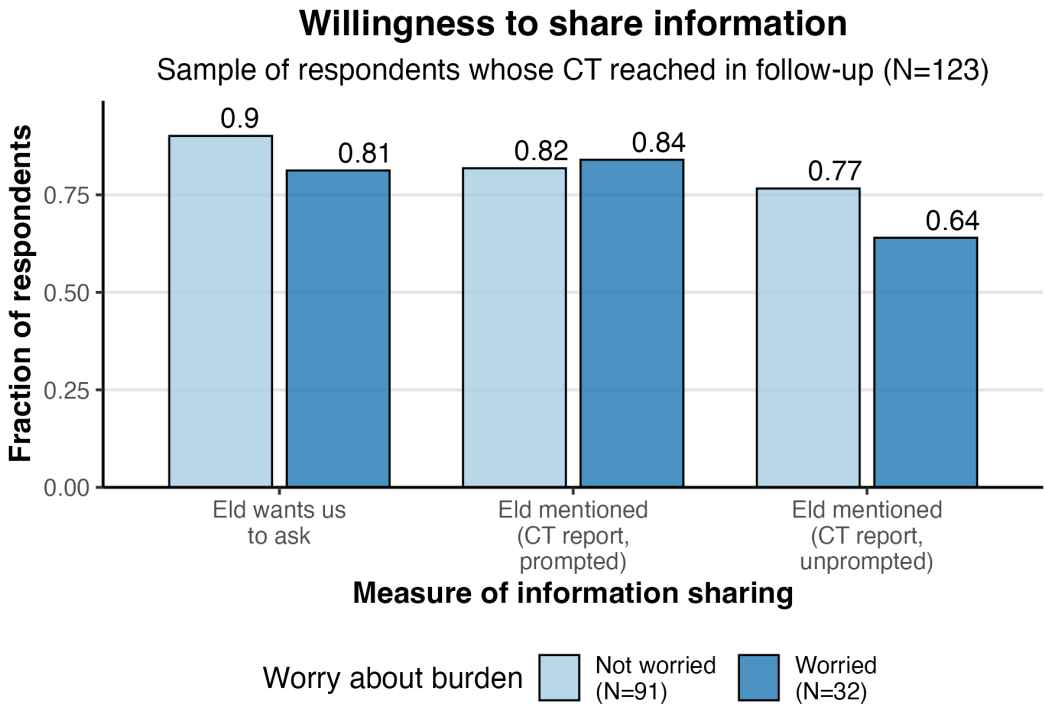


Figure A.6: Measures of willingness to share information (raw averages)



Only showing for control group (only elder told, no transport)
Fraction of respondents for whom CT was reached in follow-up: 0.68

A.7 Survey choice experiments

Table A.8: Survey choice experiment: Caregiver versus Elder Preferences

	Outcome: Chose Check-up over Cash					
	All caretakers	All elders	Worried elders	Not worried elders	CTs of worried elders	CTs of not worried elders
	(1)	(2)	(3)	(4)	(5)	(6)
High price	-0.111*** (0.033) p = 0.001	-0.067*** (0.022) p = 0.002	-0.127*** (0.044) p = 0.005	-0.041 (0.025) p = 0.105	0.039 (0.088) p = 0.660	-0.191*** (0.041) p = 0.00001
Transport Arr	0.064* (0.037) p = 0.081	0.035 (0.025) p = 0.166	0.085* (0.050) p = 0.091	0.017 (0.029) p = 0.546	0.034 (0.072) p = 0.637	0.060 (0.043) p = 0.166
Transp Reimb	0.025 (0.035) p = 0.472	0.091*** (0.025) p = 0.0004	0.101** (0.048) p = 0.038	0.088*** (0.030) p = 0.004	0.066 (0.064) p = 0.306	0.023 (0.042) p = 0.587
Control mean	0.83	0.64	0.68	0.63	0.8	0.83
Strata + Surveyor FEs	X	X	X	X	X	X
LASSO Controls	X	X	X	X	X	X
Observations	730	2,034	529	1,503	192	538
R ²	0.180	0.129	0.264	0.150	0.555	0.245

A.8 Reason for survey drop-off

Table A.9: Reason for Survey Drop-off: Caregiver Survey

Status	N	Fraction
Survey complete	800	0.55
Not attempted	303	0.21
Consent refused	110	0.08
Not home	80	0.06
No phone number	46	0.03
Did not pick up	44	0.03
Other	38	0.03
Dropped off	13	0.01
Impairments	7	0
Not age eligible	6	0
Could not find house	2	0

Table A.10: Reason for Survey Drop-off: Elder Survey

Status	N	Fraction
Survey complete	841	0.58
Not attempted	349	0.24
Door locked / Elder not home / Elder busy	111	0.08
Elder refused consent	70	0.05
Other (not elder) refused consent	32	0.02
Other	25	0.02
Elder is bedridden / sick	8	0.01
Dropped off	4	0
Elder migrated	4	0
Could not find house	3	0
Elder passed away	2	0

A.9 Elder rationale for no check-up / no ask

Table A.11: Elder reason for no interest in check-up (N=52)

Reason	Fraction
Feel fine / Too old / Not useful	0.35
Hospital too far	0.23
Did a heart check-up recently	0.21
Afraid of doctors or hospitals	0.12
I will be busy that day	0.12
No one to accompany or help me	0.12
Afraid of bad results	0.1
Don't want to trouble my family	0.06
Don't have money	0.02

Table A.12: Elder reason for not wanting to inform caregiver

Reason	All elders (N=90)	Not worried (N=68)	Worried (N=22)
I want to ask them myself	0.37	0.34	0.45
I don't want you to trouble them	0.28	0.28	0.27
I don't want to go to check-up	0.24	0.25	0.23
I don't want them to know about check-up	0.02	0.01	0.05
I think they will say no	0.01	0.01	0
Refused	0.04	0.06	0
Do not know	0	0	0
Other	0.07	0.09	0

Other reasons include: No phone (N=4), Caregiver in room (N=1), and unclear Tanglish (N=1).

Figure A.7: Frequency of hesitating to ask for help

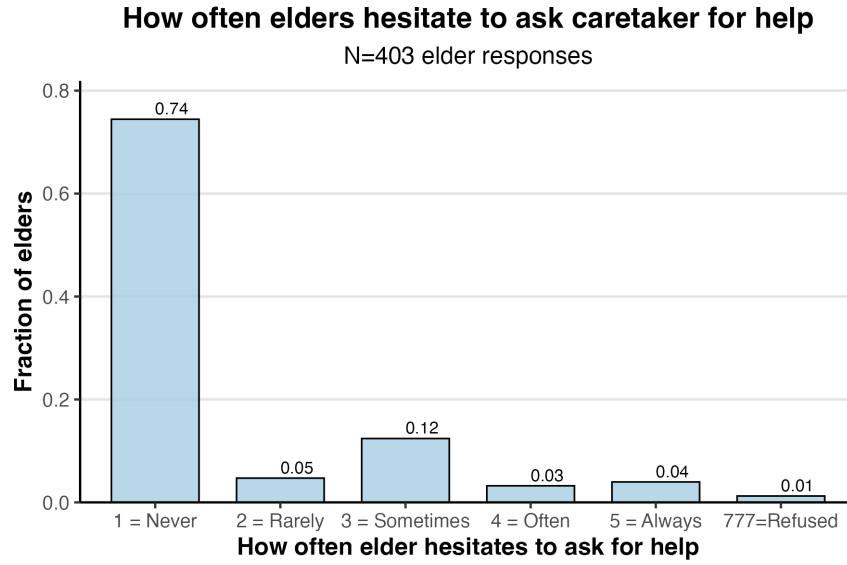


Table A.13: Elder reason for not asking for help (open-ended)

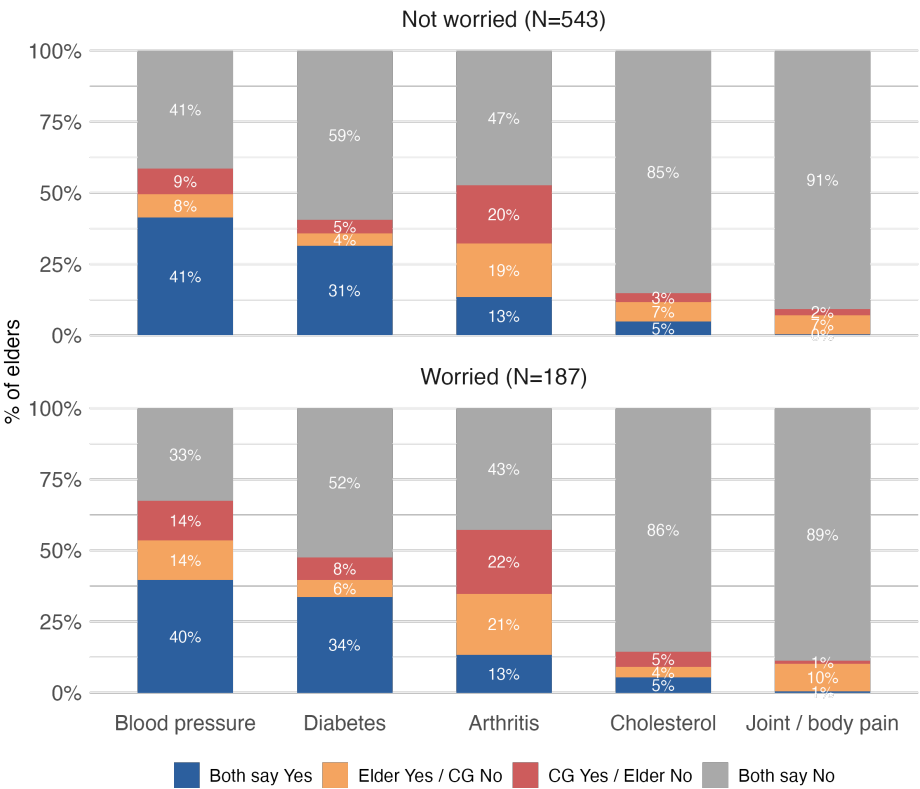
variable	N	Fraction
I do not want to burden or trouble them	68	0.69
They may be upset with me for asking	36	0.37
I think they will say no	22	0.22
I don't like to feel dependent	13	0.13
I may need help for something urgent later	8	0.08
Don't know	1	0.01
Other, specify	1	0.01

Table A.14: Elder frequency of avoiding ask due to crowd-out (close-ended)

Freq ration due to crowd-out	N	Fraction of elders
Never	10	0.1
Rarely	25	0.26
Sometimes	35	0.36
Often	16	0.16
Always	12	0.12

A.10 Agreement in Self-Reports

Figure A.8: Elder versus Caregiver Report of Elder's Health Conditions



A.11 Additional Worry vs Not-Worried Comparisons (Alternative explanations)

Table A.15: Characteristics by Worry about Burden - Awareness Beliefs and Behaviors

Variable	Overall (1)	Not worried (2)	Worried (3)	Difference (4)	p (5)
<i>Elder Characteristics</i>					
Fear of hospital	3.28	3.29	3.24	-0.06	0.84
Needs accompaniment	2.66	2.63	2.76	0.13	0.57
Mentions sickness (time)	0.49	0.50	0.48	-0.03	0.81
Mentions sickness never	0.08	0.08	0.07	-0.01	0.68
Mentions sickness wait	0.03	0.03	0.03	0.00	0.85
Mentions sickness $\leq 1w$	0.08	0.09	0.08	-0.01	0.8
Mentions sickness $\leq 2d$	0.12	0.13	0.11	-0.02	0.6
Mentions sickness $\leq 1d$	0.21	0.21	0.21	0.01	0.85
Mentions sickness immediately	0.79	0.79	0.79	-0.01	0.85
Uncomf asking for cash	0.66	0.64	0.71	0.07	0.1
Uncomf asking for transport	0.61	0.60	0.64	0.04	0.33
Asking for cash is pressure	0.96	0.96	0.95	-0.01	0.49
CT present at survey	0.14	0.14	0.13	-0.01	0.65
<i>Awareness of Elder Health</i>					
Blood pressure: Frac Disagree	0.20	0.17	0.28	0.11	0***
Of disagree, CG unaware	0.48	0.47	0.50	0.03	0.76
Diabetes: Frac Disagree	0.10	0.09	0.14	0.05	0.06*
Of disagree, CG unaware	0.46	0.47	0.42	-0.05	0.71
Cholesterol: Frac Disagree	0.10	0.10	0.09	-0.01	0.73
Of disagree, CG unaware	0.62	0.69	0.41	-0.27	0.04**
Cardiac: Frac Disagree	0.12	0.12	0.11	-0.01	0.79
Of disagree, CG unaware	0.71	0.71	0.71	0.01	0.95
Chronic pain: Frac Disagree	0.09	0.09	0.11	0.02	0.45
Of disagree, CG unaware	0.79	0.75	0.90	0.15	0.17
N	730	543	187		

Notes: This table compares additional baseline characteristics of households by whether the elder is worried about being a burden for the N=730 households in the Household Experiment. Elders are classified as “Worried” if they report having ever forgone care or avoided the hospital out of worry of being a burden to family members. One observation has missing value for worry about burden, because the respondent refused to respond to the question. Column (1) reports the full-sample mean of each row variable. Columns (2) and (3) report means for the not-worried and worried groups. Column (4) reports the difference estimated from a regression of the row variable on the worried indicator without any controls, and column (5) reports the p-value on the difference. *** p<0.01, ** p<0.05, * p<0.1.

A.12 Robustness: Effects on Take-up - Only Caregiver Calls

Table A.16: Effect of Interventions on Take-up (Restricting to Caregiver Sign-ups Only)

	Signed up		Attended	
	No controls (1)	Controls (2)	No controls (3)	Controls (4)
<u>Panel A: Inform both</u>				
Inform	0.071** (0.030) <i>0.020</i>	0.075** (0.031) <i>0.017</i>	0.059** (0.027) <i>0.027</i>	0.063** (0.027) <i>0.022</i>
Control mean	0.186	0.186	0.123	0.123
<u>Panel B: Free transport</u>				
Transport	0.054* (0.030) <i>0.072</i>	0.056* (0.033) <i>0.090</i>	0.042 (0.027) <i>0.113</i>	0.037 (0.029) <i>0.206</i>
Control mean	0.185	0.185	0.123	0.123
<u>Panel C: Interacted</u>				
Inform	0.085** (0.040) <i>0.033</i>	0.090** (0.042) <i>0.033</i>	0.061* (0.033) <i>0.065</i>	0.063* (0.034) <i>0.067</i>
Transport	0.068 (0.042) <i>0.108</i>	0.071 (0.044) <i>0.106</i>	0.044 (0.037) <i>0.233</i>	0.037 (0.039) <i>0.350</i>
Inform × Transport	-0.030 (0.064) <i>0.643</i>	-0.030 (0.066) <i>0.650</i>	-0.005 (0.054) <i>0.919</i>	0.000 (0.057) <i>0.998</i>
Control mean	0.148	0.148	0.099	0.099
<i>p: Inform = Transport</i>	<i>0.712</i>	<i>0.692</i>	<i>0.664</i>	<i>0.515</i>
<i>p: Inform+Tran = Inform</i>	<i>0.403</i>	<i>0.411</i>	<i>0.328</i>	<i>0.383</i>
<i>p: Inform+Tran = Transport</i>	<i>0.262</i>	<i>0.229</i>	<i>0.194</i>	<i>0.151</i>
Observations	731	731	731	731
Strata FE	X	X	X	X
Controls (LASSO)		X		X

Notes: This table presents estimated treatment effects of the two interventions (*Inform* and *Transport*) on take-up (sign-up and attendance rates), where the outcomes are only equal to one when the caregiver is the one who made the phone call. Panel A reports coefficients from regressing take-up on an indicator for assignment to the *Inform* treatment. Panel B on an indicator for the *Transport* treatment. Panel C reports coefficients from a regression of take-up on both treatment indicators and their interaction. Columns (1) and (2) show the effects on a sign-up indicator, which is equal to 1 when the household called to make an appointment at the check-up. Columns (3) and (4) show the effects on an attendance indicator, which is equal to 1 when the elder shows up for the health check-up after their household called to make an appointment. All regressions include strata fixed effects.

Columns (2) and (4) add baseline controls selected via post-double-selection LASSO and, in Panels A and B, an indicator for assignment to the other (cross-randomized) treatment. Standard errors are reported in parentheses, clustered at the transport-randomization cluster level in Panels B and C. P-values are reported in italics below.

The bottom rows report p-values from Wald tests based on the interacted specification in Panel C. *** p<0.01, ** p<0.05, * p<0.1.

Table A.17: Effect of Information Intervention on Take-up by Worry about Burden (Restricting to Caregiver Sign-ups Only)

	Signed up			Attended		
	(1)	(2)	(3)	(4)	(5)	(6)
<u>Panel A: Inform effect for Worry</u>						
Inform	0.108* (0.063) <i>0.086</i>	0.155** (0.064) <i>0.016</i>	0.155** (0.064) <i>0.016</i>	0.127** (0.055) <i>0.022</i>	0.163*** (0.060) <i>0.006</i>	0.164*** (0.060) <i>0.006</i>
Control mean	0.213	0.213	0.213	0.107	0.107	0.107
<u>Panel B: Inform effect for Not Worry</u>						
Inform	0.055 (0.035) <i>0.115</i>	0.046 (0.036) <i>0.204</i>	0.046 (0.036) <i>0.203</i>	0.033 (0.030) <i>0.269</i>	0.029 (0.031) <i>0.358</i>	0.029 (0.031) <i>0.354</i>
Control mean	0.183	0.183	0.183	0.128	0.128	0.128
<u>Panel C: Inform $\times$ Worry</u>						
Inform	0.055 (0.035) <i>0.115</i>	0.046 (0.036) <i>0.204</i>	0.046 (0.036) <i>0.203</i>	0.033 (0.030) <i>0.269</i>	0.029 (0.031) <i>0.358</i>	0.029 (0.031) <i>0.354</i>
Worry	0.030 (0.049) <i>0.538</i>	0.026 (0.051) <i>0.612</i>	0.021 (0.052) <i>0.686</i>	-0.021 (0.040) <i>0.604</i>	-0.032 (0.043) <i>0.459</i>	-0.040 (0.045) <i>0.367</i>
Inform $\times$ Worry	0.053 (0.072) <i>0.464</i>	0.109 (0.075) <i>0.145</i>	0.109 (0.075) <i>0.144</i>	0.094 (0.063) <i>0.139</i>	0.135** (0.069) <i>0.050</i>	0.135** (0.068) <i>0.049</i>
Observations	730	730	730	730	730	730
Strata FE	X	X	X	X	X	X
Controls (LASSO)		X	X		X	X
Controls (Worry predictors)			X			X

Notes: This table presents estimated treatment effects and treatment effect heterogeneity of the *Inform* intervention on take-up (sign-up and attendance rates), where the outcomes are only equal to one when the caregiver is the one who made the phone call. Panel A and B report coefficients from regressing take-up on an indicator for assignment to the *Inform* treatment arm, separately for Worried and Not-worried elders, respectively. Panel C reports coefficients from regressing take-up on an indicator for assignment to the *Inform* treatment arm, an indicator for whether the elder is worried about being a burden, and the interaction between the two. Worry is an indicator for whether the elder reported ever forgoing healthcare or avoiding the hospital out of worry of being a burden to family members. One observation has missing value for worry about burden, because the respondent refused to respond to the question. Columns (1) - (3) show the effects on a sign-up indicator, which is equal to 1 when the household called to make an appointment at the check-up. Columns (4) - (6) show the effects on an attendance indicator, which is equal to 1 when the elder shows up for the health check-up after their household called to make an appointment. All regressions include strata fixed effects. Columns (2) and (5) add baseline controls selected via post-double-selection LASSO. Columns (3) and (6) further add controls for characteristics that are imbalanced by worry about burden (indicators for above-median functional impairment, whether the caregiver is a daughter, large household, and whether elder is co-residing with the caregiver). Standard errors are reported in parentheses. P-values are reported in italics. *** p<0.01, ** p<0.05, * p<0.1.

A.13 Robustness: Effects on Take-up - Controlling for Imbalance

Table A.18: Effect of Interventions on Take-up (Controlling for Imbalanced Variables)

	Signed up		Attended	
	No controls (1)	Controls (2)	No controls (3)	Controls (4)
<u>Panel A: Inform both</u>				
Inform	0.060* (0.034) <i>0.080</i>	0.068* (0.035) <i>0.050</i>	0.038 (0.030) <i>0.205</i>	0.045 (0.031) <i>0.144</i>
Control mean	0.258	0.258	0.175	0.175
<u>Panel B: Free transport</u>				
Transport	0.087** (0.037) <i>0.019</i>	0.093** (0.038) <i>0.015</i>	0.060* (0.032) <i>0.058</i>	0.057* (0.034) <i>0.093</i>
Control mean	0.227	0.227	0.143	0.143
<u>Panel C: Interacted</u>				
Inform	0.071 (0.045) <i>0.118</i>	0.089* (0.047) <i>0.060</i>	0.029 (0.035) <i>0.413</i>	0.037 (0.036) <i>0.306</i>
Transport	0.097** (0.048) <i>0.044</i>	0.114** (0.049) <i>0.022</i>	0.051 (0.042) <i>0.216</i>	0.049 (0.044) <i>0.259</i>
Inform $\times$ Transport	-0.022 (0.070) <i>0.750</i>	-0.041 (0.071) <i>0.562</i>	0.018 (0.058) <i>0.763</i>	0.015 (0.060) <i>0.805</i>
Control mean	0.198	0.198	0.132	0.132
<i>p: Inform = Transport</i>	<i>0.599</i>	<i>0.628</i>	<i>0.603</i>	<i>0.784</i>
<i>p: Inform+Tran = Inform</i>	<i>0.157</i>	<i>0.188</i>	<i>0.122</i>	<i>0.169</i>
<i>p: Inform+Tran = Transport</i>	<i>0.350</i>	<i>0.356</i>	<i>0.311</i>	<i>0.272</i>
Observations	657	657	657	657
Strata FE	X	X	X	X
Controls (LASSO)		X		X

Notes: This table presents estimated treatment effects of the two interventions (*Inform* and *Transport*) on take-up (sign-up and attendance rates), where we include controls for the variables imbalanced across treatment groups (indicators for whether elder was interested in check-up and for whether elder had a check-up in the last one year). Panel A reports coefficients from regressing take-up on an indicator for assignment to the *Inform* treatment. Panel B on an indicator for the *Transport* treatment. Panel C reports coefficients from a regression of take-up on both treatment indicators and their interaction. Columns (1) and (2) show the effects on a sign-up indicator, which is equal to 1 when the household called to make an appointment at the check-up. Columns (3) and (4) show the effects on an attendance indicator, which is equal to 1 when the elder shows up for the health check-up after their household called to make an appointment. All regressions include strata fixed effects. Columns (2) and (4) add baseline controls selected via post-double-selection LASSO and, in Panels A and B, an indicator for assignment to the other (cross-randomized) treatment. Standard errors are reported in parentheses, clustered at the transport-randomization cluster level in Panels B and C. P-values are reported in parentheses below. The bottom rows report p-values from Wald tests based on the interacted specification in Panel C. *** p<0.01. ** p<0.05. * p<0.1.

Table A.19: Effect of Information Intervention on Take-up by Worry about Burden (Controlling for Imbalanced Variables)

	Signed up			Attended		
	(1)	(2)	(3)	(4)	(5)	(6)
<u>Panel A: Inform effect for Worry</u>						
Inform	0.149** (0.070) <i>0.033</i>	0.231*** (0.071) <i>0.001</i>	0.230*** (0.071) <i>0.001</i>	0.124* (0.063) <i>0.050</i>	0.172*** (0.067) <i>0.010</i>	0.171** (0.066) <i>0.010</i>
Control mean	0.266	0.266	0.266	0.151	0.151	0.151
<u>Panel B: Inform effect for Not Worry</u>						
Inform	0.026 (0.039) <i>0.507</i>	0.013 (0.040) <i>0.751</i>	0.014 (0.040) <i>0.724</i>	0.007 (0.034) <i>0.847</i>	0.002 (0.035) <i>0.956</i>	0.003 (0.035) <i>0.931</i>
Control mean	0.256	0.256	0.256	0.179	0.179	0.179
<u>Panel C: Inform $\times$ Worry</u>						
Inform	0.026 (0.039) <i>0.507</i>	0.013 (0.040) <i>0.751</i>	0.014 (0.040) <i>0.724</i>	0.007 (0.034) <i>0.847</i>	0.002 (0.035) <i>0.956</i>	0.003 (0.035) <i>0.931</i>
Worry	0.010 (0.054) <i>0.847</i>	-0.027 (0.055) <i>0.628</i>	-0.028 (0.056) <i>0.620</i>	-0.028 (0.048) <i>0.555</i>	-0.052 (0.050) <i>0.293</i>	-0.058 (0.051) <i>0.259</i>
Inform $\times$ Worry	0.123 (0.080) <i>0.126</i>	0.219*** (0.083) <i>0.008</i>	0.215*** (0.083) <i>0.009</i>	0.118 (0.072) <i>0.104</i>	0.170** (0.076) <i>0.025</i>	0.168** (0.076) <i>0.027</i>
Observations	656	656	656	656	656	656
Strata FE	X	X	X	X	X	X
Controls (LASSO)		X	X		X	X
Controls (Worry predictors)			X			X

Notes: This table presents estimated treatment effects and treatment effect heterogeneity of the *Inform* intervention on take-up (sign-up and attendance rates), where we include controls for the variables imbalanced across treatment groups (indicators for whether elder was interested in check-up and for whether elder had a check-up in the last one year). Panel A and B report coefficients from regressing take-up on an indicator for assignment to the *Inform* treatment arm, separately for Worried and Not-worried elders, respectively. Panel C reports coefficients from regressing take-up on an indicator for assignment to the *Inform* treatment arm, an indicator for whether the elder is worried about being a burden, and the interaction between the two. Worry is an indicator for whether the elder reported ever forgoing healthcare or avoiding the hospital out of worry of being a burden to family members. One observation has missing value for worry about burden, because the respondent refused to respond to the question. Columns (1) - (3) show the effects on a sign-up indicator, which is equal to 1 when the household called to make an appointment at the check-up. Columns (4) - (6) show the effects on an attendance indicator, which is equal to 1 when the elder shows up for the health check-up after their household called to make an appointment. All regressions include strata fixed effects. Columns (2) and (5) add baseline controls selected via post-double-selection LASSO. Columns (3) and (6) further add controls for characteristics that are imbalanced by worry about burden (indicators for above-median functional impairment, whether the caregiver is a daughter, large household, and whether elder is co-residing with the caregiver). Standard errors are reported in parentheses. P-values are reported in italics. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$

B Appendix: Study details

B.1 Caregiver experiment timeline

After the listing exercise, households in the final sample of respondents in the caregiver experiment were surveyed between March and April 2026. The enumerators attempted five in-person visits with the caregiver, and these were not done via phone.

B.2 Elder and caregiver survey choice experiments

In order to learn more precisely about each person’s separately expressed demand for the check-up (as opposed to the joint, single decision of take-up), we also conduct incentivized survey choice experiments for all elders (during the listing exercise) and for the caregivers in the *Inform* arm. The survey choice experiments also allow us to estimate how a richer set of factors affect demand for a check-up for the elder. In particular, we measure how demand for check-up varies with the degree of caregiver involvement required.

We measure each respondent’s demand for a check-up for the elder through a series of incentivized choice questions. Respondents make a series of choices between a free health check-up and a cash alternative. We measure respondent demand for a check-up, as well as how it changes in response to whether transport support is provided or not. We ask about two types of transportation support: arranged transportation and a transportation reimbursement. The type of support asked about is varied across participants.

Randomization of which choice is implemented is done at the pair-level (elder-caregiver pairs). Other logistical randomizations regarding the presentation of the choice questions in the survey are all independently randomized at the individual respondent-level (e.g. ordering of choices, which choices are shown, cash transfer amount).

B.2.1 Elder survey experiment

In the elder experiment, we measure the elder’s demand for the check-up through a series of incentivized choice questions. Elders make a series of four choices between a free health check-up and an alternative gift. To ensure that choices are incentive compatible, elders are informed that we will choose one of their choices to implement as long as their caregiver participates in the household survey.

We measure how elder demand for a check-up changes in response to whether transport support is provided.

We ask elders two questions in which they are asked to choose between a **check-up** and **cash**. The baseline check-up is a free heart check-up at a nearby private hospital. In one question, the check-up comes with transportation assistance of some form; in the other, the check-up is offered alone. The order of these two questions is randomized.

We ask about two types of transportation support: free arranged transportation and a transportation subsidy of 500 Rs. The type of support asked about is varied across participants.

1. **Arranged transport and accompaniment at hospital.** This option reduces the involvement needed from the caregiver in terms of both financial costs of transport, as well as time costs of providing and accompanying the elder.
2. **Transportation reimbursement.** This option reduces the financial costs of the caregiver’s involvement. The reimbursement amount was up to 500 INR, which is sufficient to cover an auto-rickshaw ride to and from the hospital in most of our study areas.

To benchmark the demand responses to transport in monetary units, we also estimate the demand response to the cash amount. For elders, we randomly vary the outside option between 500 and 1000 INR across participants. For caregivers, the cash amounts are varied between 200 and 500 INR.

Elders then make two last choices for the purposes of study implementation, which we exclude from our empirical analysis. First, to have a choice that was feasible to implement at scale, we ask elders to choose between a check-up and a small gift of herbal ointment ³⁰. The last “check-up vs herbal oil” choice is the choice that is implemented for 95% of participants. Further, in order to invite all elders in the household experiment to the check-up, we gave check-ups and ointments to all participants. If this was the choice to be implemented and they choose the ointment, they are given the ointment and surprise upgraded to also get the check-up. If they choose the check-up, they are given the check-up and surprise-upgraded to the ointment. Even under this surprise upgrade structure, it is in the elder’s best interest to select what they prefer most in each choice. Second, we have one choice that is between a non-health-related benchmark good and the cash transfer. The benchmark good is a “senior clubs” program in which elders participate in weekly meetings with other elderly women in the area to play games, do light physical exercises, and do laughter therapy. The benchmark good is a non-fungible benefit to the elder.

³⁰Herbal ointments are used for cooling and joint pain relief. They are broadly available at medical shops in our context and worth 100-150 INR at the time of the study

B.2.2 Caregiver survey experiment

We conduct a parallel survey experiment among caregivers in the *Inform* arm to measure caregiver demand for a check-up for their elder. For caregivers in households that are randomized into the *Inform* arm, we will also elicit their incentivized choices over check-ups and alternative goods, before being invited to the check-up at the end of the survey. Caregivers are asked the same set of questions as those asked to the elder, except the cash amounts are varied between 200 INR and 500 INR. At the end of the survey, we choose one of their choices to implement.

Additionally, the implementable choice is between the check-up and a one-time mobile phone recharge of 200 Rs, selected to be a gift to the caregiver but comparable to the market value of the herbal oil. Similar to the elder’s survey choice implementation, the randomly selected 95% of households whose elders received the implementable choice had caregivers also receive the implementable choice. We surprise upgrade caregivers who chose the mobile recharge so that they are offered a check-up in addition.

B.3 Survey modules

B.3.1 Crowne-Marlowe Social Desirability Bias

We measure social desirability bias for all caregivers in our study using a seven-item subset of the Crowne-Marlowe Social Desirability Bias survey questionnaire (Crowne and Marlowe, 1960). The seven yes-or-no questions are the same as the ones used in Webb ([forthcoming](#)):

1. I sometimes feel annoyed at people when I don’t get what I want.
2. No matter who I’m talking to, I’m always a good listener. (reverse coded)
3. I sometimes try to take revenge instead of forgiving and forgetting.
4. I am always polite, even to people who are not nice. (reverse coded)
5. There have been times when I was jealous of other people’s luck.
6. I am sometimes annoyed when people ask me for favors.
7. I have deliberately said something that hurt someone’s feelings.

To calculate each individual’s social desirability score, the number of socially desirable answers given is summed across the items (items 2 and 4 are reverse-coded in the scoring).

B.4 Study timeline

The household experiment was conducted over 12 weekly waves to accommodate capacity limits at the hospital hosting the check-ups. Eligible households recruited to the study via the listing survey were assigned to waves based on their village size and location³¹ For each wave, we conduct the study over four stages:

1. **Caregiver and elder visits** (Days 1-3): At the start of the week, we conduct separate home visits with elders and caregivers to conduct surveys and deliver the invitations.

- (a) **Caregiver invitation.** Enumerators conduct a 10- to 15-minute baseline survey³² to collect information on demographics, occupation, relationship with the elder (tasks to support elder, knowledge of elder’s health symptoms and status, communication patterns). We also measure social desirability bias using a seven-item subset of the Crowne and Marlowe, 1960 questionnaire (see Section B.3.1 for details) to be used in analysis to check whether caregiver-reported measures are influenced by wanting to present themselves to others in a particular way.

Caregivers in the *No Inform* arm who do not receive an invitation are asked survey questions about frequency of interaction with the elder, perceived caregiving burden (as measured with a 5-item subset of the Zarit Burden Index), their own non-communicable disease (NCD) status and knowledge, care-seeking behavior, and life satisfaction.

For caregivers in the *Inform* arm, the enumerator elicits incentivized measures of demand for check-ups for the elder³³ and delivers the check-up invitation to the randomly selected 95% subsample who were selected to receive their choice. Caregivers assigned to the *Transport* arm are informed about the option to utilize free transport to and from the check-up.

- (b) **Elder invitation.** Enumerators conduct a 15- to 20-minute follow-up survey with elders to deliver the invitation and collect information on income (receipt of pension), NCD health status and utilization. All elders are informed of the check-up and given a paper voucher with the check-up details. Elders assigned to the *Transport* arm are informed about the option to utilize free transport. All elders are offered to have a reading conducted by the enumerator using a handheld single-lead ECG device, as a way to explain the check-up (to demonstrate

³¹Each week, the remaining set of households were allocated to surveyor and supervisor teams to balance travel requirements (i.e. distance to hospital) across teams and to account for survey productivity (e.g. some weeks included local holidays, so fewer days could be allocated to delivering invitations to ensure households had time to sign up for the check-up).

³²Caregivers who cannot be reached in person after **three** attempts are surveyed by phone.

³³See details in Section B.2.

the ECG) and collect an objective measure of cardiac risk for all participants in our study. Enumerators are instructed to prioritize privacy during the survey. For visits where the survey is able to be conducted out of earshot of the caregiver and other household members, the elder responds to a survey module on relationship with the caregiver (whether the elder supports the caregiver with childcare or financial assistance, communication patterns, comfort asking for transport or financial support).³⁴

2. **Sign-up deadline** (Day 5): Households who are interested in the check-up were instructed to have the caregiver call the research team’s phone hotline by noon on the day of the deadline to reserve a slot at the check-up.
3. **Caregiver follow-up survey** (Days 5-6): Once the sign-up deadline has passed, all caregivers who have not already called to reserve a spot at the check-up receive a follow-up phone call, regardless of the treatment arm, with the purpose of collecting both a prompted and unprompted measure of whether the elder has informed the caregiver about the check-up in the *No Inform* arm. For caregivers who know about the check-up (either reported knowing from elder or were directly been informed in the *Inform* arm), they are offered a last chance to sign up for the check-up.
4. **Reminders** (Day 6): The evening before the check-up, caregivers that have confirmed attendance for their elder receive a reminder call about the camp, with information about their assigned time slot, free food at the hospital, and the transport pickup arrangements, if applicable.
5. **Check-up.** (Day 7): The check-up is held at the partner hospital. During the check-up, research team enumerators are present to collect vouchers, guide participants through the hospital, and conduct a brief exit survey on who accompanied the elder to the visit, how they traveled to the hospital, and what she learned from the consultation (diagnoses, treatments prescribed).

B.5 Check-up invitation materials

B.5.1 Video script

The video begins with a title slide: “J-PAL Heart Health Check-up Camp” This is followed by a slide that reads: “Tests and Services Provided at the Camp” The video then shows an elderly woman arriving at Thirumalai Mission Hospital and participating in the different components of the check-up camp. Visuals include:

³⁴Surveys were completely private for 86% of visits, for whom we were able to collect these measures of relationship closeness.

- Blood pressure (BP) screening
- ECG test
- Doctor consultation

The video then shows a slide highlighting: “Free Food Provided”

Doctor’s Message (Dr. Peter Samuel):

Rather than waiting for heart problems to become severe and then treating them, it is better to identify them early and seek treatment to prevent the condition from worsening. Today, our lifestyles and eating habits have changed significantly. Compared to the past, many of us do not lead healthy lifestyles. As a result, health conditions such as heart disease are becoming increasingly common.

Many people are aware that chest pain can be a warning sign of heart disease. However, there are several other symptoms. These include shortness of breath while doing small routine activities, unexplained sweating, racing heartbeat, and unexplained dizziness.

If you are above 60 years of age and experience any of these symptoms, it is encouraged to visit a hospital and undergo a medical evaluation.

The video concludes with the J-PAL and Thirumalai Mission Hospital logos displayed on screen.

C Appendix: Alternative Models

In this section, we consider four alternative theoretical frameworks. The first three are broadly about communication frictions, in which there is a wedge between elders’ willingness to inform and willingness to go (if the caregiver is informed) that scales with elder altruism. The fourth model is about sensitivity to rejection, in which informing the caregiver makes the elder vulnerable to being hurt by a rejection.

C.1 Model of rejection sensitivity

We also consider a model of rejection sensitivity, in which elders do not inform caregivers about the check-up because they are worried that the caregiver’s refusal will reveal a lack of care for the elder. Bénabou et al., 2025 is a theory paper about how people who need help may not ask because they face uncertainty about other person’s generosity and don’t want to risk finding out that the potential helper does not

care about them. In this subsection, we consider whether the alternative model of rejection sensitivity, captured empirically by worry of burden, might explain our experimental results.

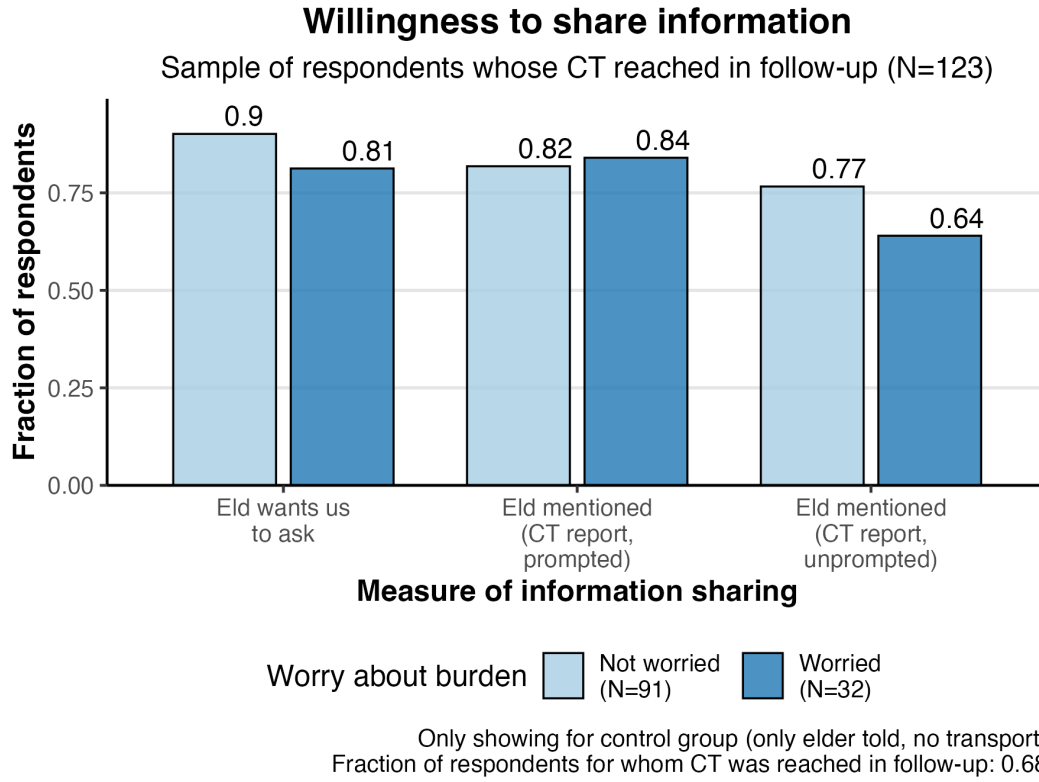
Their model has two players, a Sender, who is positioned to potentially give help, and a Receiver, who is in need of help. The Sender does not know the Receiver’s need, and the Receiver does not know the Sender’s generosity towards them. The game has three stages: the Sender can offer help. If no offer, the Receiver can ask for help. If asked, Sender can help or not. The Receiver’s payoffs are a function of their need (e.g. benefit of help), whether they got help, and their beliefs about the Sender generosity. The Sender’s payoffs are a function of generosity to the Receiver (e.g. altruism), cost, and self-image from their own generosity.

In the Bénabou et al., 2025 framework, the main intervention of informing the caregiver on behalf of elder could be modeled as changing two things: (1) changing the mapping between observed Sender actions (e.g. no help) and shifting Receiver beliefs about generosity; and (2) Giving caregivers the chance to offer first (full three-stage game, not just last two stages). Taking “worry about burden” to be a measure of rejection sensitivity of the Receiver (elder), several of our main results are consistent with this model:

- The treatment effect of informing the caregiver should be larger for more worried (sensitive) elders, because it’s closing a larger “no ask” gap
- Elder willingness to ask is positively correlated with caregiver likelihood of selecting check-up over cash (measured for the control group only) for worried-about-burden elders, consistent with rejection sensitivity story. The correlation is zero/weakly positive for not-worried elders.

However, our data is not consistent with the rejection sensitivity model in other ways. First, we note that one ideal test of the rejection sensitivity model is to show that (a) “elder willing to ask themselves” is lower than (b) “elder willing to let us ask” – and this should be especially true for worried (sensitive) elders. The closest we have to (a) is the caregiver-reported elder asking rates, and those aren’t that much lower than (b). Importantly: the gap between (a) and (b) is not larger for worried/sensitive elders. See Figure C.1.

Figure C.1: Measures of willingness to share information (raw averages)



Second, we can also check whether “elder willingness to let us ask” varies with plausible deniability. Under the rejection sensitivity model, we would expect that elders who know their caregiver was directly informed have less plausible deniability (i.e. if caregiver does not offer to help, this could just be due to experimenters not properly conveying the invitation) and thus are less willing to ask. We see that willingness to ask is NOT different (1) by whether caregiver was informed first or (2) by whether they live in the same household (assuming both makes elder more likely to know the invitation was delivered to the caregiver) or (3) by whether caregiver was informed (treatment) or not (control) Table C.1.

Table C.1: Heterogeneity of *Inform* by whether caregiver visited first

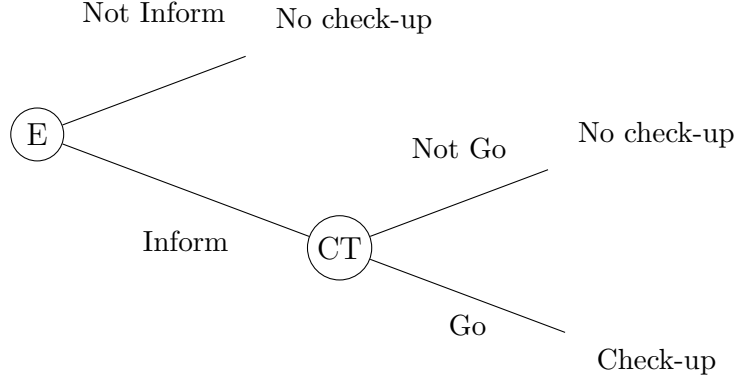
	Take-up	
	Signed up	Attended
	(1)	(2)
Treated: Tell both	0.065 (0.056) p = 0.246	0.069 (0.048) p = 0.151
Invited CT First	-0.073 (0.052) p = 0.157	0.003 (0.046) p = 0.943
Tell both*Invited CT First	0.007 (0.069) p = 0.925	-0.028 (0.059) p = 0.633
Control mean	0.32	0.19
Strata FEs (village)	Y	Y
Controls (LASSO)	Y	Y
Observations	731	731
R ²	0.245	0.247

C.2 Models of communication frictions

C.2.1 Timing

There are two stages of the game. First, the elder decides to share or hide information about check-up (inform vs not inform). If caregiver is informed, caregiver decides whether to go.

Figure C.2: Timing of game with communication frictions



C.2.2 Payoffs

If the household decides to attend the check-up, the elder e gets direct benefit B from going to a check-up, while the caregiver c incurs direct cost C of bringing the elder to check-up.

Person i has altruism for person j with weight α_i . We assume altruism only accounts for direct costs of person j , so the elder only internalizes the costs they impose on the caregiver and not the (altruistic) benefits the caregiver gets from bringing the elder.

The payoffs of going to the check-up for the elder and caregiver are:

$$U_e = B - \alpha_e C$$

$$U_c = \alpha_c B - C$$

Free transport. Suppose that if there is free transport provided, this reduces the travel costs by τ so that the cost of the check-up to the caregiver is $C - \tau$.

C.2.3 Communication Wedge

We introduce a wedge Δ_e that captures the gap between elders being willing to ask their caregiver for help and elders being willing to actually attend the check-up once the caregiver is informed.

We consider three specific ways to model the communication friction: (1) misperceptions about caregiver costs $\tilde{C} \geq C$, (2) costs of informing that scale with altruism $\alpha_e k$, and (3) costs of rejection that scale with

altruism with uncertainty $\alpha_e \tilde{\phi}$, with $\tilde{\phi} \sim f$.

Across all three models, the key idea is that the communication friction scales with the elder's altruism for the caregiver α_e , so the most altruistic elders are least likely to be willing to inform the caregiver. However, the group of compliers who attend the check-up are actually only elders who have intermediate ranges of altruism. This is because the most altruistic elders are so put off by the caregivers' direct costs of taking them C that they don't want to go to the check-up, even if offered.

Comparison to pressure model One key difference between these models and the “dislike of pressure-based helping” framework is the elder's utility function.

$$U_{e, \text{ ask}} = B - \alpha_e C \quad , \text{ models with frictions} \quad (32)$$

$$U_{e, \text{ ask}} = B + \alpha_e (\alpha_c B - C) \quad , \text{ model with pressure} \quad (33)$$

In the former, the elder's utility only internalizes caregiver costs, so elders and caregivers sometimes disagree about whether the check-up is worth it or not. In the latter, the elder's utility internalizes full caregiver's net welfare of helping so that elder's extra altruism term always makes them agree more with the caregiver.

The upside of these models is that they are simple and clear. The downsides of these models is that they are perhaps too simple. The predictions (almost) immediately fall out of the way the utility function is set up (e.g. elder's utility of asking is essentially $B - \alpha_e C - \alpha_e \Delta_e$, with all costs scaling with α_e).

The opposite issue comes up for the main model of pressure-based helping, where the predictions are almost always ambiguous due to different opposing forces. This works nicely in some places to have interesting, empirically testable ambiguity for predictions - for example, for the first prediction that elder altruism only decreases asking when elders think caregiver involvement comes from pressure rather than genuine willingness. But then it feels a little more disingenuous to then later simplify predictions to say that transport generally increases willingness to inform (even though there's *technically* ambiguity in the comparative static).

In all of the models, altruism makes elders sensitive to some costs, but we generally need another friction (e.g. misperception of pressure) in order to generate the main predictions and the core inefficiency (gap between asking and going rates). For example, altruism exacerbates misperceptions or costly asking frictions, but the misperception and/or uncertainty is the core reason for a gap between willingness to go and willingness to ask.

C.2.4 Take-up by Treatment Arm

Without any communication frictions, the check-up happens as long as both agree to go. This is the scenario that corresponds to the *Inform* arm in which both the caregiver and elder are directly informed about the check-up. However, with communication frictions, the friction acts as an extra wedge that reduces willingness to inform by the elder. This captures the situation in the *Control* arm, in which only the elder is informed.

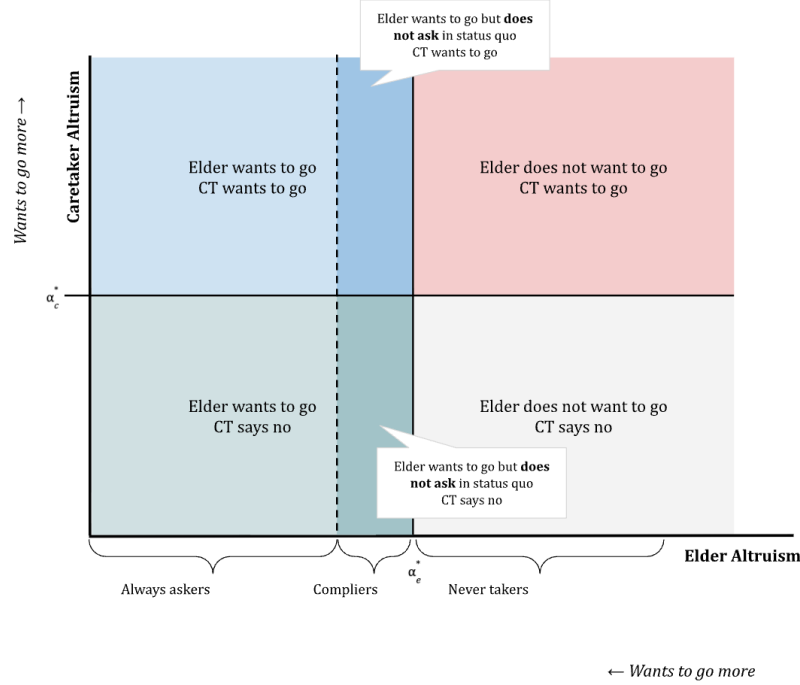


Figure C.3: **Diagram of take-up by elder and caregiver altruism.** The darker shaded blue region corresponds to the range of elder and caregiver altruism values (α_c, α_e) for whom the intervention of directly informing the caregiver will increase take-up. This region corresponds to households where the elder's altruism is in the intermediate range (so that she is not so altruistic as to not be willing to go but altruistic enough to be worried about imposing asking costs on the caregiver) and the caregiver's altruism is high to agree to go when informed.

Compliers. The intervention increases take-up for elders who would go but do not inform, who have caregivers that would have wanted to go. In other words, compliers are the households where:

1. Elders do not ask but would go: $\frac{B}{C+\Delta_e} < \alpha_e < \frac{B}{C}$
2. Caregivers help³⁵ when they know: $\alpha_c > \frac{C}{B}$

³⁵Note that our definition of altruism means caregiver's decision to help is independent of elder altruism α_e .

Welfare of experimenters informing. We assume actual costs associated with the communication are incurred even when “we” inform the caregiver, at rate $\lambda \in [0, 1]$. For example, if the cost of rejection in the control group for elders is $-\alpha_e\phi$ and for caregivers is $-\phi$, then we assume that our intervention induces costs $-\lambda\alpha_e\phi$ and $-\lambda\phi$, respectively.

Welfare effects of treatment are ex-ante ambiguous. There is an upside to increasing take-up among the group of compliers, but there is a downside from potentially introducing informing or rejection costs to elders and caregivers who do not agree to go to the check-up. We outline the take-up and welfare differences for different groups in Table C.2 below.

Table C.2: Effect of intervention on take-up and welfare - by household type

Elder decision	CT help	Inform elder (control)	Inform both (treated)	Δ take-up	Δ welfare
Elder would ask anyway	Yes	Go; get U_e	Go; get U_e	0	0
	No	Don't go; get $-\alpha_e\phi$	Don't go; get $-\alpha_e\phi$	0	0
Elder would go but does not ask	Yes	—	Go; get U_e	+1	$U_e > 0$
	No	Don't go; get 0	Don't go; get $-\alpha_e\phi$	0	$-\alpha_e\phi < 0$
Elder would not go	Yes	—	Don't go; get 0	0	0
	No	Don't go; get 0	Don't go; get $-\alpha_e\phi$	0	$-\alpha_e\phi < 0$

C.2.5 Predictions

For all three of the alternative models, we take worry about burden to be an empirical measure of elder altruism for the caregiver. There are three main empirical predictions that are consistent with all three models.

1. Worried elders are less willing to inform caregiver
2. Worried elders inform more when caregiver costs are lowered
3. Removing communication friction increases take-up for households where
 - Elders face high perceived asking costs (which increase with α_e)
 - And not-asked caregivers would be willing to go (which increases with α_c)

Importantly, the framework must have that elders might not inform but would still potentially go if caregiver is informed. Further, we would expect larger treatment effects when there is a positive correlation

between elder altruism (because this decreases asking so the wedge Δ_e is larger) and caregiver altruism (because this makes it more likely that help is granted when the wedge is closed)

C.2.6 Model 1: Misperceived costs to caregiver

Setup. Suppose the elder does not know the caregiver’s costs of bringing them to the check-up C and forms belief $\tilde{C}$ with expectation $E[\tilde{C}]$. If the elder underestimates or correctly perceives the costs $\tilde{C} \leq C$, then there is no effect of the intervention because they ask regardless. If the elder overestimates the costs $\tilde{C} > C$, then we get the main prediction that the elder may not ask but would go if the caregiver knew and offered (upon which she updates the perceived costs to $C^* \equiv E[\tilde{C}|\alpha_c B - \tilde{C}] \geq E[\tilde{C}]$).

Take-up predictions. From this model, we get all of our main predictions. First, informing the caregiver increases take-up more for more altruistic elders: the treatment increases take-up by $\alpha_e(E[\tilde{C}] - C^*)$. The magnitude of effect is elder altruism interacted with misperceptions: the “do not ask but would go” gap is the magnitude of the misperception, scaled by elder altruism. Second, more altruistic elders’ asking rates are more responsive to (perceived) costs $E[\tilde{C}]$.

In this model, there is no interaction effect of the two treatments.

Welfare. Compliers (those who would not ask but would go) are better off by $B - \alpha_e C^*$. Households where the elder would not have asked in control *and* the caregiver says no are worse off by some costs (ϕ_c, ϕ_e) for the caregiver and elder, respectively.

C.2.7 Model 2: Asking costs, scaled by elder altruism

Setup. Suppose there is a asking costs k that gets incurred once the caregiver is informed, regardless of whether the caregiver agrees to go or not. The caregiver being informed causes cost k to the caregiver and cost $\alpha_e k$ to the elder (who internalizes this asking cost at the same rate that they internalize actual costs of check-up).

Take-up predictions. We get all our main predictions. First, informing the caregiver improves take-up more for altruistic elders: it increases take-up by $\alpha_e k$. The “do not ask but would go” set of compliers is driven by altruistic elders being more sensitive to the asking costs (rather than in model 1, in which altruism is just scaling up misperceptions). Second, more altruistic elders are more responsive to costs C .

There is no interaction effect of the two treatments.

Welfare. Compliers are better off by $B - \alpha_e C$. Welfare is worse for all households where the elder would not have asked in control, by $(\alpha_e k, k)$ for elders and caregivers, respectively.

C.2.8 Model 3: Uncertain rejection costs, scaled by elder altruism

Setup. Let $-\phi$ be a cost to the caregiver if they are informed but do not help. This could capture awkwardness of rejection, guilt from not helping, or relationship strain.

The elder internalizes this cost of informing at rate $-\alpha_e$. But the elder faces uncertainty about this cost and have belief $\hat{\phi} \sim \hat{f}$. Note that we need uncertainty in order for the rejection cost to factor directly into the elder's asking decision, otherwise it is only ever an off-equilibrium cost that never gets incurred.

Now the elder's asking condition is:

$$\hat{p}(B - \alpha_e C) + (1 - \hat{p})(-\alpha_e \hat{\phi}) > 0$$

where $\phi^* = C - \alpha_e B$ is the cost that makes caregiver indifferent; $\hat{p} = 1 - \hat{F}(\phi^*)$ is the likelihood of caregiver saying yes; and $\hat{\phi} = E[\phi | \phi < \phi^*]$ is the expected rejection cost.

Take-up predictions. We get all our main predictions, but only in some cases. First, informing the caregiver improves take-up more for altruistic elders *only if* elders are more sensitive to rejection costs than costs of the check-up $\underline{\phi} > C$. When elders are more sensitive to rejection costs than actual costs (i.e. if the elder thinks that the caregiver is subject to a lot of pressure when subject to an ask that leads them to help when they would rather not be asked), then as altruism α_e varies, the willingness-to-ask is decreasing faster than the willingness-to-go. Otherwise we have that the treatment effect on take-up actually decreases with altruism α_e . Also, elder willingness to inform is more responsive to caregiver costs C when elder is more altruistic, just as in the other models.

In this model, there is a positive interaction effect between the transport and inform treatments: $(1 - p)\alpha_e \tau > 0$. **We don't find positive TEs empirically, though.**

C.3 Model extensions

C.3.1 Allow uncertainty about pressure

In this section, we discuss the implications for the model when we allow the elder to have uncertainty over their belief about the caregiver's pressure term ϕ , in addition to uncertainty over their belief about the caregiver's altruism term α^c .

Suppose that the elder believes that the caregiver's preferences (α^c, ϕ) follow the distribution $H(\hat{a}, p)$ with marginal distributions $\alpha^c \sim \hat{F}$ and $\phi \sim \hat{G}$. Then the elder's expected value of informing is given by the following expression:

$$V_{inform} = \int \int \mathbf{1}\{aB - \kappa > -p\} [B + \alpha^e(aB - \kappa)] d\hat{H}(a, p)$$

In the case that $\hat{F}, \hat{G}$ are independent, this expression simplifies to:

$$\begin{aligned} V_{inform} &= \int [B + \alpha_e(aB - \kappa)] \cdot Pr_{\hat{G}}(p > K - aB) d\hat{F}(a) \\ &= \int [B + \alpha_e(aB - \kappa)] \cdot [1 - \hat{G}(\kappa - aB)] d\hat{F}(a) \end{aligned}$$

This is the same expression as in the case where the elder has only point belief $\hat{\phi}$, except that now the elder's utility is weighted by the likelihood that the caregiver has a pressure term such that they would agree to attend the check-up. The elder evaluates and integrates this at each potential value of the caregiver's altruism parameter.

We can also differentiate this expression with respect to the elder's altruism to understand how willingness to inform varies with elder altruism. When we do this, as before, we can again decompose the expression into two portions: the range of caregiver altruism values that are high enough that the caregiver would want to help absent pressure ($\alpha^c > \frac{\kappa}{B}$) and values that are low such that the caregiver helps only in the presence of pressure ($\alpha^c \leq \frac{\kappa}{B}$):

$$\frac{\partial V}{\partial \alpha_e} = \underbrace{\int_{\frac{\kappa}{B}}^{\bar{\alpha}} (aB - \kappa) d\hat{F}(a)}_{\text{Caregiver agrees out of altruism}} + \underbrace{\int_{\underline{\alpha}}^{\frac{\kappa}{B}} (aB - \kappa)(1 - \hat{G}(\kappa - aB)) d\hat{F}(a)}_{\text{Caregiver agrees out of pressure}}$$

As before, the sign of the comparative static is still diagnostic of the elder's beliefs about the caregiver's motives when agreeing to help facilitate the check-up.

C Effect of Transport on Take-up

C.0.1 Treatment: Provide free transport

We consider the effect of providing free, arranged transport to and from the check-up on the take-up of the intervention, assuming only the elder is informed about the check-up.

On the elder's side, the effect of lowering caregiver costs is to increase elder willingness to inform the caregiver about the check-up (as long as selection effects do not dominate the direct effect of reducing caregiver costs). On the caregiver's side, lowering caregiver costs has a first order effect, increasing their willingness to attend the check-up.

Taken together, reducing transport costs has the impact of increasing elder willingness to inform, as well as directly increasing caregiver agreement, which overall increases take-up.

Under the free transport intervention, the check-up occurs when:

$$\int_{\frac{\kappa - \hat{\phi} - \tau}{B}}^{\bar{\alpha}} [B + \alpha_e(aB - (\kappa - \tau))] d\hat{F}(a) > 0$$
$$\alpha_c B - (\kappa - \tau) > -\phi$$